

Hematology

Saturday, November 29, 2025

Dr. Clarissa Skorupski



BONUS

Read on own

Acronyms Used in this Talk

AA	<i>Aplastic anemia</i>
ASH	<i>American Society of Hematology</i>
BMT	<i>Bone marrow transplantation</i>
CLL	<i>Chronic lymphocytic leukemia</i>
FFP	<i>Fresh frozen plasma</i>
FLC	<i>Free light chains</i>
LN	<i>Lymph node</i>
LPD	<i>Lymphoproliferative disorder (eg lymphoma)</i>
MGUS	<i>Monoclonal gammopathy of unknown significance</i>
DOAC	<i>Direct oral anticoagulants (eg dabigatran, rivaroxaban...)</i>
PB	<i>Peripheral blood</i>
PCC	<i>Prothorombin complex concentrates</i>
Plt	<i>Platelets</i>
PNH	<i>Paroxysmal nocturnal hemoglobinuria</i>
R-CHOP	<i>Rituximab - Cyclophosphamide - Vincristine - Doxorubicin – Prednisone</i>

Hematology Outline

- **Malignant hematology**
 - Myeloid disorders (AML, MPN)
 - Bone marrow failure disorders (AA, MDS, PNH)
 - Lymphoid disorders (ALL, LPD, plasma cell/MM)
- **Classical hematology**
 - Thrombosis
 - Bleeding
 - High yield anemia & thrombocytopenia topics
 - Transfusion medicine
 - Sickle cell disease
- **Bonus Content**
 - Blood films, Transfusion, Approaches to Cytopenias, Bonus Bleeding and Clotting, SCD and thalassemia
 - Choosing Wisely
 - Bonus MCQs + Oral scenarios

This is an overview!

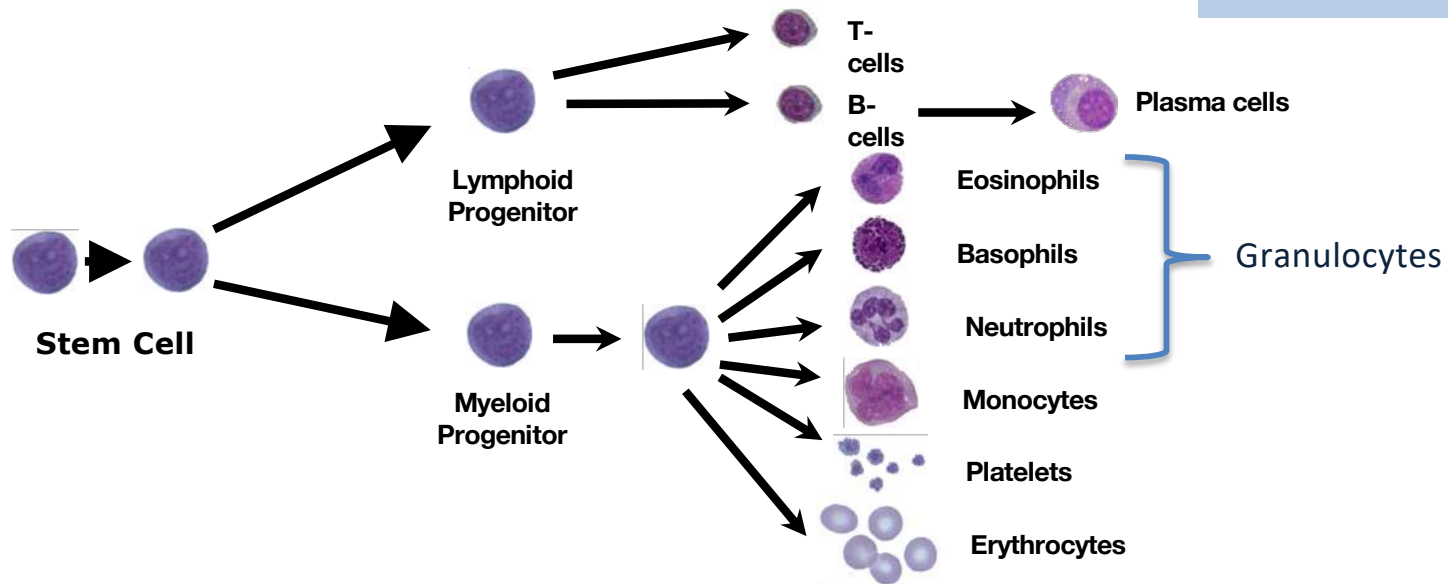
(eg) Nuances of Stem Cell Transplant and induction chemo for Leukemia etc are beyond scope of GIM.

Objective is to highlight relevant, high yield, and challenging topics in hematology that will help you pass the RC exam

Approach to Malignant Hematology

- **Myeloid vs. lymphoid** disorders
 - Myeloid = RBCs, platelets, monocytes, granulocytes
 - Lymphoid = B-cells, T-cells, Plasma cells

In 2023 the WHO and ICC updated Heme malignancy classifications, but this should not have an impact for the exam



Myeloid Disorders

1. Acute Myeloid Leukemia (AML)

- too many *immature* myeloid cells (i.e. myeloid blasts)

2. Myeloproliferative Neoplasms (MPNs)

- too many *mature* myeloid cells
 - too many RBCs = Polycythemia vera (PV)
 - too many platelets = Essential thrombocytosis (ET)
 - too many granulocytes = Chronic Myelogenous Leukemia (CML)

3. Chronic Myelomonocytic Leukemia (CMML)

- features of BOTH MPN and MDS
 - too many monocytes
 - myeloid cells are *dysplastic*

4. Myelodysplastic Syndrome (MDS)

- not enough myeloid cells (cytopenias) → bone marrow failure
 - ineffective hematopoiesis
 - myeloid cells are *dysplastic*

**All myeloid
disorders can
evolve into AML**

Lymphoid Disorders

1. Acute lymphocytic leukemia (ALL)

- too many *immature* lymphoid cells (i.e. lymphoid blasts)
- Blasts may arise from **B** cell (**B-ALL**) or **T** cell (**T-ALL**) lymphoid lineages

2. Lymphoproliferative disorders (LPDs)

- too many *mature* lymphoid cells
 - CLL: burden of disease is in the peripheral blood
 - Lymphoma: burden of disease is in lymph node
 - Waldenstrom Macroglobulinemia/Lymphoplasmacytic lymphoma

3. Plasma cell dyscrasias

- Too many plasma cells → overproduction of a single antibody / light chain
 - MGUS → smoldering myeloma → multiple myeloma
 - Primary AL amyloidosis

Malignant Hematology

Myeloid disorders

- **Acute Myeloid Leukemia (AML)**
 - too many immature myeloid cells (i.e. myeloid blasts)
- Myeloproliferative Neoplasms (MPNs)
 - Too many mature (normal) myeloid cells
 - Polycythemia vera = RBCs
 - Essential thrombocytosis = Platelets
 - Chronic Myelogenous Leukemia = Granulocytes
- Myelodysplastic Syndrome (MDS)
 - ineffective hematopoiesis
 - cytopenias, dysplastic cells

* All myeloid disorders can potentially evolve into AML

• Lymphoid disorders

- **Acute Lymphocytic Leukemia (ALL)**
 - too many immature lymphoid cells (i.e. lymphoid blasts)
 - Lymphoproliferative disorders (LPDs)
 - Too many mature lymphoid cells
 - CLL: burden of disease is in the peripheral blood
 - Lymphoma: burden of disease is in lymph node
 - Plasma cell dyscrasias
 - too many plasma cells → overproduction of single antibody
- * MGUS → smoldering myeloma → multiple myeloma
- * Primary amyloidosis

Acute Leukemia

AML (myeloid blasts, common in adults) vs **ALL** (lymphoid blasts, common in peds)

Etiology:

- *de novo* vs. 2° - prior chemo, transformed MDS/MPN, congenital d/o (e.g. Fanconi Anemia, Down Syndrome), benzene

– Organ dysfunction

- Skin → rash
- Mucosa → gum hypertrophy
- CNS → Neuro symptoms

Presentation:

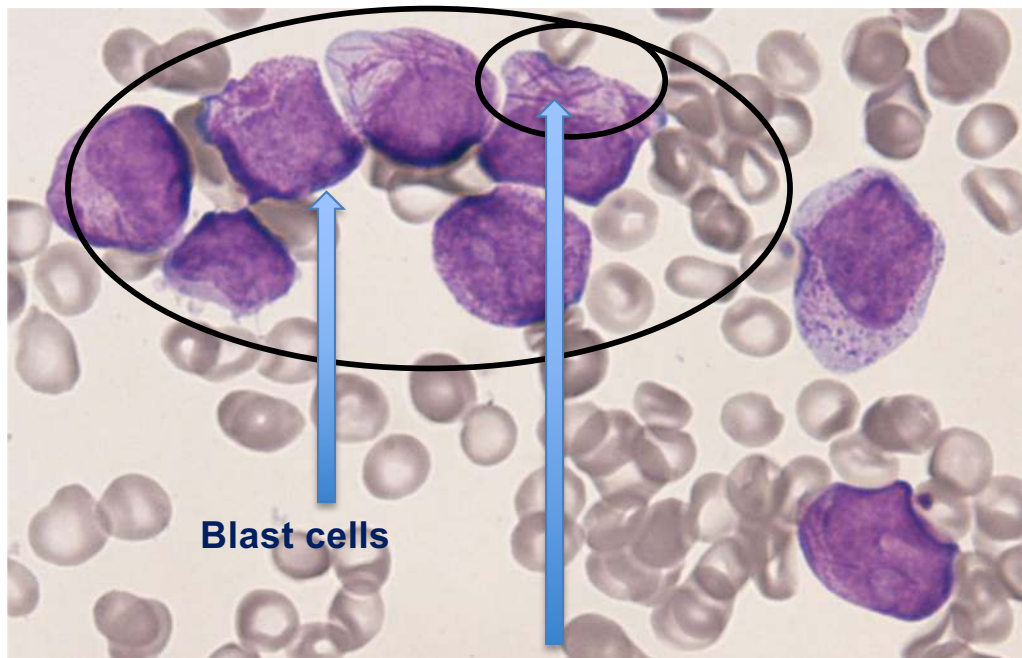
1. BM failure (loss of normal hematopoiesis)
 - Cytopenias:
 - Anemia (fatigue), thrombocytopenia (bleeding), neutropenia (infection)
2. Systemic symptoms (due to blasts)
 - Leukostasis, DIC, tumor lysis syndrome

Diagnosis:

- $\geq 20\%$ blasts in peripheral blood or marrow*
- WHO classification relies on genetic mutations
- Distinguishing AML from ALL:
 - Based on BM biopsy, flow cytometry, cytogenetics, molecular
 - **Auer rods** only in **myeloid neoplasms** → *Auer rods never normal*

RC CLINICAL PEARL:

Blast cells + Auer rods +/- DIC → think APL



Blast cells

Auer rods: red-stained, needle-shaped structures

APL (Acute Promyelocytic Leukemia)

- Subtype of AML
- T(15;17) PML-RARA
- Most curable AML type
- ↑early mortality from DIC:
 - Intracranial hemorrhage
 - Blood clots
- Urgent treatment: ATRA
all-trans retinoic acid (tretinoin)

Acute Leukemia Associated Emergencies

1. Tumor lysis syndrome

- ↑K, PO₄, uric acid (UA), ↓ Ca
- Renal failure, seizures, arrhythmias

Treatment:

1. Manage hyper-K, correct extended lytes
2. IVF (target UO 80-100 mL/m²/h)

AND EITHER:

3. Allopurinol

OR

4. Rasburicase if: AKI, ↑↑ uric acid (>535 umol/L), no response to allopurinol (**NOT in G6PD def**)

Zuckerman et al. How I Treat Hematology Emergencies in Adults with Acute Leukemia. Blood, 2012.

2. Leukostasis (AML>ALL>CML>>CLL):

- Rigid sticky blasts ↓flow → organ dysfunction
- Can affect any/all organ systems
- Lungs (dyspnea/hypoxia)
- CNS (confusion/visual changes)
- In AML if WBC >50-100, higher WBC in CML/ALL, rare in CLL

Treatment:

1. IVF
2. Cytoreduction (e.g hydroxyurea, leukapheresis, induction chemotherapy)
3. TLS prophylaxis
4. Avoid transfusion

Clinical Practice Guideline From the AABB; Ann Int Med 2015

Wada et al. Journal of Intensive Care 2014

Rolling and Ehninger. How I Treat Hyperleukocytosis in AML. Blood, 2015.

Acute Leukemia Associated Emergencies (2)

3. DIC

Non-APL:

1. Platelets (see “cytopenias”)
2. FFP (15 cc/kg)
Indication: If bleeding & PT or PTT >1.5X ULN
3. Fibrinogen concentrate (4g)
Indication: If bleeding & fibrinogen <1.5

APL:

1. ATRA
2. Platelets (see “cytopenias”)
3. Fibrinogen concentrate
Indication: fibrinogen <1.5
4. FFP (15 cc/kg)
Indication: if bleeding
No PT/PTT targets unless bleeding

4. Cytopenias

Platelet transfusion thresholds:

Non-APL:

- No bleeding: plt < 10
- Bleeding: plt < 50

APL:

- Plt < 30-50

5. Febrile neutropenia (see med onc lecture)

6. Extensive thrombosis (low threshold to investigate)

Stein et al. Best Practice and Research Clinical Hematology 2009.
Choundhry et al. Am J Hematol. 2012.
Wada et al for ISTH. J Thromb Haemost. 2013.
International Collaboration for Transfusion Medicine Guidelines.

Treatment of Leukemia

*specific regimens likely beyond GIM exam scope except for **APL**. Others included here as FYI

General Principles - Treatment of acute leukemias is intensive and comes with risks of treatment related morbidity (EX: bleeding, infection, hospitalizations/ICU admissions) and mortality.

- All patients (especially older) should undergo assessment of their **comorbidities/frailty** to determine **fitness** and **GOC**. Significant frailty may preclude induction chemotherapy and/or HSCT (allo-transplant)

AML

Curative intent treatment

1. Achieve remission: Induction chemotherapy *(EX: 7+3)
2. Prevent relapse: Consolidation +/- HSCT +/- Maintenance

Less fit adults → Hypomethylating agents (Azacitidine) +/- Venetoclax

Targeted therapies (EX: FLT3 inhibitors) can be incorporated based on disease subtype

APL

EARLY EMERGENT MANAGEMENT IS KEY ++ upfront mortality from hemorrhage/coagulopathy

Urgent Therapy

1. **ATRA** (all trans retinoic acid) – give **ASAP** if suspected APL (even before confirmation).
- Causes differentiation of APL blasts + reverses coagulopathy

Treatment → ATRA + Arsenic

ALL

B-ALL (~75% of adult cases), **T-ALL** ~1/3 – 1/2 of patients will be positive for t(9;22) (Philadelphia chrn)

Curative intent Treatment

Long (~2yr) pediatric adapted protocols (EX: Dana-Farber)

1. Induction
 2. Consolidation/Intensification
 3. Maintenance
- +Tyrosine kinase inhibitor if Ph positive
Intrathecal chemo included given high risk for CNS relapse/involvement

Internist's treatment of Acute Leukemia



ATRA initiation if APL is suspected (Auer rods, DIC)



Supportive transfusion

(Hgb >70, Plt > 10 *see APL slides for differences, give irradiated products. All blood in Canada is leukoreduced. See slides on indications for FFP, Fibrinogen if DIC)



Emergencies – TLS, DIC, Leukostasis

DIC, TLS labs always part of your workup

Assess for leukostasis especially when WBC > 50 - 100



Antimicrobials

Broad spectrum antimicrobials for febrile neutropenia



Goals of Care



Transfer to acute leukemia treatment center

MCQ #1 - 2026

A 51 year old male is referred to the clinic for evaluation of leukocytosis. He had presented to his family doctor with a 4 month history of worsening fatigue and abdominal discomfort. He otherwise has been previously healthy with no prior medical conditions, and no medications. He has a 5 pack year smoking history. On workup his CBC demonstrated a Hb of 110 (MCV 81), PLT of 219, and WBC of 28 with an increase in myelocytes (0.3), metamyelocytes (0.1), neutrophils of 18, eosinophils of 3.3 and basophils of 1.9. An abdominal US demonstrated normal liver contour and size, with an enlarged spleen at 17.3cm.

Which of the following is the least likely diagnosis?

1. Myelofibrosis
2. CML
3. MDS
4. CMML

Myeloproliferative Neoplasms (MPNs)

Definition: too many mature myeloid cells due to a marrow (clonal/malignant) process

- 4 types that you need to know for RC:

1. Chronic Myeloid Leukemia (CML)

→ too many granulocytes (neuts, eos, baso)

2. Polycythemia Vera (PV)

→ too many RBCs

3. Essential Thrombocytosis (ET)

→ too many platelets

4. Primary Myelofibrosis (PMF)

→ too much fibrosis (in the bone marrow)

Philadelphia-chromosome +ve
(BCR-ABL+)

Philadelphia-chromosome -ve
JAK2 mutation +

- >95% PV
- ~50% ET/PMF

Calreticulin (CALR) mutation +
• 30% of ET & PMF

****All MPNs can transform to acute leukemia and myelofibrosis****

Chronic Myeloid Leukemia (CML)

Diagnosis:

- Peripheral blood PCR for BCR-ABL (t(9;22) aka “**Philadelphia chromosome**”)
- Bone marrow biopsy to confirm phase of CML/prognosis (cytogenetics)
- 3 phases: chronic → accelerated → blast

Chronic phase CML

- ↑ WBC (mostly neutrophils + its precursors)
- Splenomegaly, mild constitutional sx
- BM + peripheral blasts < 10%
- Peripheral blood: left shift + basos + eos
- Treatment: **tyrosine kinase inhibitors (TKI)** (e.g. imatinib), life expectancy ≈ gen population
- Response assessments: monitoring of BCR-ABL transcript levels

TKI adverse effects – Vary based on specific TKI (1st gen – Imatinib, 2nd gen – Dasatinib, Bosutinib, Nilotinib, 3rd gen – Ponatinib, 4th gen – Asciminib).

Common AEs – myelosuppression, headache, arthralgias, diarrhea (B), pleural effusions/ pulm HTN (D), ↑ CV events (N, P)

Chronic Myeloid Leukemia (CML)

B) Accelerated phase CML*

- 10-19% blasts in peripheral blood or bone marrow
- Basophils $\geq 20\%$
- Presence of additional clonal cytogenetic abnormality in Ph+ cells

C) Blast phase CML

- $\geq 20\%$ blasts peripheral blood or BM (= acute leukemia, usually AML, 20-30% ALL), myeloid sarcoma
- Treatment as per acute leukemia + Tyrosine Kinase Inhibitors

*Other definitions exist as well

Polycythemia Vera (PV)

Presentation:

- CBC: ↑Hb +/- WBCs +/- PLTs
- Erythromelalgia, aquagenic pruritus
- Arterial / venous thromboembolic complications

Diagnosis: 3 major OR 2 major + 1 minor

Major:

1. Hb > 165/160 g/L (M/F) or HCT > 49%/48% (M/F)
2. BM: hypercellular for age, trilineage growth
3. JAK2 V617F (or JAK2 exon 12) mutation

Minor: Low serum EPO

Risk stratification (for **thrombosis**)

LOW RISK: Age < 60 **AND** no thrombosis history

HIGH RISK: Age ≥ 60 **OR** thrombosis history

Polycythemia Vera (PV)

Treatment:

ALL Patients → ASA 81 mg daily (ECLAP NEJM 2004)

- Optimize CV risk factors
- Hct < 45% with phlebotomy (CYTO-PV NEJM 2013)
- Monitor for new thrombosis/bleeding

LOW RISK (<60yo, no thrombosis)

Consider cytoreduction if
new clot/bleeding, frequent or intolerant
of phlebotomy, symptomatic
splenomegaly, disease related symptoms
not relieved by ASA

Hydroxyurea side effects – myelosuppression,
mucositis, skin ulcers, non-melanoma skin cancers

Interferon SEs – myelosuppression, retinopathy,
hepatotoxicity, rash, flu-like symptoms.

**Caution/avoid use in patients with history of
mood disorders**

HIGH RISK (≥60y or thrombosis)

Cytoreduction – Hydroxyurea or Interferon (if
young or pregnant)

2nd line – JAK inhibitors (Ruxolitinib) if
intolerant/resistant to previous cytoreduction
(RESPONSE trials)

Venous Thrombosis - Therapeutic anticoagulation
(consider dual therapy with ASA if thrombosis
occurs while on AC and if acceptable bleeding risk)

Essential Thrombocythemia (ET)

Presentation

- CBC: ↑ PLTs
- Vasomotor sx, thrombosis, bleeding (patients with ET are at risk of acquired VWD with plt $\geq 1000^*$)

***Acquired VWD:**

Increased risk of acquired von Willebrand Disease in MPN patients with extreme thrombocytosis. In bleeding patients or PLT > 1000 – 1500 send VWD screen to ensure VWD activity > 30%

Diagnosis: 4 major or first 3 major + minor

Major

1. Sustained ↑ PLT (≥ 450)
2. BMBx: ↑ mature hyperlobulated megakaryocytes
3. R/O other MPN/myeloid neoplasms
4. Presence of JAK2 (50%) or CALR (25%) or MPL (5%) mutation

Minor: NO reactive cause

Essential Thrombocythemia (ET)

Risk Stratification for Thrombosis (Revised-IPSET)

Very Low	No thrombosis AND age $\leq$ 60 AND JAK2 NEG
Low	No thrombosis AND age $\leq$ 60 AND JAK2 POS
Intermediate	No thrombosis AND age $>$ 60 AND JAK2 NEG
High	Thrombosis OR age $>$ 60 AND JAK2 POS

Hold ASA if PLT $>$ 1500 + suspected/confirmed acquired VWD

NCCN MPN Guidelines;
ASH-SAP; AJH 2024 Update

Treatment **ALL** patients $\rightarrow$ modify CV risk factors, monitor for thrombosis / bleeding

VERY LOW, LOW, INTERMEDIATE RISK

ASA 81mg daily

- Or observation alone in VL risk with no CV RFs

Consider cytoreduction if new clot/bleeding, symptomatic splenomegaly, uncontrolled ET sx, or extreme PLT $>$ 1500 / aVWD

HIGH RISK

ASA 81mg daily ⁺

Cytoreduction to target PLT $<$ 400 – 600.

- Hydroxyurea, Interferons (young or pregnant). 2nd line – Anagrelide.

Venous Thrombosis - Therapeutic anticoagulation

⁺ BID ASA – emerging evidence that BID dosing may improve PLT COX-1 inhibition in vivo. Consider if refractory ET sx

Myelofibrosis (MF)

Pathophysiology:

Clonal ↑ megakaryocytes/granulocytes
→ BM fibrosis → extramedullary hematopoiesis
- etiology: *de novo* (primary MF) vs. post-ET/PV

Presentation:

- Massive spleen, early satiety, B symptoms
- CBC: cytopenias
- Peripheral blood film:
 - Leukoerythroblastic = nucleated RBC + left-shift
 - “Teardrop” RBC cells
- 50% of Budd Chiari have an MPN

Diagnosis: all 3 major & 1 minor

Major	Minor
1. BM: fibrosis, megakaryocyte proliferation, atypia 2. JAK2 > CALR > MPL mut * 3. Rule out other MPN/MDS	1. Leukoerythroblastic film 2. LDH > ULN 3. Anemia 4. Leukocytosis ≥ 11 5. Palpable splenomegaly

Risk stratification: DIPSS, MIPSS70 scores

Treatment (based on risk stratification)

Low risk: supportive (Ex: anemia – EPO, danazol, transfusion)

High risk: allogeneic hematopoietic stem cell transplant is only curative option.

Splenomegaly → Hydroxyurea, JAK-inhibitors (e.g. Ruxolitinib), splenectomy

B Symptoms → JAK inhibitors

Radiotherapy for extra-medullary hematopoiesis

MPN Summary

	Mutation(s)	Peripheral Blood	Features	Treatment
Chronic Myeloid Leukemia (CML)	t(9;22) – BCR-Abl	Neutrophils/left shift, basophils , eos	Splenomegaly Mild B symptoms	Tyrosine Kinase Inhibitors (e.g. imatinib, dasatinib)
Polycythemia Vera (PV)	JAK2	Hemoglobin/Hct +/- WBC/PLT	Arterial/venous thrombosis Erythromelalgia Aquagenic pruritus	ASA Phlebotomy Hydroxyurea (high-risk)
Essential Thrombocytosis (ET)	JAK2, CALR, MPL	Platelets	Thrombosis Bleeding (PLT > 1000)	ASA (low-risk) ASA + hydroxyurea (high-risk)
Primary Myelofibrosis (PMF)	JAK2, CALR, MPL	Leukoerythroblastic Teardrop cells	Cytopenias Splenomegaly B symptoms	Supportive (transfusion, EPO, danazol) JAKi (splenomegaly, B symptoms) Allo-HSCT (high-risk)

Chronic MyeloMonocytic Leukemia (CMML)

Both an MPN and MDS (i.e MDS/MPN Overlap Syndrome)

Diagnosis:

- Peripheral blood **monocytosis** $\geq 1 \times 10^9/\text{L}$ and $\geq 10\%$ of total WBC for $>3\text{mo}$
- Blasts $<20\%$ (blasts $\geq 20\%$ = acute leukemia)
- Exclusion of other MPN (e.g. CML, ET, PV)
- Supporting criteria: dysplasia, cytogenetic/molecular abnormalities, partitioning of monocyte subsets

Prognosis:

- **15-30% get AML**
- Most important prognostic factor = % blasts

Treatment (likely beyond scope of GIM exam)

- Risk stratification → High risk patients refer for consideration of allo-HCT (only curative approach)
- Dysplasia: treat as MDS
- Proliferative: treat as MPN

Bone Marrow Failure Syndromes

- **Myelodysplastic Syndrome (MDS)**
 - Cytopenia(s) + dysplastic cells/genetic abnormalities/blasts
- **Aplastic anemia (AA)**
 - Hypocellular bone marrow
 - Primary vs. secondary
- **Paroxysmal nocturnal hemoglobinuria (PNH)**
 - Bone marrow failure
 - Hemolysis
 - Thrombosis

NOTE: THESE CAN OVERLAP

Myelodysplastic Syndrome (MDS)

Pathophysiology:

- Clonal, dysplastic cells → dysfunctional blood cells → risk of progression to AML
- Etiology: idiopathic vs 2° to DNA damage (radiation/chemo/hydrocarbons)

Definition:

Cytopenia **AND** <10-20% Blasts

AND at least one of:

- Dysplasia in ≥10% of cells in one myeloid lineage

OR

- Characteristic cytogenetic/ molecular features (ex TP53 mutation, del(5q))

Presentation:

- Older pt. w/ a cytopenia (often ↓Hb + ↑MCV)
- Blood film: hypolobated/hypogranular neutrophils, hypogranular platelets, macrocytosis, ↓ retics

Risk stratification (IPSS-M): - based on:

- 1) Clinical
- 2) Cytogenetics
- 3) Molecular

Mimickers of MDS:

- Vit B12/Folate def
- ETOH
- Cu deficiency or Zn toxicity
- Meds: MTX, azathioprine

Treatment

• **Low risk:**

- Supportive (EPO, transfusions, Fe chelation)
- Lenalidomide (5q deletion), immunosuppression, luspatercept

• **High risk:**

- Supportive (EPO, transfusions)
- Cure: allogeneic stem cell transplant
- Ineligible for transplant: hypomethylating agents (Azacitidine)

Aplastic Anemia (AA)

misnomer... should be called aplastic pancytopenia

Definition: stem cell disorder, ↓ blood cell production and ↓ marrow cellularity

Presentation: Previously healthy pt with pancytopenia, low reticulocytes, +/- ↑ LFTs

Etiology:

- Idiopathic
- 2°: pregnancy, autoimmune disease, drugs (NSAIDs, propylthiouracil, carbamazepine)
- Inherited BM failure syndromes (IBMFS)

*Inherited BM failure syndromes → Eg. Fanconi Anemia, Telomere Biology Disorders, Schwachman-Diamond syndrome. These will present in childhood typically.

Diagnosis

- BM biopsy: hypocellular (<25%)
- Severe: ANC <0.5, Plt <20, Retics <60
- **Workup** – Exclude IBMFS (skin fibroblast testing, chromosomal fragility, telomere length testing), eval for etiology, PNH flow cytometry

Treatment

- Required for severe AA
- Supportive transfusion (**irradiated**), immunosuppression (Cyclosporine + anti-thymocyte globulin (ATG) +/- Eltrombopag, or allo-hematopoietic stem cell transplant (HSCT)
- If drug-related: may obtain remission with cessation of drug alone

Paroxysmal Nocturnal Hemoglobinuria (PNH)

Definition: **Acquired clonal stem cell disorder** arising from mutations in *PIGA* leading to loss of GPI-anchored complement chaperone proteins (CD55 and CD59). Characterized by triad of:

1. **Bone marrow failure**
2. Chronic intravascular (+extravascular) **hemolysis** → uncontrolled complement activation, *DAT* negative
3. **Thrombosis** (venous + arterial). Common sites = **splanchnic, hepatic**. Also cause **multi-site** clots.

Presentation: thrombosis in **atypical locations** (ie. abdominal or cerebral veins), **anemia Sx**, **hemolysis Sx** (jaundice, red urine), pulmonary hypertension, signs of decreased Nitric Oxide (erectile dysfunction, dysphagia)

*Beware risk of infection/sepsis on complement inhibitor - Meningococcal vaccination (ACWY + Men B) mandatory when starting complement inhibitor therapy → start at time of 1st dose of complement inhibitor

Work-up

- Hemoglobinuria (rare), hemolysis labs, Cr (renal failure from pigment nephropathy)

Diagnosis

- Peripheral blood **PNH flow cytometry**
- Overlap w/ aplastic anemia → if pancytopenia → perform BM biopsy/aspirate

Treatment

- Therapeutic anticoagulation for thrombotic events
 - Anticoagulation alone not enough to ↓ risk of thrombosis
- Primary thromboprophylaxis → consider if high thrombosis risk/high disease activity (stop once hemolysis under control on complement inhibitors)
- Eculizumab/Ravulizumab (1st line) or Pegcetacoplan (complement inhibitors*) → ↓ hemolysis + transfusion needs + improved overall survival (not curative)

MCQ #1 - 2025

A 51 year old male is referred to the clinic for evaluation of leukocytosis. He had presented to his family doctor with a 4 month history of worsening fatigue and abdominal discomfort. He otherwise has been previously healthy with no prior medical conditions, and no medications. He has a 5 pack year smoking history. On workup his CBC demonstrated a Hb of 110 (MCV 81), PLT of 219, and WBC of 28 with an increase in myelocytes (0.3), metamyelocytes (0.1), neutrophils of 18, eosinophils of 3.3 and basophils of 1.9. An abdominal US demonstrated normal liver contour and size, with an enlarged spleen at 17.3cm.

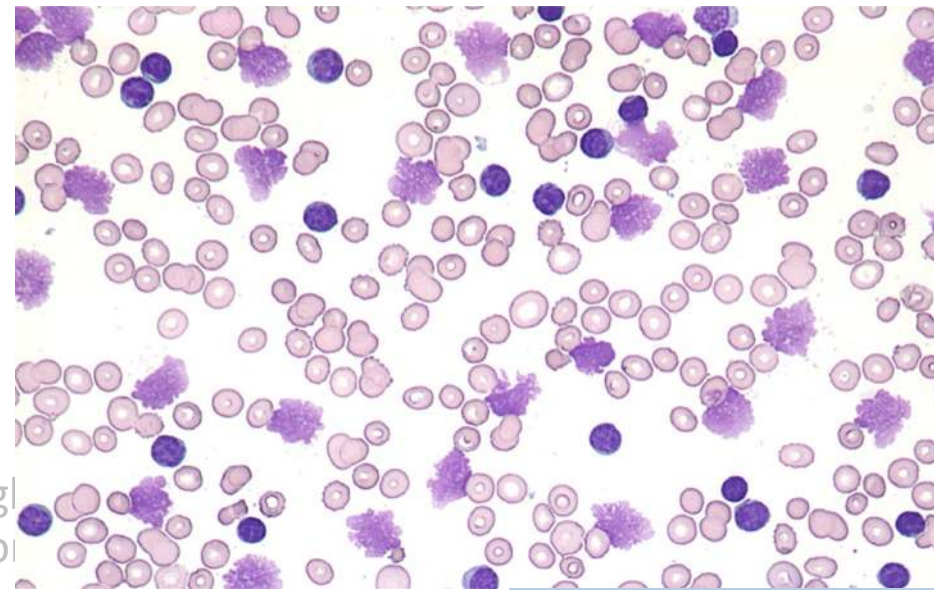
Which of the following is **the least** likely diagnosis?

1. Myelofibrosis
2. CML
3. **MDS**
4. CMML

This patient is presenting with a constellation of findings including constitutional symptoms, splenomegaly, as well as leukocytosis with evidence of a left shift. These features can be seen in various MPNs including MF, CML as well as CMML (MDS/MPN overlap). Presence of basophilia and eosinophilia here are most suggestive of CML > CMML or MF. MDS however more likely to present with single or multi-lineage cytopenias, macrocytosis, and less likely to cause splenomegaly

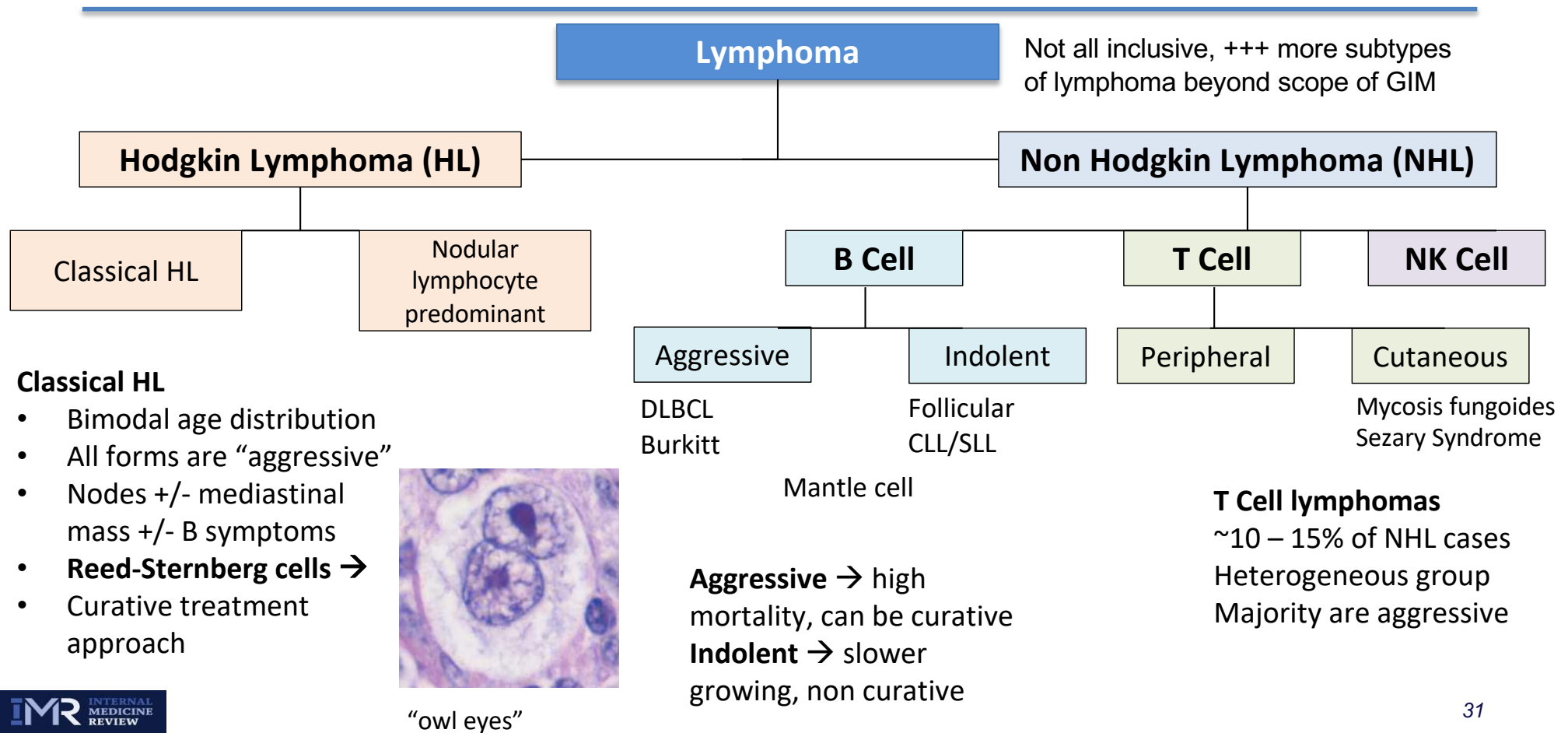
Lymphoid Disorders

- Acute Lymphocytic Leukemia (ALL)
 - Too many immature lymphoid blasts
- **Lymphoproliferative disorders (LPDs)**
 - Too many mature lymphoid cells
 - CLL: burden of disease is in the peripheral blood
 - Lymphoma: burden of disease is in lymph node
- Plasma cell dyscrasias
 - Too many plasma cells → overproduction of single
 - * MGUS → smoldering myeloma → multiple myeloma
 - * Primary amyloidosis



Source :ASH bank

Lymphoma Classification



Chronic Lymphocytic Leukemia (CLL)

Presentation:

- Older pt, well, isolated ↑WBC (lymphocytes)
- Peripheral blood: ++ small mature lymphocytes & “smudge” cells *

Diagnosis:

- **peripheral blood** flow cytometry
- **peripheral blood CLONAL** B-cell lymphocyte count ≥5 *Do not need BMBx to make dx

Complications:

- Lymphadenopathy / splenomegaly
- Cytopenias – crowding in marrow and/or autoimmune disorders (ITP / AIHA or both - *Evan's syndrome*)
- Richter's transformation → **sudden** ↑LN size → aggressive lymphoma (DLBCL)
- Secondary malignancies
- Hypogammaglobulinemia (↑infection risk)

Treatment:

- If asymptomatic: watch & wait
 - Ensure vaccines up to date, avoid LIVE vaccines
- Indications for treatment
 - Lymphadenopathy > 10 cm or symptomatic
 - Splenomegaly > 6 cm below CM or symptomatic
 - Progressive cytopenias → BM failure (Hb < 100, plts < 100)
 - AIHA/ITP with poor steroid response
 - Extranodal involvement (skin, lung, kidney, spine)
 - Significant constitutional symptoms
 - Treatment choices based on pt values, disease risk stratification
 - Chemoimmunotherapy (V-O) vs. tyrosine kinase inhibitors i.e. Ibrutinib, zanubrutinib (SE = bleeding, **A.fib**)

History/Exam for Lymphoma

B symptoms

- Weight loss (>10% in 6 mo)
- Unexplained fevers ($>38.3^{\circ}\text{C}$ x ≥ 2 weeks, no infection)
- Night sweats (drenching)

Adenopathy → Timeline (slow vs. rapid onset), pain/painless, compressive symptoms (EX: leg edema from large groin nodes, hydronephrosis) + hepatosplenomegaly symptoms/exam

CNS Involvement → focal motor/sensory neuro deficits, altered mental status, ↑ ICP, headaches, vision changes, bladder/bowel dysfunction

Risk Factors

- Infections
 - HIV (Hodgkin's, NHL: DLBCL, 1^oCNS lymphoma, Burkitt, 1^o effusion lymphoma)
 - Others → EBV (Burkitt's), H. pylori (gastric MALT), C. psittaci (ocular MALT), Hepatitis C (NHL), HTLV-1 (ATLL – Adult T cell leukemia-lymphoma)
- Immunosuppression: Post-transplant lymphoproliferative disease (PTLD), ↑ risk with certain DMARDs and biologics
- Chronic inflammation: Sjögren's (salivary gland MALT), Celiac or Crohn's (GI lymphoma), SLE & RA associated with NHL

Lymphoma Diagnosis/Work Up

***FNA NOT sufficient – need CORE or ideally EXCISIONAL LN BIOPSY FOR DIAGNOSIS*
Notify Pathology Dept – Surgeon to send for “Lymphoma protocol”**

Staging:

- (PET)-CT neck/chest/abdo/pelvis
- BMBx/aspirate **not routinely performed in all patients*
- +/- LP **depends on symptoms / risk factors*

Lugano Staging

- Stage 1: Single lymph node region/structure*
- Stage 2: 2+ lymph node region/structures on same side of diaphragm
- Stage 3: lymph node regions/structures on both sides of the diaphragm
- Stage 4: 1+ Extranodal organs (includes bone marrow)
- Disease-specific prognostic scores

Pre-Chemo Workup also includes:

- Baseline labs (CBC, chem, LDH, Uric acid)
- HIV status
- HBV status (HBsAg, HBcAb, HBsAb)
- TB status (depending on risk factors)
- LV function testing with ECHO or MUGA
- +/- Fertility preservation consult

Lymphoma Treatment

General Principles (*specific regimens likely beyond scope of GIM exam)

- Aggressive lymphomas (EX: HL, DLBCL) → **Curative** intent treatment
 - EX: R-CHOP chemotherapy for DLBCL (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisone), ABVD for HL (adriamycin, bleomycin, vinblastine, dacarbazine)
 - Radiation to sites of bulky disease
 - Steroids may be used for symptoms up front (*ideally post staging PET, watch for TLS)
 - TLS prophylaxis
- Indolent lymphomas (EX: Follicular) → **Non-curative** intent treatment
 - **Watch and wait strategy** usually appropriate unless developing symptoms/complications
 - EX: cytopenias, bulky nodal/extranodal disease, symptomatic/massive splenomegaly (16cm), multiple large nodal sites, B symptoms, organ compromise/compression
 - Treatment → Chemotherapy (EX: Bendamustine-Rituximab) +/- radiation

MCQ #2 2025

A 54-year-old gentleman is referred for assessment of recently diagnosed plasmacytoma following biopsy of a rib lesion. This was identified on CT chest after several months history of bony pain. Additional findings on imaging confirmed lytic lesions involving the ribs bilaterally. He is otherwise healthy with no relevant medications and has no symptoms aside from bony pain. Initial blood work demonstrates a hemoglobin of 120, leukocytes 5.3, platelet 400, creatinine 75, and normal calcium. These findings are most consistent with which of the following:

- 1) Multiple myeloma
- 2) Waldenstrom's macroglobulinemia (i.e. lymphoplasmacytic lymphoma)
- 3) Solitary plasmacytoma
- 4) Smoldering myeloma

Lymphoid Disorders

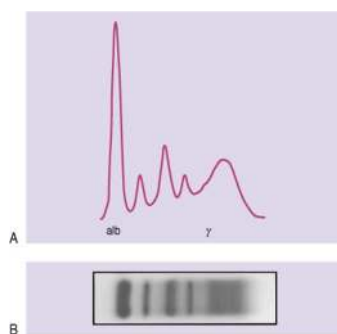
- Acute Lymphocytic Leukemia (ALL)
 - Too many immature lymphoid blasts
- Lymphoproliferative disorders (LPDs)
 - Too many mature lymphoid cells
 - CLL: burden of disease is in the peripheral blood
 - Lymphoma: burden of disease is in lymph node
- **Plasma cell dyscrasias**
 - Too many plasma cells → overproduction of single antibody
 - * MGUS → smoldering myeloma → multiple myeloma
 - * Primary amyloidosis

Work up (1)

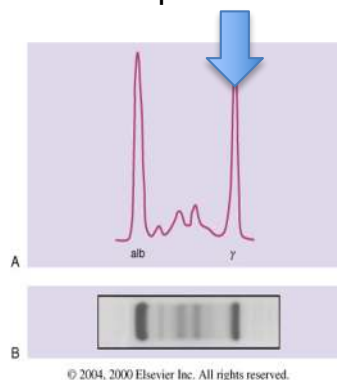
1. Protein electrophoresis

- Serum (SPEP) and 24 hr urine (UPEP)
- Quantity of M protein present

Normal protein electrophoresis



M spike
= M protein



2. Immunofixation

- Qualitative: determine “type” of M-protein (eg: IgG /IgA/IgM, κ vs. λ)

3. Serum free light chain (FLC) assay

- Normally κ & λ light chains present in ~ equal amount
 - » κ : λ ratio varies around 1 (0.26-1.65)
 - » Free light chains filtered by kidney but resorbed
 - » Monoclonal excess light chains supercede resorption capacity → urinary excretion
- Renal failure (*not MM*): ↓light chain excretion, κ ↑ ↑, λ ↑
 - » E.g. κ : λ ratio ~2
- Myeloma: clonal FLC will skew ratio
 - » E.g. IgG-κ myeloma → κ : λ ratio = 35

4. BMBx, aspirate

- Plasma cell burden; normally <1%
- Prove clonality (flow cytometry)
- Cytogenetics (risk stratify myeloma)

5. Other tests: Blood film (rouleaux), serum immunoglobulins, albumin, LDH and beta-2-microglobulin (prognostic)

Work up (2)

6. Imaging

Low Risk MGUS: no imaging needed

High Risk MGUS: low dose whole body CT (*more sensitive than skeletal survey*)

- if IgM – no bone imaging, do CT abdo/pelvis

Solitary Plasmacytoma: PET/CT or MRI

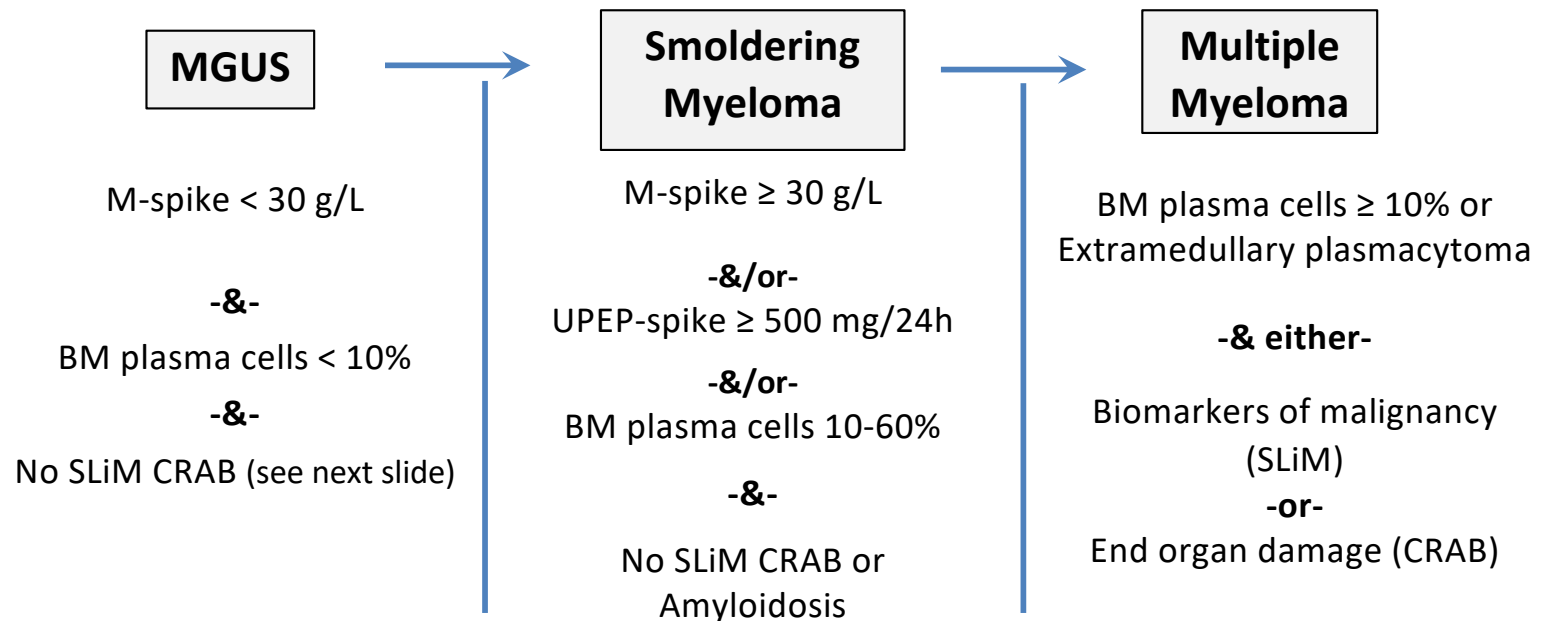
Smoldering Myeloma: low dose whole body CT

Myeloma: low dose whole body CT or PET/CT

Plasma Cell Dyscrasias

- **Clonal** proliferation of plasma cells
 - (usually) results in excess production of a **SINGLE** antibody & its associated light chain (either K or λ)

The MGUS-Myeloma Spectrum IMWG 2014 Criteria



For MM diagnosis → no cut-off level for M-spike, UPEP, SFLC etc.. → key to MM dx is plasma cell clone (on BM biopsy) AND end organ damage OR biomarkers (see next slide for details)

Myeloma - Diagnosis

BM plasma cells $\geq 10\%$ or plasmacytoma AND any SLiM CRAB = myeloma

Sixty: BM clonal plasma cells $\geq 60\%$

Light chains: Involved / uninvolved FLC ratio ≥ 100 , involved light chain should be $>100\text{mg/L}$

MRI: > 1 focal bone lesions that are $\geq 5\text{ mm}$ on MRI

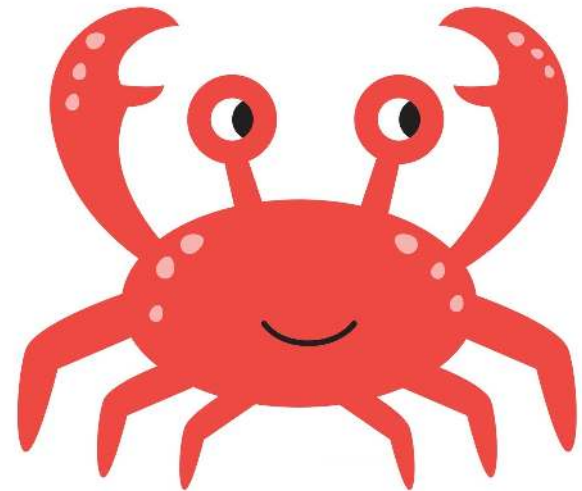
*Or any of the following if attributable to myeloma:
(eg if pt has diabetic CKD w baseline Cr 180 would not meet criteria)*

[Ca²⁺]: > 2.75 or $> 0.25\text{ mmol/L}$ above ULN

Renal: Cr $> 177\text{ }\mu\text{mol/L}$ or CrCl < 40

Anemia: Hb $< 100\text{ g/L}$ or 20g/L below the LLN

Bone: ≥ 1 osteolytic lesion (skeletal survey, CT or PET/CT)



Management

MGUS - based on risk of transformation into MM/waldenstrom/amyloidosis

High Risk (any 1 of):

1. M-protein ≥ 15 g/L
2. Non-IgG M-protein
3. Abnormal free light chain (FLC) ratio

Low risk MGUS: 2% risk of transformation @ 20y → no BM biopsy/bone imaging → CBC, SPEP, FLC, Ca, Cr in 6 mo → if stable FU if new symptoms or q2-3y

High risk MGUS: >20% risk of transformation @20 yrs → BMBx, whole body CT, CT abdo if IgM → CBC, SPEP, FLC, Ca/Cr in 6 mon → stable → rpt q1y lifelong

Smoldering Myeloma

- BM biopsy/aspirate & imaging to rule out bony lesions (CT or MRI) → rule out MM
- If no MM → monitor closely (+ annual re-imaging)

Myeloma

- No cure, but front line and second+ line treatments **rapidly** evolving in 2025
- Transplant (auto-HSCT) eligible ($\sim \leq 70$ yo, fit): chemo → auto-HSCT → maintenance therapy
- Transplant ineligible: chemo → maintenance therapy
- 2nd + lines of treatment → chemo combinations, BiTEs, CAR-T

BONUS

Read on own

Monoclonal Gammopathies of Clinical Significance (MGCS)

MGCS → Heterogeneous group of clinical syndromes where the presence of a monoclonal protein or FLC leads to **symptoms/end organ dysfunction** without meeting diagnostic criteria for MM or other lymphoproliferative disorder.

Treatments → Vary based on condition; may include plasma cell directed therapy, immunosuppression, PLEX

Renal	Neurologic	Hematologic	Cutaneous / Other
MGRS – Presentations depending on underlying subtype. <u>Need renal biopsy</u> - Proteinuria / nephrotic syndrome - Hematuria - Fanconi syndrome - Extrarenal sx	Peripheral neuropathy (PN) most common. May have isolated neurologic sx, or in setting of multisystem dx. POEMS * - DADS-M / IgM associated PN - CANOMAD - AL amyloid (PN, carpal tunnel, autonomic sx)	Cold agglutinin disease - Red cell agglutination, DAT positive hemolytic anemia (C3d), usually IgM related. MGTS (thrombotic significance) – Recurrent, unexplained thrombosis secondary to monoclonal anti-PF4 abs.	TEMPI syndrome (Telangiectasias, ↑Epo/Hb, M protein, P erinephric fluid, I ntrapulmonary shunting) Schnitzler syndrome (urticaria plus monoclonal IgG or IgM) Cyroglobulinemia Scleromyxedema Necrobiotic xanthogranuloma

***POEMS** – Polyneuropathy, **O**rganomegaly, **E**ndocrinopathy, **M**onoclonal plasma cell disorder, **S**kin Changes. Other manifestations include sclerotic bone lesions, edema, papilledema, ↑PLT, ↑VEGF. Diagnostic criteria/treatment likely beyond scope of GIM RC.

AL (Light Chain) Amyloidosis

Definition: Plasma cell dyscrasia (or lymphoma) with 2° monoclonal light chain deposition in organs (e.g. renal, cardiac, liver, peripheral nerves, GI, soft tissue infiltration, coagulopathy)

Diagnosis: IMWG criteria [diagnosis requires all 4]

1. Clonal plasma cell disorder (eg: evidence of M protein or abnormal FLC or clonal plasma cells in BM)
2. Presence of amyloid related systemic syndrome (eg renal, cardiac, GI, liver, neuropathy)
3. Amyloid on bone marrow or tissue biopsy → “apple green birefringence” on Congo red stain
4. Prove amyloid is light chain restricted (mass-spectrometry or electron microscopy)

Treatment: Similar to myeloma

Rajkumar et al. Lancet Oncol 2014

Waldenstrom's Macroglobulinemia

Definition:

1. Lymphoplasmacytic lymphoma
(an indolent lymphoma with features between B-cells & plasma cells)

AND

2. Monoclonal IgM in the peripheral blood

Presentation:

1. Symptoms 2° to infiltration
 - Anemia (BM infiltration), (**order CT Abdo**)
hepatosplenomegaly, lymphadenopathy
2. Symptoms 2° to autoimmune processes
 - Neuropathy
 - Autoimmune hemolytic anemia
 - Cold agglutinin disease

3. Symptoms 2° to monoclonal **IgM**:

- Type I cryoglobulinemia
- Hyperviscosity syndrome
 - Headaches, neurologic symptoms, mucocutaneous bleeding
- Bleeding

Management:

- Treat when symptomatic
- PLEX for hyperviscosity syndrome (removes serum IgM)
- Chemoimmunotherapy with rituximab

MCQ #2

A 54-year-old gentleman is referred for assessment of recently diagnosed plasmacytoma following biopsy of a rib lesion. This was identified on CT chest after several months history of bony pain. Additional findings on imaging confirmed lytic lesions involving the ribs bilaterally. He is otherwise healthy with no relevant medications and has no symptoms aside from bony pain. Initial blood work demonstrates a hemoglobin of 120, leukocytes 5.3, platelet 400, creatinine 75, and normal calcium. These findings are most consistent with which of the following:

- 1) **Multiple myeloma**
- 2) Lymphoplasmacytic lymphoma
- 3) Solitary plasmacytoma
- 4) Smoldering myeloma

This patient meets criteria for multiple myeloma based on **biopsy-proven plasmacytoma AND evidence of end-organ involvement (lytic lesions)**. LPL (i.e. Waldenstrom's) is a lymphoproliferative disorder and often presents with lymphadenopathy and elevated IgM. Solitary plasmacytoma and smoldering myeloma would not have CRAB symptoms.

Malignant Hematology Concepts

Acute Leukemias

- **Blasts** are the hallmark (peripheral blood or marrow)
- 20% blasts is diagnostic in most cases (PB or BM)
- Any myeloid cancer (MPN or MDS) can progress to AML

Lymphomas

- Small lymph nodes, slow growth, no B symptoms – think indolent lymphoma (e.g. follicular)
- Larger lymph nodes, fast growth, B symptoms – think aggressive lymphoma (e.g. DLBCL)

Aggressive lymphomas aim for cure

MDS vs. MF

- MDS – macrocytosis, dysplasia (hypersegmented neutrophil), cytopenias
- MF – leukoerythroblastic smear, tear drop cells, splenomegaly, cytopenias/cytosis

Chronic Leukemias

- Lymphocytes – mature lymphoma (e.g. CLL)
- Myeloid cells (neutrophils, basophils, eosinophils) – think CML

Splenomegaly

- Cancers of the lymphoid system (e.g. CLL, marginal zone, etc)
- Extramedullary hematopoiesis when the BM is unable to produce cells (e.g. MF)

Leukoerythroblastic film

- Nucleated RBCs (immature) + left shift (immature WBCs including myelocytes, metamyelocytes)
- DDx – marrow infiltration (MF, solid tumor), high stress (severe shock, G-CSF)

BONUS

Read on own

Malignant Hematology Summary

	Acute Leukemias / MDS	MPNs	LPDs	Plasma Cell Dx
Clinical Presentations	Cytopenias + complications → bleeding/DIC, fever/infection, fatigue, transfusions TLS, leukostasis CNS symptoms if neuro involvement	Cytoses (↑Hb, PLT, WBCs – eos, basos, left shift, tear drops) or cytopenias (MF) Thrombosis (arterial, venous, unusual sites), Splenomegaly Vasomotor sx, B sx	Aggressive (more sx, rapid onset) Indolent (slow) Lymphadenopathy Hepatomegaly and/or splenomegaly B symptoms, cytopenias	Asymptomatic (MGUS) Anemia, ↑ Ca ²⁺ , Cr Bony pain, fractures Amyloid – Cardiac dx, GI, hypotension, skin, neuropathy
Key Investigations	CBC + blood film (blasts , dysplasia), TLS labs, coags Flow cytometry Bone marrow – morphology, molecular studies +/- imaging (EX: CT head)	CBC + blood film (left shift, tear drops, blasts) JAK2 / CALR / MLP / BCR:ABL Bone marrow biopsy Imaging - EX: Abdo US, workup for thrombosis	Flow cytometry (blood) Lymph node biopsy (excisional preferred) CBC, TLS labs, LDH, Pan CTs, PET scan +/- bone marrow or LP	CBC, chem, Cr, trop, BNP SPEP, sFLC, UPEP , Immunoglobulins Bone marrow biopsy Imaging – CT myeloma, MRI spine/Pelvis, PET scan
Treatment Principles	Treatment stratified by disease risk + pt fitness Acute leukemias → APL (ATRA) AML/ALL chemo +/- HSCT MDS → transfusions, ESAs, luspatercept, HMA +/- HSCT (higher risk)	PV/ET – Treatment stratified by thrombosis risk. ASA +/- Cyto reduction, phleb (PV) MF – Risk stratify. Low risk – ESA Transfusions, JAKi. High – HSCT CML – TKIs (ex: Imatinib) CMML – MDS/MPN overlap	Treatment stratified by lymphoma subtype Aggressive (Ex: DLBCL) – curative intent chemo Indolent – watch/wait, treat if symptomatic or complications CLL – Observe or tx (EX: BTKi)	MGUS/Smouldering MM – observation MM – Systemic therapy +/- ASCT (if candidate). 2 nd line – BITEs/CART Amyloidosis – similar to MM WM – Chemo/Ritux

Classical Hematology

- **Thrombosis**
 - Anticoagulant choice + reversal
 - Cancer associated thrombosis
 - Thrombolysis + IVC filters
- Bleeding
 - Approach to abnormal PTT, INR
 - Von Willebrand disease
 - Immune thrombocytopenia (ITP)
- High yield anemia and thrombocytopenia topics
 - Thrombotic microangiopathies (TMAs)
 - Heparin induced thrombocytopenia (HIT)
 - Hemolytic anemias
- Transfusion medicine
- Sickle cell disease

MCQ #3 - 2026

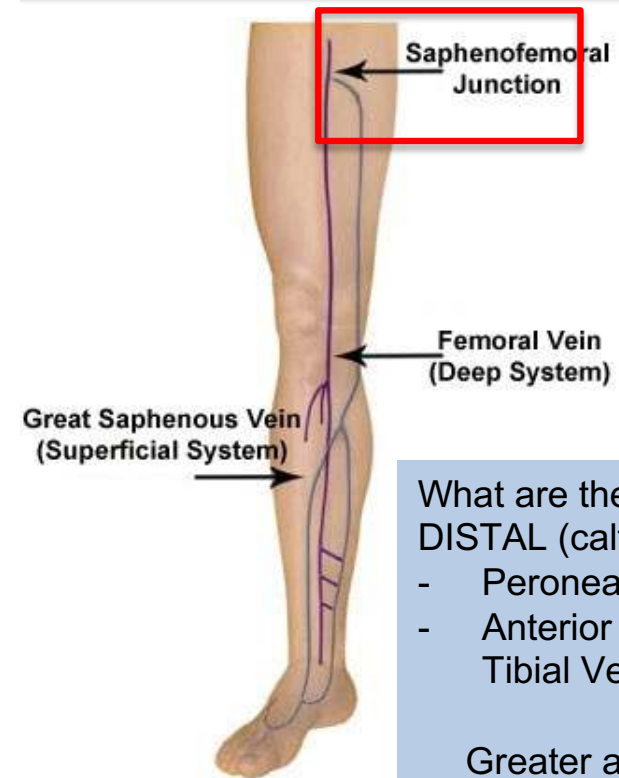
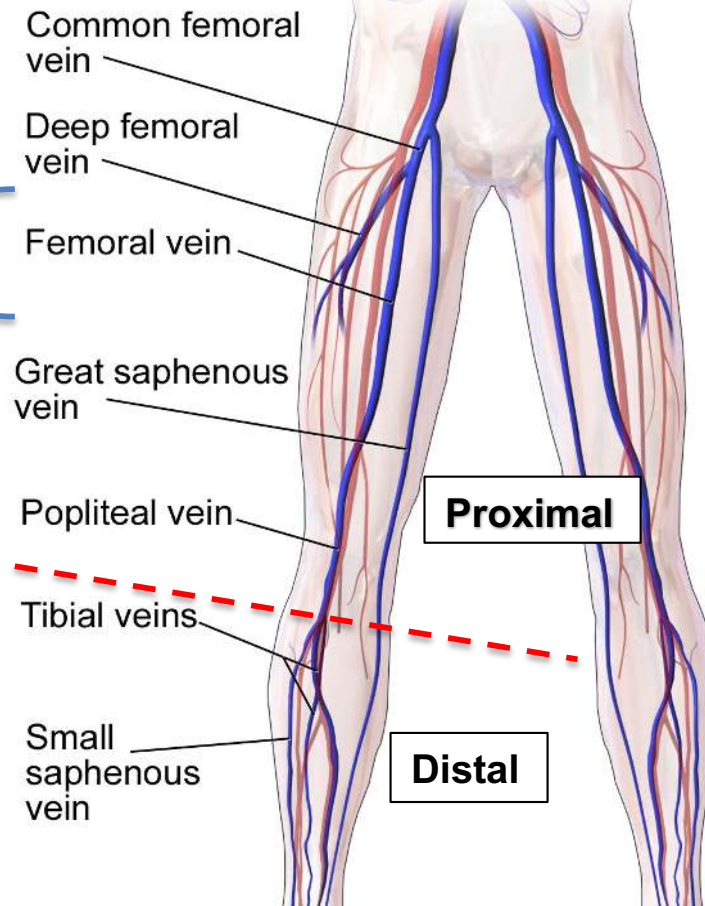
A 63 year old female is admitted to hospital for treatment of community acquired pneumonia and exacerbation of COPD. Her medical history includes hypertension, COPD, and dyslipidemia, and her home medications included amlodipine, inhaled fluticasone/vilanterol and atorvastatin. She is treated with intravenous antibiotics and steroids. On day 8 of her admission she develops pain, redness, and swelling to her right upper extremity near the IV site. A doppler US is performed that demonstrates a 2cm segment of thrombus in the cephalic vein.

What is your proposed management?

1. Remove IV. Start low dose Rivaroxaban 10mg daily x 45 days
2. Remove IV. Apply warm compresses to the area and topical Voltaren PRN
3. No change to IV, start Apixaban 5mg BID x 3 months
4. Remove IV, Rivaroxaban 10mg daily until discharge

Anatomy of the Deep Veins

Femoral vein sometimes called "superficial femoral vein" = a misnomer for this DEEP PROXIMAL vein



What are the DEEP DISTAL (calf) veins?

- Peroneal vein
- Anterior & Posterior Tibial Vein

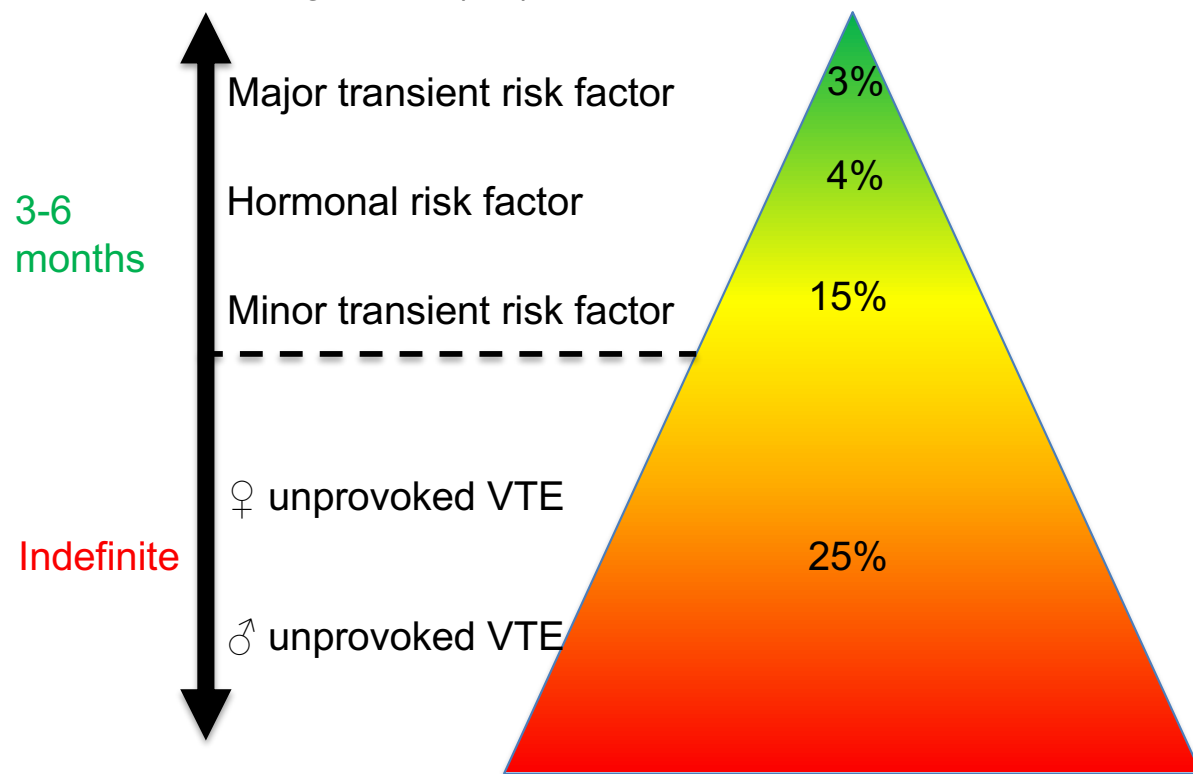
Greater and lesser saphenous veins are **SUPERFICIAL**

Risk Factors and Recurrence

Risk Factor	Examples
Major transient	- Surgery with general anesthetic for >30 min - Admission to hospital for acute illness with confinement to bed for 3+ days - C-section
Minor transient	- Surgery with general anesthetic for <30 min - Admission to hospital for acute illness for <3 days - Confined to bed out of hospital for 3+ days with acute illness - Leg injury associated with reduced mobility for 3+ days
Chronic	- Active cancer - High risk hereditary thrombophilia - Antiphospholipid Antibody Syndrome

Duration of anticoagulation (a/c)

Recurrence Risk at 5 yrs off A/C



Adapted from Moll, *The Hematologist*, 2018;
Thrombosis Canada; ASH 2020 VTE Guidelines (Oct 2020)

Treatment of Proximal DVT and PE

NON-CANCER Associated Rx – DOAC 1 st line if no contraindications*	
Provoked (or transient RF)	Unprovoked (or chronic RF)
Duration – minimum 3 months of anticoagulation (up to 6 months) Second provoked event – duration depends on the provoking factor. • If recurrent strong transient RF, treat 3-6 mo as above • If two weak transient provoking factors indefinite treatment may be appropriate	Duration – Indefinite[^] [^] - minimum 3-6 months of treatment with reassessment of bleeding risk and decision to treat indefinitely based on balance of risk:benefit. Prognostic models – HERDOO2, DODS Dose reduction – At 6-12 mo if clinical equipoise to continue vs. stop AC, dose reduction of apixaban (2.5mg BID) or rivaroxaban (10mg OD) can be used for long-term treatment <i>AMPLIFY-EXT, EINSTEIN-CHOICE, RENOVE (2025)</i>



Which DOAC? 2025 COBRRA Trial - Practice Changing Update

Apix vs. Riva for acute VTE → Apix sig ↓ clinically relevant bleeding vs. Riva with no diff in VTE recurrence. *Pending final published results.*

***LMWH preferred in high bleeding risk / if DOAC contraindicated – See “DOAC not for *all* VTE ” slide**

Treatment of Proximal DVT and PE

CANCER Associated

Rx – LMWH OR DOAC if no contraindications*

Anticoagulation – DOAC or LMWH*. 3 trials supporting DOACs: [HOKUSAI-VTE](#) (edoxaban), [SELECT-D](#) (rivaroxaban), [CARAVAGGIO](#) (apixaban)

- Use Apix/Riva if using upfront DOAC. Edoxaban trial had 5d therapeutic LMWH run in

Duration – At least **6 months**. If acceptable bleeding risk, **indefinite** therapy while active cancer, or on systemic treatment.

Dose reduction – Emerging evidence. [API-CAT \(2025\)](#) – After 6 mo AC, dose reduction (**Apix 2.5mg BID**) non inferior for VTE recurrence + ↓ bleed risk.



Which DOAC? 2025 COBRRRA Trial - **Practice Changing Update**

Apix vs. Riva for acute VTE → **Apix sig ↓ clinically relevant bleeding vs. Riva** with no diff in VTE recurrence. *Pending final published results.*

***LMWH preferred in high bleeding risk / if DOAC contraindicated – See “DOAC not for *all* VTE ” slide**

	Dabigatran	Rivaroxaban	Apixaban	Edoxaban
Target	Thrombin	Factor Xa	Factor Xa	Factor Xa
Standard Dose	150mg BID	20 mg OD	5 mg BID	60 mg OD
Minimum CrCl for A. Fib	>30	>15	>15	>30
Minimum CrCl for VTE	>30	>30 (caution if 15-29)	>30 (caution if 15-29)	>30
Antidote?	Idarucizumab (PRAXBIND)	Adenexanet alfa if available PCC 2000 units	Adenexanet alfa if available PCC 2000 units	Adenexanet alfa if available PCC 2000 units
Health Canada Approved Indications (Clinical Trial)				
Non-valvular Afib	(RELY) *dose 110bid if: age ≥80, or ≥75 w/ bleeding risk incl CrCl30-50, P-gp inhib	(ROCKET-AF) *dose 15mg OD if CrCl 30-49	(ARISTOTLE) *dose 2.5bid if 2 of: Cr≥133, age≥80, wt ≤ 60kg for AFIB – not studied to dose reduce for <u>acute</u> dvt	(ENGAGE –AF) 60mg OD, see 30mg dose restriction below as it applies here too.
Acute DVT	(RE-COVER) After 5-10d of parenteral therapy	(EINSTEIN) 15 mg BID x 3 wk then 20 mg OD <i>At 6-12 mo, if clinical equipoise can decrease dose to 10mg PO daily for extended duration treatment (EINSTEIN-CHOICE)</i>	(AMPLIFY) 10 mg BID x 7d then 5 mg BID <i>At 6-12 mo, if clinical equipoise can decrease dose to 2.5mg BID for extended duration treatment (AMPLIFY-EXT)</i>	(Hokusai-VTE) After 5d LMW/heparin CrCl >60= 60mgOD. CrCl 30-60 or weight <60kg or on P-glycoprotein inhibitors (eg dronedarone!) = 30mg OD
Prophylaxis post TKA/THA	(REMODEL) (RENOVATE)	(RECORD)	(ADVANCE)	(STARS – TKA) not licensed in Canada yet for this indication.

Caution : DOAC not for *a//* VTE

CAUTION IN:

1. **Cancer** – Unresected intraluminal GI or GU malignancy, high risk CNS (glioma, RCC, melanoma)
2. **Renal** – CrCl < 30 (Apix < 25; up to < 15 but limited data), or dialysis
3. **High clot burden** – Extensive/sub-massive PE, post-thrombolysis, iliofemoral DVT
4. **Bariatric surgery** – LMWH for first 4 wks–3 months | other malabsorption syndromes = consider LMWH
 - Riva absorbed stomach/duodenum, Edoxa proximal small bowel, Apix mostly absorbed in prox and distal small bowel.
5. **Drug-drug interactions:** (eg - amiodarone ↑ levels of all DOACs [see Thrombosis Canada DDI guide])
6. Recent **bleeding**

Rivaroxaban or Apixaban appropriate for any BMI/weight (except post bariatric sx.)

CONTRAINDICATION IN:

1. **Liver failure:** Child Pugh B and C *some may use DOAC with caution in CP B cirrhosis, avoid use in CP C
2. **Anti-phospholipid antibody syndrome:** arterial thrombosis, triple positive, small vessel thrombosis (livedo reticularis, nephropathy)
3. **Pregnancy AND Breastfeeding**
4. **Drug-drug interactions**

(e.g. doxorubicin, cyclosporine, carbamazepine, phenytoin, azoles, protease inhibitors – bleeding risk): apix/riva CYP3A4/P-gp; dabi/edox P-gp)

5. **Platelets < 50**

PAXLOVID (ritonavir) = contraindicated to use w/ apix and riva

When to Use Warfarin? “Be CALM”

Breastfeeding

- INR generally 2-3 depending on indication for anticoagulation. DOACs contraindicated, LMWH is OK (but patients don't like injections and \$\$\$ compared to warfarin!).

CKD (stage 4/5)

- INR generally 2-3 depending on indication for anticoagulation

Antiphospholipid Ab Syndrome

- APLA Syndrome associated VTE [INR 2-3]

LV Thrombus* (+ “Likes warfarin” – pt on warfarin stable for years, especially FRAIL ELDERLY (FRAIL-AF Trial))

- ***LV Thrombus:** 2023 AHA guideline suggests DOACs may be non-inferior to VKA based on MA of pooled small RCTs and retrospective studies. DOAC may be of particular benefit for those in whom INR checks difficult/impractical

Mechanical Valve

- Mechanical valves [INR 2.5-3.5 – mitral, INR 2-3 - aortic]

OTHER INDICATIONS:

Bubonic plague (for the extermination of rats!)



Role of Thrombolysis in VTE



DVT

- Limb-threatening DVT (phlegmasia cerulea dolens)
- ASH: selected younger patients with iliofemoral DVT at high risk of Post thrombotic syndrome (and low risk for bleeding)

PE

- Hemodynamic instability (sBP <90 for **over 15 mins**) with no high bleeding risk (Gr 2B)
- IV thrombolysis with tPA
- PEITHO trial (NEJM 2014): thrombolysis vs. placebo in patients with “intermediate risk PE” (i.e.: RV dysfunction on ECHO/CT, elevated troponin)

	Thrombolysis	Placebo	
Hemodynamic decompensation within 7d	2.6%	5.6%	P=0.02
Extracranial hemorrhage	6.3%	1.2%	P<0.001
Stroke	2.4%	0.2%	P=0.003
Mortality within 7d	1.2%	1.8%	P=0.42

Bottom line:

Only thrombolyse PEs with hemodynamic instability, not for RV strain or ↑ trop

IVC filters: likely the wrong answer!

Choosing Wisely: “Don’t use IVC filters routinely in patients with acute VTE”

Thrombosis Canada: “Use of a (vena cava filter) VCF should be considered judiciously given the lack of high quality evidence.”

CHEST 9th ed/10th ed guidelines:

IVC filter only if acute prox DVT (or PE*) & contraindication to anticoagulation (grade 1B)

- *If there is no prox DVT, do not insert an IVC filter; serial US’s can be used
- IVC filters should not be used as adjunct treatment for DVT or PE on anticoagulation
- Uncertainty of benefit in anticoagulated pts with severe PE (e.g. with hypotension)

Complications of IVC filters:

- Filter placement – bleeding, infection, malposition of filter, guidewire entrapment
- Post-procedure – DVT, hematoma, AV fistula
- Long-term – filter erosion/migration/embolization, chronic/recurrent VTE
 - Filters may decrease risk of PE but increase risk of DVT and have no effect on mortality (Bikdeli, et al., 2017)

Reversal Agents for Anticoagulants

Warfarin

Non-bleeding:

- INR >9: hold warfarin + VitK 2.5-5 mg po
- INR 4.5-9: hold warfarin + decrease dose

Non life-threatening bleed:

- Vit K + supportive

Life threatening bleed/imminent procedure:

- IV Vit K + prothrombin complex (PCC)
 - PCC dosing per INR: INR 1.5-3: PCC 1000U;
INR 3-5: PCC 2000U; INR > 5: PCC 3000U

LMWH

Life threatening bleed:

- Protamine 25-50 mg

Heparin

- Protamine dose based on last admin

DOACs

Consider TXA

Non life-threatening: supportive

Life-threatening bleed: supportive +

- Dabigatran:
 - Idarucizumab (Praxbind) 2.5g x 2 doses;
consider dialysis (~65% removal in 4 h)
- Apixaban/Rivaroxaban/Edoxaban:
 - 4 factor PCC (e.g. Octaplex, Beriplex);
2000U (can be repeated)
 - Andexanet alfa (Ondexxya) - Health Canada approved
but not widely available - ~\$27k per patient)

Thrombosis Canada: "Warfarin: Management of Out-of-Range INRs"

Thrombosis Canada: "NOACS/DOACS: Management of Bleeding"

Thrombosis Canada: UFH and LMWH

ASH 2018 Guideline Blood Advances 2018 2:3257-3291

The AABB Choosing Wisely Campaign of ABIM, RE-VERSE AD. NEJM. 2015;373:511-20

UPRATE Trial. ASH 2017; 130: 1706-12.

Distal DVTs, Subsegmental PEs, Superficial Vein Thrombosis

Distal (calf) DVT	Consider full-dose anticoagulation for: • Severe symptoms or multiple deep veins involved • Active cancer or irreversible risk factor • ≥ 5 cm long or close to popliteal vein • +D-dimer • Progression on repeat U/S If high bleeding risk or no indication to treat → monitor with serial U/S
Subsegmental PE (with negative leg dopplers for DVT*)	Controversial, but favor treatment • Consider treatment if: active cancer/other thrombotic risk, symptomatic, high D-dimer • Recurrent VTE incidence 2.1% if single SSPE, 5.7% multiple SSPE
Superficial Vein Thrombosis	Questions to ask yourself → <u>Size</u> of clot? + <u>distance</u> to SFJ or SPJ? • ≤ 3 cm from SFJ or SPJ → FULL dose anticoag x 3 months • > 3 cm from SFJ/SPJ and ≥ 5 cm long → prophylactic anticoagulation x 45d (fondaparinux 2.5 mg sc daily or Rivaroxaban 10 mg po daily or LMHW proph/intermed dose) • > 3 cm from SFJ/SPJ + < 5 cm long → NSAIDs and monitor with serial U/S. ** Exceptions: prophylactic anticoag in pregnancy, cancer, surgery, trauma, prior hx of SVT/DVT • SVT due to IV cannula – remove IV, supportive NSAID/warm compresses (no anticoag)

Unusual Sites of Thrombosis (1/2)

Anatomic location	Symptom	Risk factor(s)	Anticoagulation duration
Upper extremity DVT: brachial, axillary, subclavian, internal jugular NOT basilic or cephalic (these are considered superficial veins)	Upper extremity pain, erythema, arm swelling May see venous collaterals develop	Majority provoked (ie. Catheter related) Unprovoked: “effort” thrombosis ie thoracic outlet syndrome	Duration: 3-6 months *in catheter associated VTE consider prolonged anticoagulation while catheter remains in situ Thoracic outlet: vascular surg
Cerebral venous sinus thrombosis - EX superior sagittal sinus, transverse sinus, sigmoid sinus, cavernous sinus Need CT-Venogram or MR-Venogram for diagnosis	Sinus thrombosis (mass effect): headache, visual loss, papilledema, sz, hydrocephalus Cortical vein thrombosis (venous ischemia/hemorrhage): neuro deficit May need neurosurgery if severe ↑ICP	Smoking Estrogen: OCP/pregnancy MPN; PNH; Heritable thrombophilia; APLA; VITT Parameningeal infection Neurosurgery Trauma	Upfront anticoagulation with LMWH or UFH is preferred *NOTE – ICH as a consequence of CVST is not a contraindication to anticoagulation Long term → DOAC or warfarin (LMWH preferred in pregnancy) Provoked: 6 months Unprovoked: consider indefinite Thrombophilia: indefinite

Unusual Sites of Thrombosis (2/2)

Anatomic location	Symptom	Risk factor(s)	Anticoagulation duration
Splanchnic : portal, mesenteric	Often clinically silent Acute with rapid extension: portal hypertension, acute GI bleeding, decompensation of previously stable cirrhosis, intestinal ischemia (severe pain)	Cirrhosis; Malignancy; APLA; PNH; IBD; intra-abdominal infection; trauma or GI surgery <i>Up to 30% of non-cirrhotic PVT have MPN, JAK2 testing recommended*</i>	Acute PVT: 3-6 months or longer. Consider thrombolysis, TIPS. Chronic PVT: dependent on RFs, case by case, consider treatment to avoid extension and ↑ portal hypertension (expert opinion)
Splanchnic: splenic	Abdominal pain most common	Embolic: Afib, endocarditis, aortic dissection, atherosclerosis/thrombi, PFO In situ: APLA, sickle cell, cancer, MPN Congestion: heme cancer, Gaucher's	Treat underlying cause; anticoagulant if embolic
Splanchnic: hepatic vein (Budd Chiari)	Fulminant liver failure: acute liver failure, encephalopathy, ascites, portal HTN	MPN; PNH; APLA; IBD; cancer	Acute: consider thrombolysis, TIPS; indefinite anticoagulation

FYI: ASH 2023 Thrombophilia Updates

Low certainty of evidence (and controversial)! These guidelines not widely adopted in Canada. Thrombosis Canada still does not recommend routine Thrombophilia testing

ASH Suggests thrombophilia testing (if positive, consider **indefinite** treatment/prophylaxis)

Specific scenario –Test for FVL, PT, ProtC/S, AT3, APLA

Non-surgical major transient risk-factor after short-term treatment

VTE in pregnancy or post-partum after initial treatment

Combined OCP-related VTE after initial treatment (Rx)

Cerebral Venous Sinus Thrombosis after initial Rx

Splanchnic venous thrombosis without cirrhosis after initial treatment

See OB Medicine lecture for pregnancy-VTE considerations

Family History

FHx of VTE **and** thrombophilia prior to minor provoking risk factor - test if AT/PC/PS deficiency [NOT FVL or PT]

FHx of thrombophilia but **no** VTE prior to minor provoking risk factor - test if AT/PC/PS deficiency

FHx of VTE **and** thrombophilia prior to combined OCP - test if AT/PC/PS deficiency

FHx of VTE **and** thrombophilia prior to combined HRT - test if AT/PC/PS deficiency

FHx of VTE **and** thrombophilia and receiving systemic therapy for cancer - test for all hereditary thrombophilias

MCQ #3 - 2026

A 63 year old female is admitted to hospital for treatment of community acquired pneumonia and exacerbation of COPD. Her medical history includes hypertension, COPD, and dyslipidemia, and her home medications included amlodipine, inhaled fluticasone/vilanterol and atorvastatin. She is treated with intravenous antibiotics and steroids. On day 8 of her admission she develops pain, redness, and swelling to her right upper extremity near the IV site. A doppler US is performed that demonstrates a 2cm segment of thrombus in the cephalic vein.

What is your proposed management?

1. Remove IV. Start low dose Rivaroxaban 10mg daily x 45 days
2. **Remove IV. Apply warm compresses to the area and topical voltaren PRN**
3. No change to IV, start Apixaban 5mg BID x 3 months
4. Remove IV, Rivaroxaban 10mg daily until discharge

This patient would be considered to have an upper extremity SVT (**cephalic = superficial vein**) secondary to IV cannulation. No role for low or therapeutic dose anticoagulation. Management supportively with removal of IV, topical NSAIDs and warm compresses is recommended. This patient should otherwise be on routine DVT ppx while admitted for their AECOPD.

Classical Hematology

- Thrombosis
 - Anticoagulant choice + reversal
 - Cancer associated thrombosis
 - Thrombolysis + IVC filters
- **Bleeding**
 - **Approach to abnormal PTT, INR**
 - **Von Willebrand disease, Acquired Hemophilia**
 - **Immune thrombocytopenia (ITP)**
- High yield anemia and thrombocytopenia topics
 - Thrombotic microangiopathies (TMAs)
 - Heparin induced thrombocytopenia (HIT)
 - Hemolytic anemias
- Transfusion medicine
- Sickle cell disease

Hemostasis

Primary Hemostasis

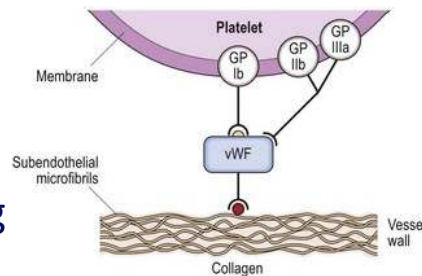
Clinical Presentation:

- Mucocutaneous bleeding
- Easy bruising
- Heavy menstrual bleeding
- Petechiae (low plts)

Labs: +/- ↓ plts, often coags normal

DDx

1. VWF Deficiency
Von Willebrand Disease (VWD)
2. Plt Disorder
↓ Quantity → e.g. ITP
↓ Quality → e.g. ASA, uremia, congenital



Secondary Hemostasis

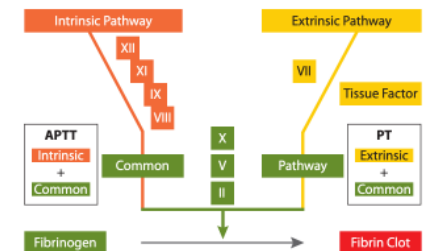
Clinical Presentation:

- Hemarthrosis
- Intramuscular bleeding
- Retroperitoneal bleeding

Labs: abnormal coag. tests

DDx [see slides specific to INR/PTT]

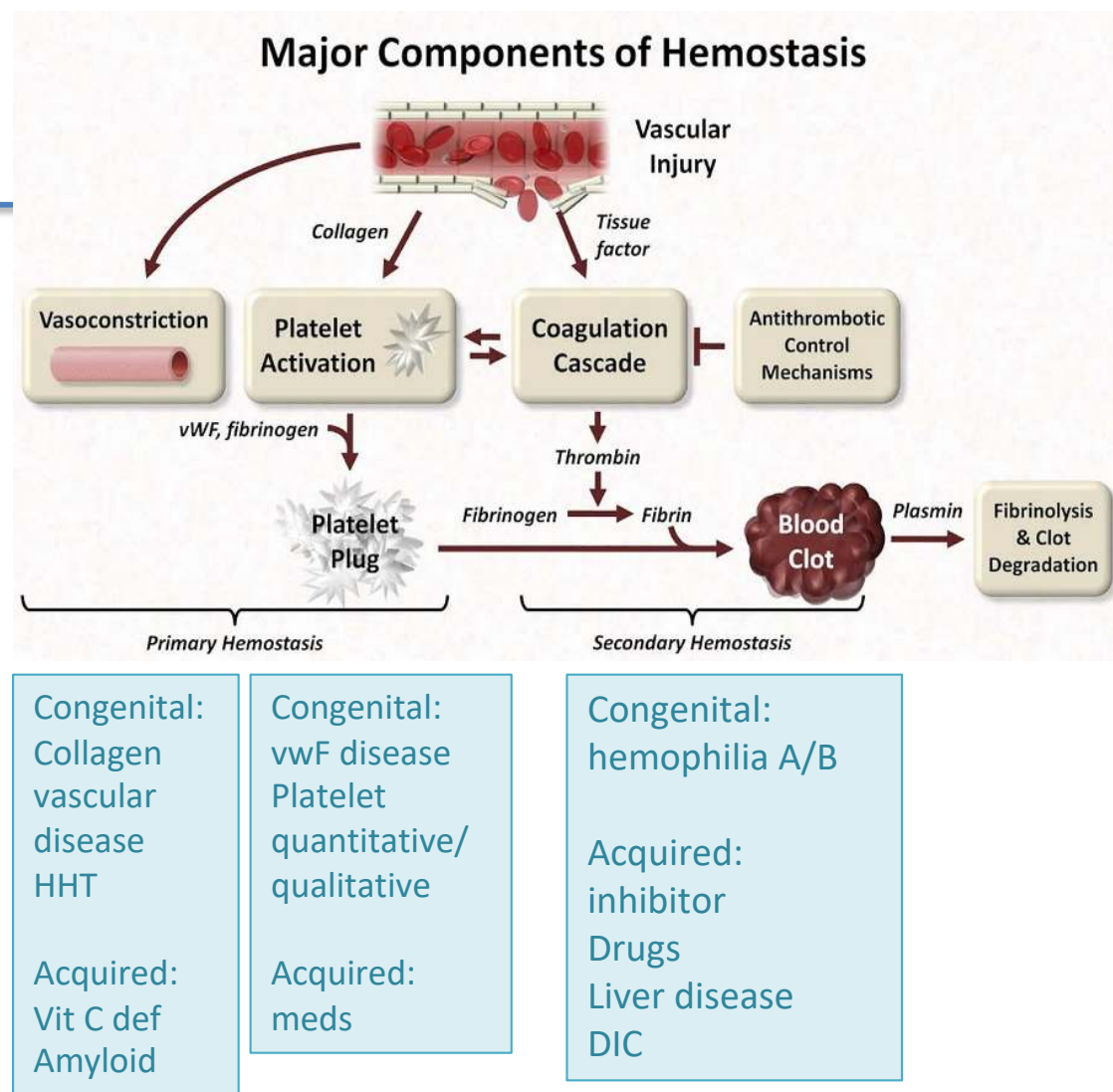
1. Coagulation factor deficiency
 - Hemophilia A (Factor VIII)
 - Hemophilia B (Factor IX)
 - Warfarin (II, VII, IX, X)
2. Coagulation factor inhibitor
 - Idiopathic/various diseases
 - Medications: DOACs/non VKA anticoagulants



Bleeding Disorders: Approach to Abnormal Bleeding

	Congenital	Acquired
Primary Hemostasis	Gray platelet Bernard-Soulier vWD	Uremia Drugs: antiplatelets (e.g. ASA) Acquired vWD
Secondary Hemostasis	Hemophilia Dysfibrinogenemia	FVIII Inhibitor Drugs: A/C
Structural Defect	HHT Ehler-danlos Collagen disorder	Trauma Iatrogenic

REMEMBER – A normal INR/PTT does NOT rule out a bleeding disorder.



Approach to Isolated Elevated PT or aPTT

Elevated PT (INR)

- Step 1 - rule out pre-analytical error, medication contamination etc.
- Step 2 = **50:50 mixing study**:
 - A. **PT corrects = factor *deficiency***
 - If FVII low (<50%) → factor deficiency confirmed
 - DDx: congenital/acquired FVII deficiency, vit K def., liver disease
 - B. **PT does NOT correct = *inhibitor present***
 - Check FVII level
 - If VII low, check inhibitor level

Elevated aPTT

- Step 1 - rule out pre-analytical error, medication contamination
- Step 2 - **50:50 mixing study**:
 - A. **aPTT corrects = *factor deficiency***
 - Measure factors VIII, IX, XI, XII (intrinsic pathway)
 - » Etiologies: Acquired vs. congenital factor deficiency vs. VWD
 - » Confirm by measuring relevant factor levels (<50% is low)
 - B. **aPTT does NOT correct = *inhibitor present***
 - Lupus anticoagulant (LA) testing
 - LA positive → r/o APLA (see rheum lecture)
 - » Presence of LA is a risk for THROMBOSIS not bleeding!
 - LA negative → factor inhibitor
 - » Measure factors VIII, IX, XI **then** inhibitor for low factor



Von Willebrand Disease

Most common bleeding disorder

- ↓quantity or quality of Von Willebrand Factor
- VWF: 1^o - plt adhesion and 2^o – protects FVIII

Presentation: mucocutaneous bleeding, heavy menstrual bleeding, easy bruising

Screening tool: **Bleeding assessment tool** (e.g. MCMDM1: score ≥4 significant)

REMEMBER – A normal INR/PTT does NOT rule out a bleeding disorder!!
If suspicious for abnormal bleeding, you must perform a validated bleeding assessment tool

Diagnosis:

- VWF Ag (quantity), VWF:Ristocetin Cofactor or VWF:GP1bR (VWF activity), FVIII level
- aPTT can be normal in VWD, not a good screening test
 - VWF level: ↑ with stress, estrogens, pregnancy ; ↓ *levels in blood group O*

Von Willebrand Disease

Diagnosis (Continued)

Type 1: Quantitative deficiency

↓VWF:Ag [$<30\%$] = ↓VWF Rco [$<30\%$] ie. concordant; **ratio** >0.7

VWF $< 30\%$ or VWF 30-50% + abnormal bleeding = diagnostic

Type 2: Qualitative deficiency of VWD ... multiple subtypes (2A, 2B, 2M, 2N) beyond scope of GIM exam

↓/↔ VWF Ag [$<30-2200\%$], ↓↓VWF:Rco [$<30\%$] ie. discordant; **ratio** <0.7

Type 3: No VWF produced

↓VWF:Ag [$<5\%$] = ↓VWF:Rco [$<5\%$]; ↓↓ factor VIII = Behaves like Hemophilia A

Treatment:

- Acute Bleeding/Peri-Op: TXA for all
 - DDAVP: boosts vWF level (can work for Type 1; **not** in type 2B/2N or 3)
 - Blood products:
 - plasma derived concentrates of vWF and FVIII (e.g. Humate P)
- Heavy menstrual bleeding: TXA +/- OCP/IUD

Acquired Factor VIII Deficiency/Hemophilia A

Pearl - Commonly missed diagnosis. Maintain high degree of suspicion especially in elderly pt with new unexplained bleeding/bruising

Definitions:

Hemophilia A – Congenital ↓ in Factor VIII

Hemophilia B - Congenital ↓ in Factor IX

- X linked recessive bleeding disorders
- Hemophilia A more common > Hemophilia B
- ~4,000 Canadians with Hemophilia A/B

Acquired Hemophilia A – Acquired antibodies against Factor VIII. **Medical Emergency**

- Rare ~1 in 1 mil, more common in elderly
- Causes → Autoimmune dx, Malignancy (AdenoCa, Heme malig), Pregnancy/Postpartum, Medications, Idiopathic
- Suspect in cases of **spontaneous bleeding / bruising** (see image) - subcutaneous bleeding most common > muscle > retroperitoneal/intra abdo bleeding and when **isolated ↑ in PTT**

Investigations

- Isolated ↑ in PTT
- Mixing study
- Factor VIII level and inhibitor assay
- + eval for underlying triggers/etiology

Treatment

- **Bleeding Control** – Bypassing agents (EX: FEIBA, rFVIIa, recombinant porcine Factor VIII). *Initial choice may depend on product availability in Blood Bank*
- FEIBA (factor eight inhibitor bypassing activity) contains non-activated FII, IX, X and activated FVII. Initial dosing usually 50 – 100 units / kg q8-12H
 - TXA contraindicated when using FEIBA ↑ thrombosis risk
- Inhibitor elimination – Steroids +/- Ritux +/- Cyclophosphamide
- **Consultation with Hemophilia Centre**



BONUS

Read on own

Hemophilia A and B

Pearl - Always consult patient treatment wallet card + notify patients' Hemophilia treatment centre of acute bleed episodes

Definitions:

Hemophilia A – Congenital ↓ in Factor VIII

Hemophilia B - Congenital ↓ in Factor IX

- X linked recessive bleeding disorders
- Hemophilia A more common > Hemophilia B
- ~4,000 Canadians with Hemophilia A/B

Severity + Clinical Presentations

Mild Factor levels > 5 IU/dL / > 5%	Provoked bleeding (EX: surgery, trauma)
Moderate Factor levels 1 – 5 IU/dL / 1-5%	Intermediate bleeding phenotype
Severe Factor levels < 1 IU/dL / < 1%	Severe spontaneous life threatening bleeding Frequent joint/MSK bleeds

Canadian Hemophilia Society; ISTH Congenital Hemophilia A and B Guidelines 2024

Treatment Principles: **Bleeding Management** –

Factor First! i.e Treat first and investigate later

1. Assess bleed severity (examples below)

- Minor bleed – EX: epistaxis, mild bruising
- Major bleed – ICH, spinal cord, airway compromise, hemarthrosis, major GI/GU bleeding, intra-abdo/retroperitoneal bleed, major vaginal/PPH

2. Bleed Treatment

- Factor replacement (Factor VIII or Factor IX product – Canadian Blood Services good resource for specifics)
 - Dosing based on severity of deficiency, but in general ~50 units/kg (Hemophilia A) or 100 units/kg (Hemophilia B) to reach ~100% Factor activity
- DDAVP may be used for some mild Hemophilia A
- TXA
- Monitor factor levels *FYI for patients on Emicizumab, cannot use clot based FVIII assay, must use chromogenic

3. Prophylaxis – Emicizumab (Hemlibra), Factor replacement

Immune Thrombocytopenia (ITP)

Definition: Plt < 100 in the absence of other causes

- Presentation: mucosal bleeding, petechiae/purpura, wet purpura (oral blood blisters)
- Workup:
 - CBC, blood film, HCV, HIV serology, H pylori
 - Bone marrow biopsy not necessary
 - H. pylori – test if eradication therapy would be used (grade 2C¹; conditional/low quality ²)

Other testing depending on history: Med review, Liver function testing, autoimmune panel, VWD panel.

Immune Thrombocytopenia (ITP): Treatment

1) NOT BLEEDING/MILD MUCOCUTANEOUS BLEEDING & PLT >30

- Watch and wait

2) NOT BLEEDING & PLT < 30

1st line: Pred 0.5-2mg/kg or Dex 40 mg X 4d, IVIG
(caution: hemolysis), anti-D (RhD+; caution: hemolysis)

— FLIGHT: MMF + steroids

2nd line: splenectomy, rituximab, TPO-R agonists

3) ACTIVE BLEEDING

Steroids + Tranexamic Acid +/- IVIG

Life threatening: Plt transfusion +/- splenectomy

4) **PREGNANCY:**

Tx if bleed, plt <30, delivery or plt<50+≥36 wk GA

- steroids or IVIG (**Pred**>Dex →Dex crosses placenta)
- plt>50 for delivery, >50-80 for neuraxial anesthesia
- Consult peds: Check baby's platelet count

Onset of Response:

- IVIG: 1-3d → FASTEST
- Corticosteroids: 2-14d
- Rituximab: 7-56d

Secondary ITP:

1. If significant bleeding and HCV/HIV:
HCV→ use IVIG; HIV→ steroids or IVIG
2. **Evan's Syndrome** = ITP + hemolytic anemia
 - 2^o Autoimmune/Connective tissue diseases
 - Difficult to treat, frequent relapse

If new diagnosis and plt <20 → admission

Classical Hematology

- Thrombosis
 - Anticoagulant choice + reversal
 - Cancer associated thrombosis
 - Thrombolysis + IVC filters
- Bleeding
 - Approach to abnormal PTT, INR
 - Von Willebrand disease
 - Immune thrombocytopenia (ITP)
- **High yield anemia and thrombocytopenia topics**
 - **Hemolytic anemias**
 - **Thrombotic microangiopathies (TMAs)**
 - **Heparin induced thrombocytopenia (HIT)**
- Transfusion medicine
- Sick cell disease

Hemolytic Anemia Approach

Based on why the hemolysis occurs

1) *Intrinsic RBC issues*

- Membrane defect: hereditary elliptocytosis / spherocytosis
- Enzyme defect: G6PD deficiency
- Hemoglobinopathy: thalassemia, sickle cell
- Complement-mediated: PNH

2) *Immune-mediated*

- Autoimmune (warm/cold)
- Alloimmune (i.e. hemolytic transfusion reaction)

3) *Mechanical*

- TMA
- Mechanical heart valves

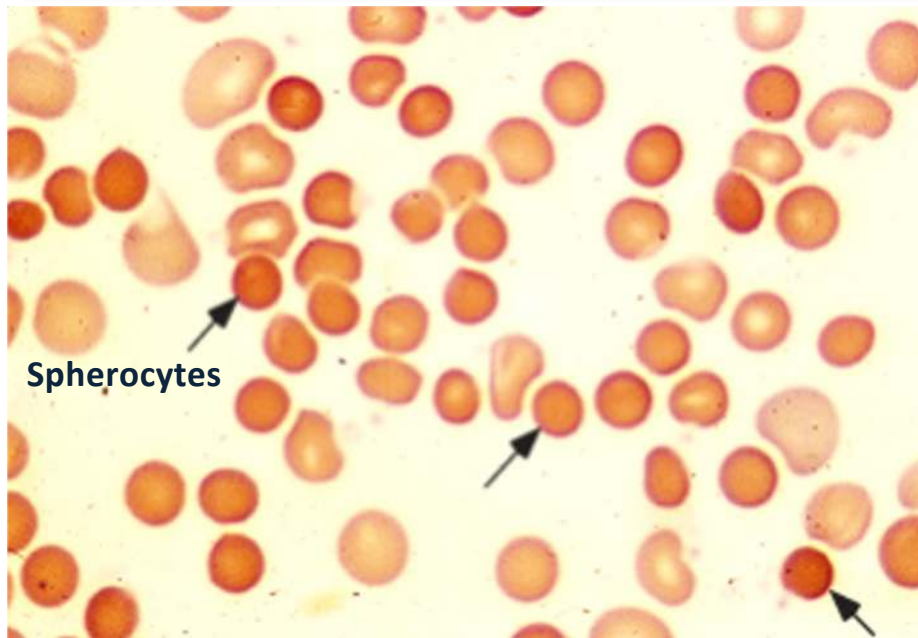
4) *Other*

- Malaria, babesia, *C. perfringens*, Wilson's disease

Hemolysis labs

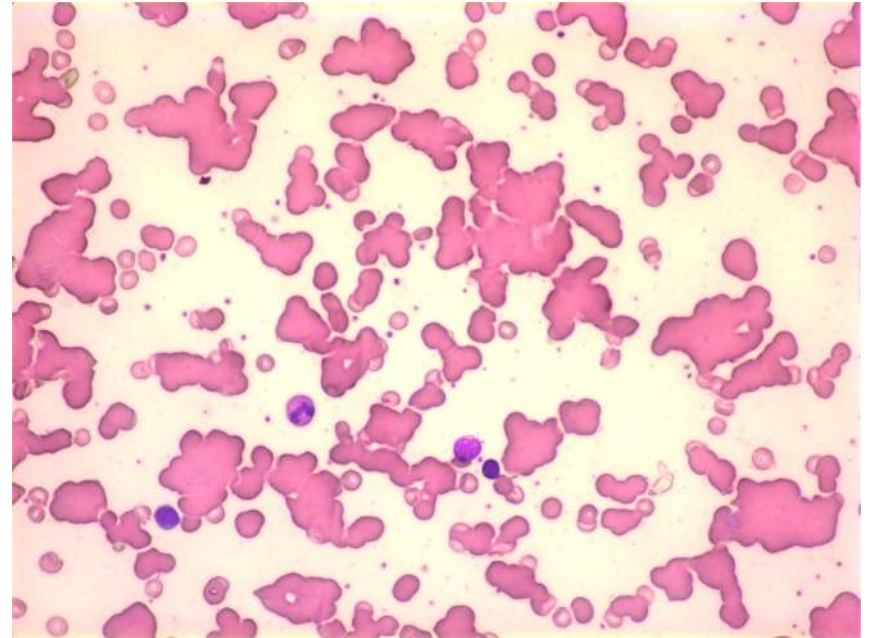
- CBC (anemia)
- LDH ↑
- Bilirubin (indirect) ↑
- Haptoglobin ↓
- Reticulocytes ↑
- DAT i.e Coombs (immune)
- **Blood film**
- *Others* → plasma free hemoglobin, urine hemosiderin

Hemolytic Anemia



Spherocytes:

- Warm autoimmune hemolytic anemia (IgG)
- Hereditary spherocytosis
- Delayed hemolytic transfusion reaction



Agglutination:

- Cold autoimmune hemolytic anemia (IgM)
- Paroxysmal cold hemoglobinuria

Warm Autoimmune Hemolytic Anemia

IgG-mediated → reticuloendothelial system → hemolysis at body temp

Diagnosis:

- Hemolysis labs
- Blood film: spherocytes
- DAT: +IgG (+/- C3d)

Etiology: 1° vs 2°

- 2°: Lymphoproliferative disease (e.g. CLL), autoimmune (e.g. SLE), drugs (e.g. methyldopa, NSAIDs, PCN, cephalosporins piperacillin)

Treatment:

1st line: Prednisone 1 – 1.5 mg/kg/day

2nd line:

- Splenectomy (>8-12 mon)
- Rituximab

Supportive:

Folic acid, transfuse for symptoms (requires investigations → call blood bank)

Cold Autoimmune Hemolytic Anemia

IgM fixes complement → intravascular hemolysis at temp < 37°C

Diagnosis:

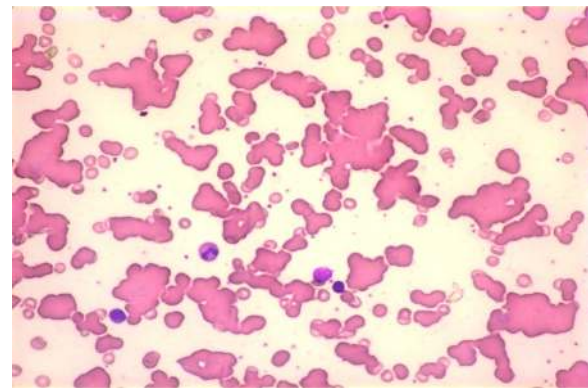
- Hemolysis labs
- Blood film: agglutination
- DAT +ve C3d (= IgM antibody)
- Thermal amplitude: significant if $\geq 28^{\circ}\text{C}$

Etiology: 1° vs 2°

- 2°: Infection (mycoplasma pneumonia, EBV), lymphoproliferative disorder (MGUS, Waldenstrom's, lymphoma)

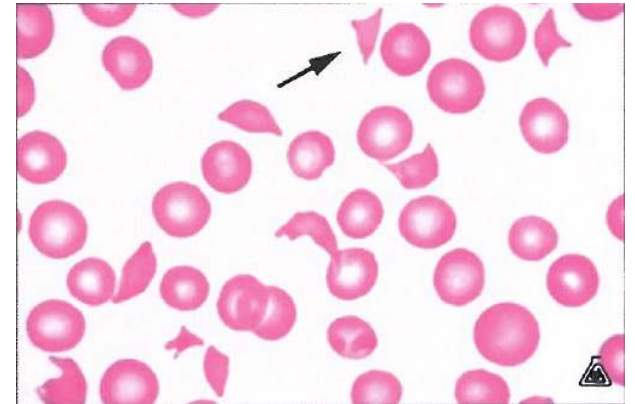
Treatment:

- **#1: Warm patient!**
- Treat underlying cause
- **Steroids not helpful**
- Rituximab for refractory cases



Thrombotic Microangiopathy (TMA)

- Schistocytes (aka fragments) on blood film: think TMA!
- TMA =
 - Micro-angiopathic hemolytic anemia (MAHA) → schistocytes
 - ↓ Plt
 - Microvascular thrombi → end organ damage
- Look for triggers: see next slide



Adamski, Thrombotic microangiopathy and indications for therapeutic plasma exchange. ASH education, 2014
George and Nestor, Syndromes of thrombotic microangiopathy. NEJM 2014

Thrombotic Microangiopathy (TMA)

Primary TMA

1. TTP (ADAMTS13 deficiency)
 - Congenital vs acquired
2. Atypical HUS (complement mediated)
 - Congenital vs acquired
3. HUS (Shiga toxin) adult << peds

Secondary TMA

1. Preg: pre-eclampsia/eclampsia, HELLP
2. Malignancy
3. Infection: HIV, HepC, H1N1, etc.
4. Malignant HTN
5. Autoimmune: scleroderma renal crisis, APLA, SLE, vasculitis
6. Transplant (BMT + solid organ) associated
7. DIC
8. Drug associated (cocaine, calcineurin inhibitors, VEGF inhibitors)

For RC, treat all non-pregnant TMAs as TTP until proven otherwise

Adamski, Thrombotic microangiopathy and indications for therapeutic plasma exchange. ASH education, 2014
George and Nestor, Syndromes of thrombotic microangiopathy. NEJM 2014

TTP – A Hematologic Emergency

- $\downarrow$ ADAMTS13 $\rightarrow$ $\uparrow$ multimer VWF $\rightarrow$ plts aggregate $\rightarrow$ + + micro-thrombi
- Mortality: without treatment >90%, with treatment 10 – 20% mortality
- **Classic pentAD is a TAD too late! (only 5% have all 5)**
 - $\downarrow$ plt, DAT negative hemolytic anemia, fever, neuro sx, renal failure
 - Only require **first 2** to suspect the diagnosis
- **Work-up:**
 - CBC, blood film, + hemolysis markers (LDH, bili, haptoglobin, retic)
 - Creatinine
 - ADAMTS13 activity level (**send before FFP/PLEX; do not wait for result**)
 - R/O other causes of TMA
 - PTT and INR will be normal
 - ECG, $\uparrow$ trop (Tnl>0.25) $\rightarrow$ 3X death/refractory disease*

ADAMTS13 activity level
<10% diagnostic of TTP

*Benhamou et al. JTH 2014
ISTH TTP Guidelines 2020,
Update - 2025 83

TTP – A Hematologic Emergency

Treatment:

- PLEX within 4-8h
 - FFP to bridge: 3-4u then 1u q2h
- No platelet transfusion unless life-threatening bleed
- Steroids (Pred 1 mg/kg/d or Solumedrol IV 1g/day)
- Folic acid 5 mg daily
- +/- ASA if ↑ trop/CNS symptoms once platelets >50
- DVT prophylaxis when plt >50
- Caplacizumab or Rituximab can be considered in first acute event or refractory cases

Oral Scenario Pearl

You may be at a community hospital- While you arrange transfer to centre with PLEX capability, don't forget to start plasma & steroids!

Heparin-induced Thrombocytopenia

- Autoantibodies vs. heparin-PF4 complex → platelet activation → thrombosis & platelet consumption
- Calculate pre-test probability via 4T score:

	2 points	1 point	No points
Thrombocytopenia	50% drop & nadir > 20	30-50% drop or nadir 10-19	<30% drop or nadir <10
Timing on heparin therapy	Day 5-14 or <1d with recent heparin in past 30 days	> Day 14; < 1 day with heparin in past 31-100 days, or timing unclear	Day <4, no recent heparin
Thrombosis	Proven clot, skin necrosis, anaphylactoid rx after bolus	Progressive thrombus, recurrent clot or skin redness	None
Other causes excluded	No cause	Possible	Definite

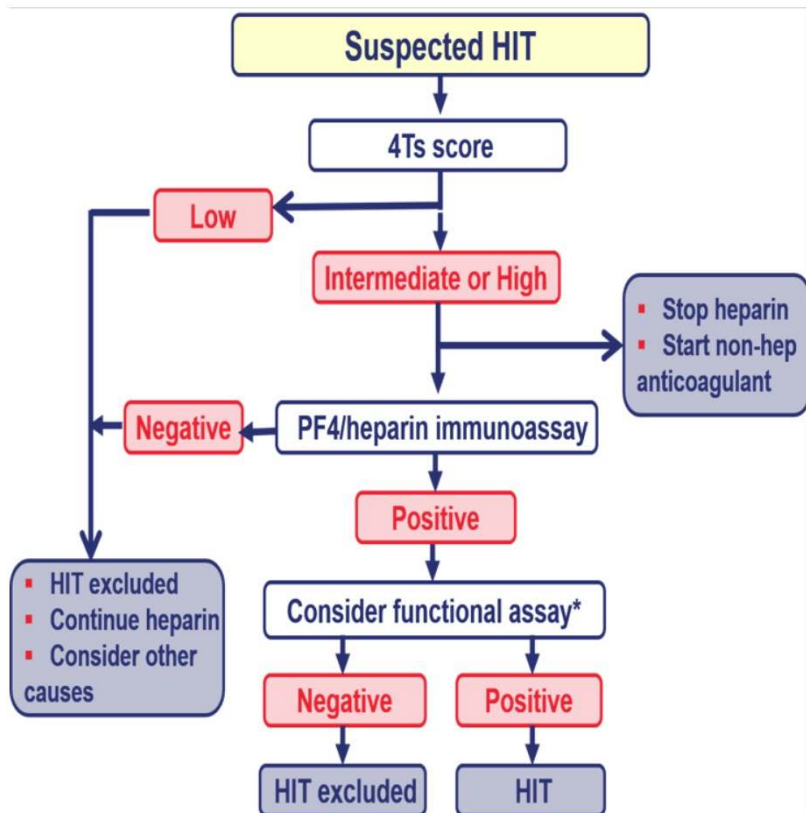
Score ≤ 3: low probability; no further investigations (NPV is 99.8% => good for ruling-out HIT)

Score 4-5: intermediate probability (PPV 14%)

Score 6-8: high probability (PPV 64%)

(Cuker et al. Blood 2012;120:4160; ASH 2018 HITT Guidelines)

Figure 1. Approach to suspected HIT



*Consider functional assay in patients with intermediate 4T score (4 or 5) or ELISA OD <2.0

Diagnostic Tests

PF4/heparin immunoassay i.e **HIT ELISA**

Functional assay i.e **SRA** (serotonin release assay)

Management of HIT:

Anticoagulant Choice:

- IV: Argatroban (short ½ life, ok in ESRD), Danaparoid
- SC: Fondaparinux (long ½ life, caution in renal impairment)
- PO: Warfarin, DOACs

- Warfarin: start when PLT ≥ 150 + overlap with non-heparin anticoagulation ≥5d once INR 2 – 3.

Contraindicated during acute HIT.

- DOAC (rivaroxaban 15 BID x 3 wks if VTE or until plt recovery if no VTE → Rivaroxaban 20 mg daily)
- Pregnancy: Danaparoid preferred

Duration:

- No thrombosis = min. until platelet recovery (>150), max 3 mon
- Thrombosis = 3-6 months

Bilateral Leg Dopplers: r/o silent DVT in all pts

Upper Extremity Dopplers: only if UE central venous line in situ

Platelet Transfusion: AVOID

Classical Hematology

- Thrombosis
 - Anticoagulant choice + reversal
 - Cancer associated thrombosis
 - Thrombolysis + IVC filters
- Bleeding
 - Approach to abnormal PTT, INR
 - Von Willebrand disease
 - Immune thrombocytopenia (ITP)
- High yield anemia and thrombocytopenia topics
 - Thrombotic microangiopathies (TMAs)
 - Heparin induced thrombocytopenia (HIT)
 - Hemolytic anemias
- **Transfusion medicine**
- Sick cell disease

Iron Deficiency Anemia

IV iron risks – Hypersensitivity (Fishbane) reactions (~1:200), Anaphylaxis (<1 in 200,000), low phosphate

- **Signs & Symptoms:**
anemia-related, restless legs, pica, fatigue, glossitis, koilonychia
- **Blood film:**
microcytosis, hypochromia, pencil cells, target cells, ↑plt
- **Diagnosis:**
Ferritin <30
Ferritin >30 + Tsat <20% + inc. TIBC
Soluble transferrin receptor sTFR ↑
- **Treatment:**
 - Identify & treat etiology of Fe loss
 - Treat if Ferritin <50 in pregnancy
- **PO**
 - Ferrous fumarate: 100 mg Fe
 - Ferrous sulfate: 60 mg Fe
 - Ferrous gluconate: 30 mg Fe
 - Daily vs. q2d
 - Continue for 3 months after ferritin normalizes
- **IV**
 - Who?
 - Mod/severe iron def. anemia (Hb <80-100)
 - No response/intolerance to oral
 - Limited time (e.g. pre-op or pregnancy <1 mon from OR/delivery)
 - What?
 - Iron sucrose (e.g. 300 mg IV x 3)
 - Monoferic (e.g. 500mg - 1g IV x 1)
 - How much?
 - Ganzoni equation to calculate Fe deficit
 - Pregnancy – IV iron can be used in the 2nd or 3rd trimester for those intolerant of PO



Which blood product & when? – RBCs and Components

Blood product	Contents	Indications / Transfusion thresholds
RBCs	Red blood cells 1U pRBC expect ~ 10g/L ↑ in Hb	- Most hospitalized pts: Hb <70 g/L (TRICC, AABB 2023) - CVD or MI or undergoing major ortho surgery: Hb <80 g/L (REALITY, FOCUS, AABB2023) - CV surgery: <75 g/L (TRICS III, AABB 2023) - Stable acute GI bleed: Hb <70 g/L (NEJM 2013)
Platelets	Platelets suspended in plasma DOSE = 1 pool of platelets (=4 units of buffy coat derived platelets) 1 PLT transfusion expect ~ ↑ 15-25 x 10 ⁹ /L	- Thrombocytopenia (non-immune) and NO bleeding : plt <10 - Central line placement (compressible anatomic site) : plt < 10 - Lumbar puncture, low-risk IR procedure : plt < 20 - Patient on anticoag. that can't be held: plt <30 - Significant bleeding, major surgery/high risk IR, epidural: plt <50 - CNS surgery, ICH, brain trauma: plt <100 - ITP – only if life threatening bleeding
Plasma *FFP/plasma has an INR of 1.6-1.7	Plasma = clotting factors + fibrinogen Dose = sml adult 3U, large adult = 4 U	- TTP - Microvascular bleeding (DIC clinically) or massive transfusion and pts clinical status precludes waiting for INR/PTT/fibrinogen results. - Liver disease or DIC and: INR >1.8 with Bleeding OR significant procedure

Which blood product & when? – Fractionated Products

Blood product	Contents	Indications
Fibrinogen concentrate	Fibrinogen Dose = 2-4g	Bleeding + fibrinogen <1.5 [<2 if obstetric or Cardiovascular emergency] Acute promyelocytic leukemia + fibrinogen <1.5 * note: this largely <u>replaces</u> cryoprecipitate
Prothrombin complex concentrate (PCC)	Factors 10, 9, 7, 2, ProtC, ProtS, ATIII, heparin*	Warfarin reversal (INR<3 = 1000 U INR 3-5 = 2000 U INR >5 = 3000 U) (*PCC has heparin = do not use in warfarin reversal w HIT, give FFP instead) DOAC reversal (2000 U)
IVIG	IgG pooled from many donors Dosed based on adjusted body weight *use dose calculator	Replacement (e.g. for patients with hypogammaglobulinemia, CVID) Immunosuppression (e.g. ITP) Adverse reactions → Hemolysis (highest risk in non-group O patients), aseptic meningitis, headaches, infusion/hypersensitivity reaction, anaphylaxis

Transfusion Reaction Differentials: Symptom-based

1) Fever (+/- chills, rigors)

- Febrile non-hemolytic transfusion rxn (FNHTR) (most common)
- Sepsis (bacterial contamination)
- Acute hemolytic transfusion rxn (AHTR)

2) Hypotension

- Sepsis (bacterial contamination)
- Acute Hemolytic Transfusion Reaction
- Anaphylaxis
- TRALI
- Bradykinin-mediated
 - No fever or allergy
 - 50% of patients on ACE-inhibitor

3) Dyspnea

- Transfusion Associated Cardiac Overload (TACO)
- Transfusion Related Acute Lung Injury (TRALI)
- Anaphylaxis

4) Urticaria/Allergic reaction

- Urticarial → anaphylaxis

5) Delayed cytopenias

- **RBC:** delayed hemolytic transfusion rxn
- **PLT:** post-transfusion purpura (PTP)
- **All lineages** + fever, rash, diarrhea
→ transfusion-associated graft vs. host disease



Transfusion Reactions – Low Risk Situations

1) Febrile non-hemolytic transfusion reaction (FNHTR)

- Temperature $\leq 39^{\circ}$ + no other concerning features
- Due to (A) pyrogens in product
(B) recipient Abs vs. donor WBCs
- When fever develops: stop, assess patient, check for clerical error
- If no concerning features*, give acetaminophen & slowly re-start

2) Urticaria

- Stop, assess patient, check for clerical error
- If $< 2/3$ body surface area + no other concerning features*
- Give Benadryl & slowly re-start

***Concerning features** – hypotension, tachycardia, rigors/chills, anxiety, dyspnea, back/chest pain, bleeding from IV sites, N/V

****Abort transfusion in all other scenarios**

Classical Hematology

- Thrombosis
 - Anticoagulant choice + reversal
 - Cancer associated thrombosis
 - Thrombolysis + IVC filters
- Bleeding
 - Approach to abnormal PTT, INR
 - Von Willebrand disease
 - Immune thrombocytopenia (ITP)
- High yield anemia and thrombocytopenia topics
 - Thrombotic microangiopathies (TMAs)
 - Heparin induced thrombocytopenia (HIT)
 - Hemolytic anemias
- Transfusion
- **Sickle cell disease**

MCQ #4 - 2026

A 23 year old female with a known history of Hb SS Sickle Cell Disease presents to the ED with pain in her lower extremities, back and chest. She is afebrile, with O₂ sats of 97% on RA, and initial CXR is normal. Her Hb is 72 from baseline of 75g/L, and she has never been previously transfused. She is admitted to the GIM ward and receives standing and PRN IV Hydromorphone, Ibuprofen and Tylenol. On day 3 of her admission she develops hypoxia with O₂ sats of 89% and shortness of breath. Her CXR demonstrates new consolidation in the LLL. She is started on 3L of supplemental O₂, and IV Ceftriaxone. 24 hours later her dyspnea is improving, but she remains on 3L of O₂. Remainder of her vitals are stable. Her Hb is 70g/L.

What are your recommendations for transfusion?

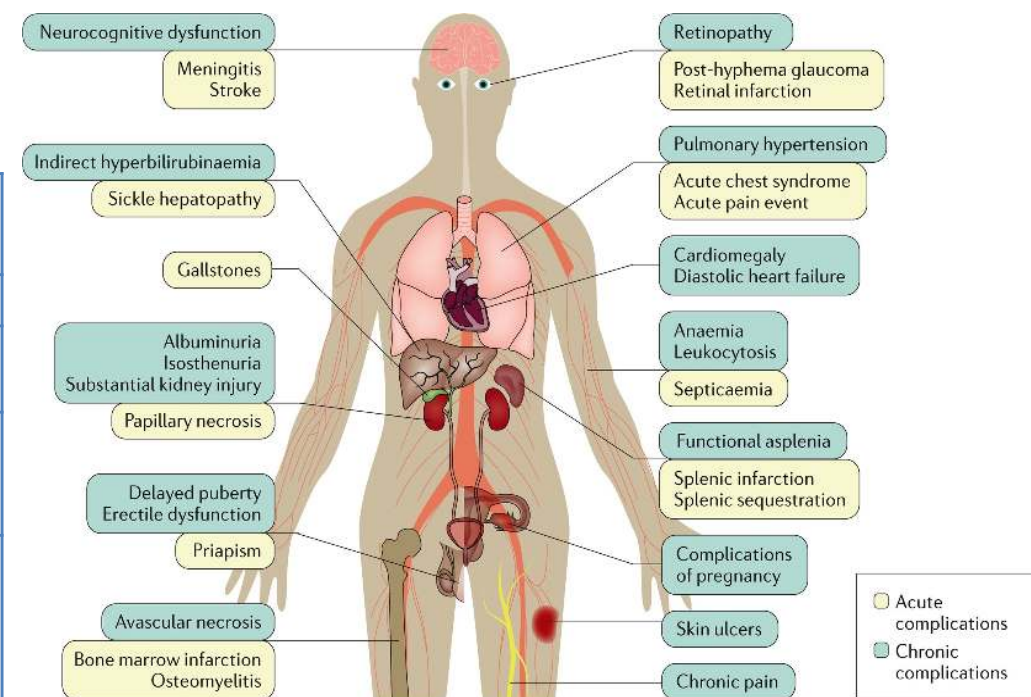
1. Arrange for exchange transfusion
2. No transfusion. Continue current management
3. Arrange for 2 units RBC simple top up transfusion
4. Arrange for 1 unit RBC simple top up transfusion

Sickle Cell Disease

- **Most common hemoglobinopathy** affecting **millions** of individuals globally
- **Pathophys:** substitution 6th position on Ch11 glutamine to valine in the β -globin gene → RBC polymerization/sickling when deoxygenated

Genotype	Clinical Syndrome	HbA	HbS	Symptoms
AA	normal	97-99%	0%	-
AS	Sickle Cell Trait	60%	40%	-
SS	Sickle Cell Anemia	0%	86-98%	ischemic perfusion hemolysis
SC (↑baseline Hgb)	Hemoglobin SC	HbC 50%	50%	Usually milder than SS

Complications → occur due to **anemia/ischemic perfusion injury** and/or **hemolysis**



Sickle Cell Disease

Investigations (Inpatient)

- CBC and reticulocytes
- Hb electrophoresis (assess HbS %)
- Blood group and screen – blood bank to phenotype blood (and consider genotyping)
- Chemistries, troponin, BNP, lactate
- Chest Xray +/- CTPA

Treatment (Outpatient)

Therapies (with approval)

- Hydroxyurea (↓VOC, ↓ACS, ↓transfusions, ↑survival, ↓ hospitalizations)
- Exchange transfusion (if indications)
- Allo stem cell transplant
- CASGEVY – CRISPR-based gene editing therapy

Non Pharmacologic → Vaccinations (asplenia). Pain plan
Complication screening → Eye exam, MRI brain, TTE, BMD
CBC, metabolic panel, U/A, ferritin, MRI liver, TSH

Treatment (Inpatient – common admissions)

Presentation	Management
<u>Vaso-occlusive crisis</u> • Pain (commonly back, extremities, chest) • May have low grade temp Triggers – temperature, dehydration, infxn, stress	<u>Rapid and sustained multimodal analgesia</u> → Parental/oral opioids, Tylenol, NSAIDs standing + PRN Adjuncts – ketamine/regional anesthesia Incentive spirometry (prevent ACS) IV fluids, DVT prophylaxis
<u>Acute Chest Syndrome</u> Pulmonary infiltrate + fever, chest pain, and/or respiratory sx Trigger: infection , VOC, fluid overload, fat embolism	Supplemental O ₂ (>95%), IVF, treat pain, antibiotics, incentive spirometry +/- transfusion (<i>exchange or simple guided by severity</i>). Low threshold to r/o PE if not improving (may mimic)
<u>Aplastic Crisis</u> ↓↓ Hb and retics. Parvo B19 = most common trigger	IVIG, Folic acid 5mg daily, ESAs, supportive RBC transfusions if needed

Transfusion in Sickle Cell Disease

General Principles → Transfusion for patients with SCD should always be guided by a Hematologist and/or with Blood Bank. **Why?** Require extended RBC matching + high risk for hemolytic reactions

Exchange Transfusion

- **Severe acute chest syndrome** (rapid deterioration, multi-organ failure, progressive hypoxia with rising O2 needs)
- **Acute ischemic stroke**
- **Secondary stroke prophylaxis**
- **Primary stroke prophylaxis** (if high-flow transcranial doppler US)
- **Pre-operative for high risk surgery** (neurosurgery, cardiac) or low-mod risk surgery with Hb $\geq$ 90g/L.
- **Others** - Progressive cholestasis, Post-solid organ transplant, High-risk pregnancy, other SCD complications (pulmonary hypertension, priapism, renal disease, ulcers), complications despite top-op transfusion, prevention of pain crises/ACS if failed hydroxyurea

Simple (Top Up) Transfusion

- Hemoglobin <50g/L ***unless from hyperhemolysis****
- Mild-moderate acute chest syndrome
- Pre-operative prophylaxis in patients with Hb <90 undergoing low-moderate risk surgery

***Be aware → Hyperhemolysis**
Severe and often fatal transfusion reaction leading to Hb drop below pretransfusion levels. **Avoid further transfusion, call Heme/blood bank**

MCQ #4 - 2026

A 23 year old female with a known history of Hb SS Sickle Cell Disease presents to the ED with pain in her lower extremities, back and chest. She is afebrile, with O2 sats of 99% on RA, and initial CXR is normal. Her Hb is 72 from transfused. She is admitted to the GIM. Hydromorphone, Ibuprofen and Tylenol. O2 sats of 89% and shortness of breath. She is started on 3L of supplemental O2 improving, but she remains on 3L of O2. What are your recommendations for transfusion?

1. Arrange for exchange transfusion
2. No transfusion. Continue current management
3. Arrange for 2 units RBC simple top up transfusion
4. Arrange for 1 unit RBC simple top up transfusion

This patient with SCD is admitted with an initial VOC episode. During hospitalization develops features of acute chest syndrome (new resp sx + pulmonary infiltrate) FYI up to 50% of SCD patients may develop ACS during their hospital admissions! In the absence of features of severe ACS (EX: dropping O2 sats, multi-organ dysfunction, escalation of care, dropping Hb) would consider this more **mild-mod in severity**, thus no clear role for an exchange. As her **Hb is < 90g/L** simple transfusion is appropriate to improve her oxygenation. Best practices – always transfuse 1U RBC at a time unless actively hemorrhaging.

Thank You!

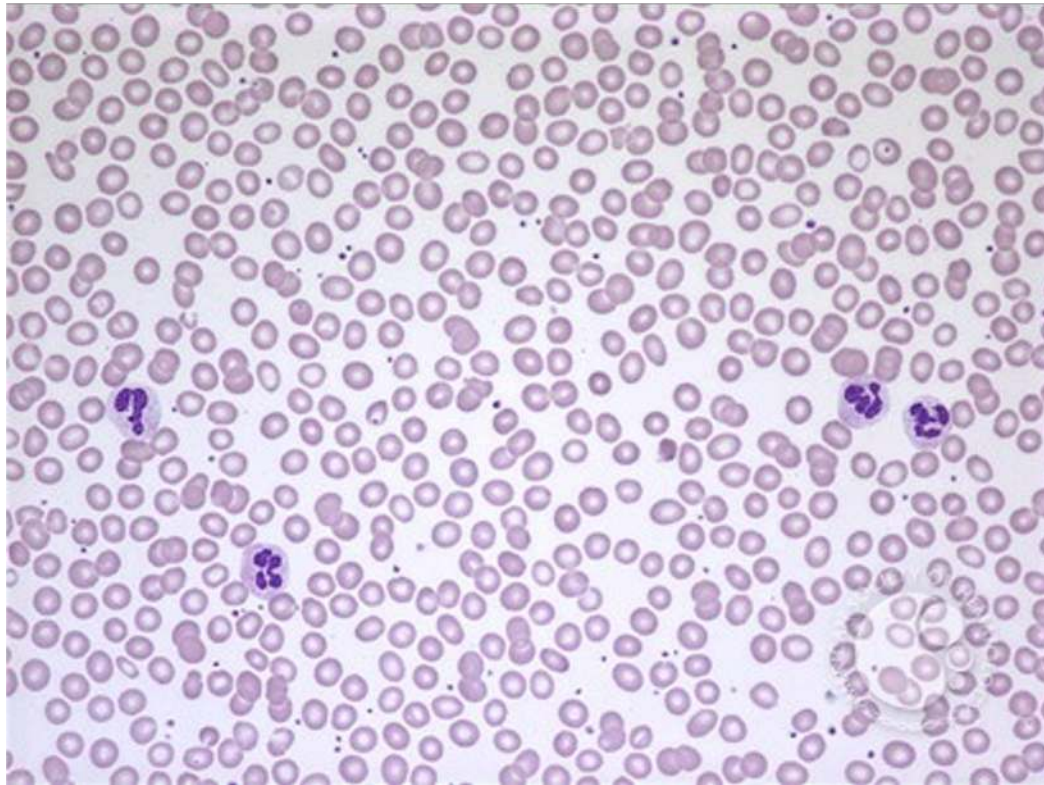
See Bonus Slides in your course manual for:

- Representative blood films → Less emphasis in recent years
- Transfusion Medicine → **high yield for MCQs**
- Choosing Wisely guidelines
- Approach to cytopenias
- Approach to lymphadenopathy
- B12 deficiency vs. syphilis → **high yield** (on exam, not in real life)
- Bonus bleeding and clotting slides
- Thalassemia
- **Practice oral scenario stems**
- **BONUS MCQs**
- Consent for a Bone Marrow biopsy (VIDEO 2018)
- Pre-op consult for bleeding disorder (VIDEO 2019)

BONUS CONTENT: Extra BLOOD FILMS

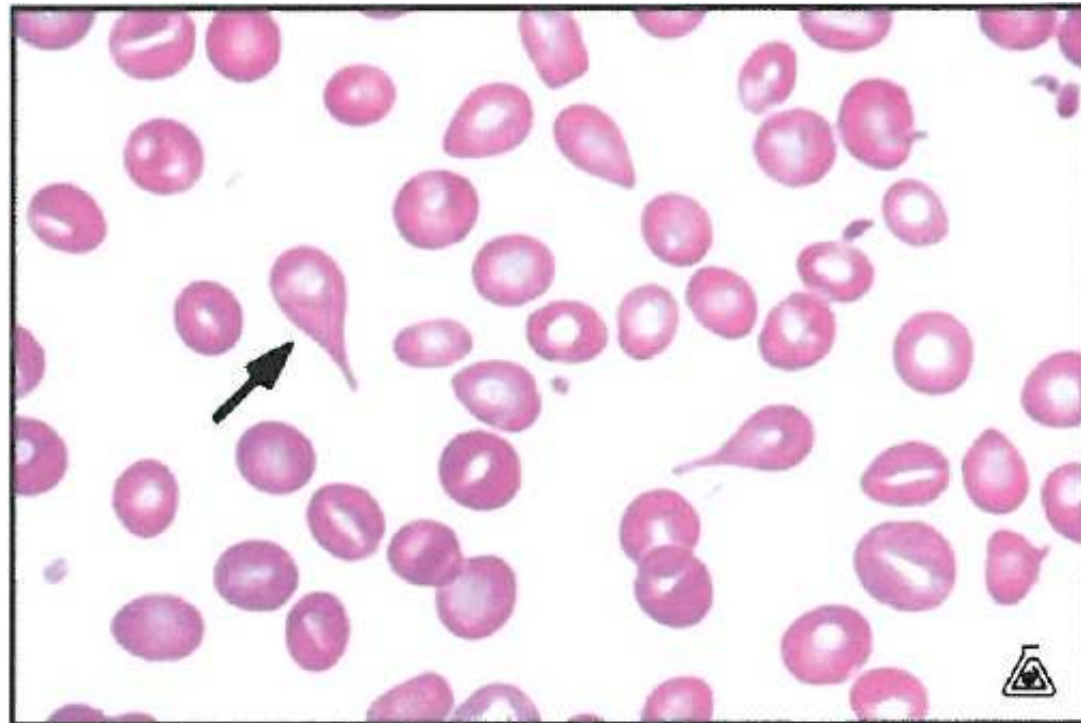


Normal Blood Smear - RBCs



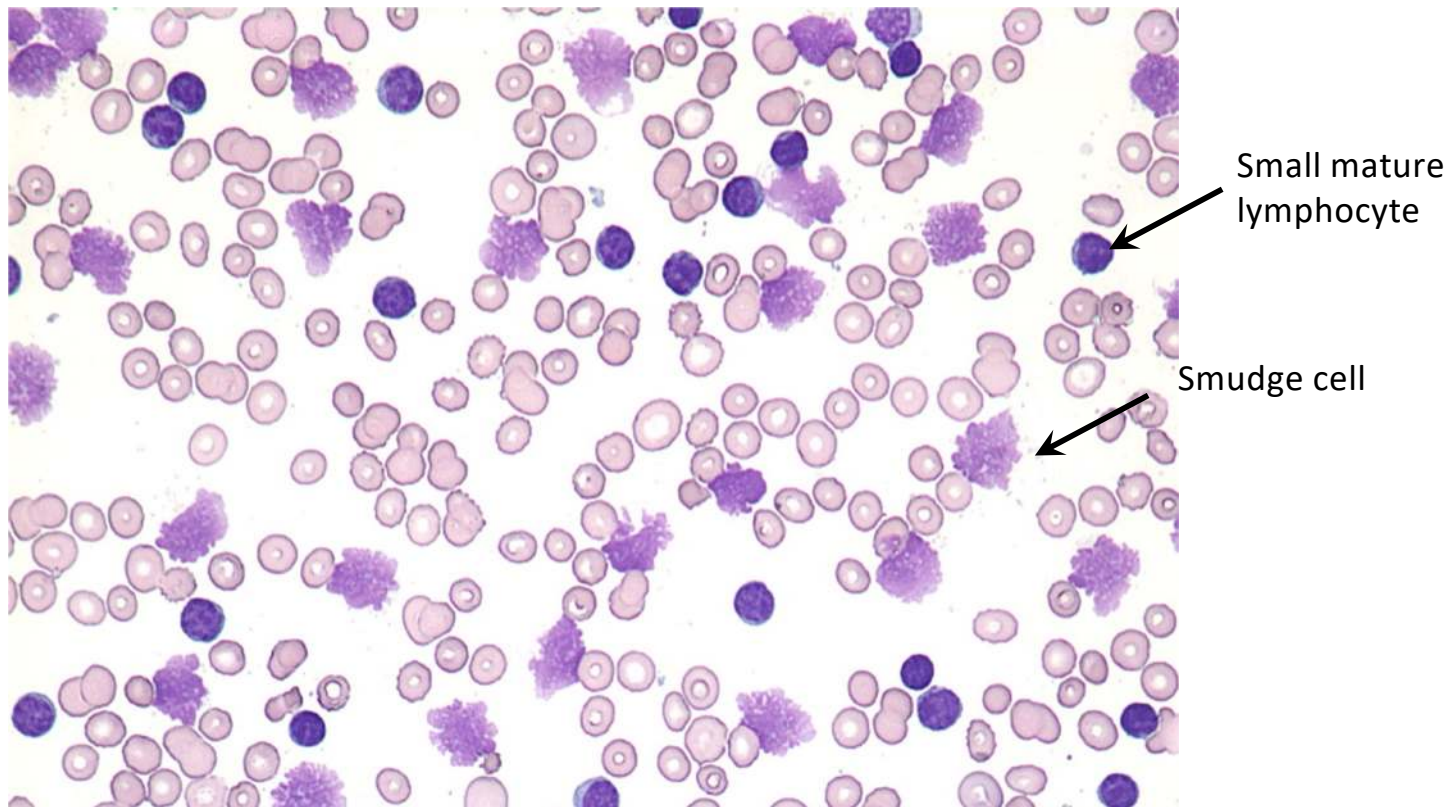
ASH image bank

Teardrops

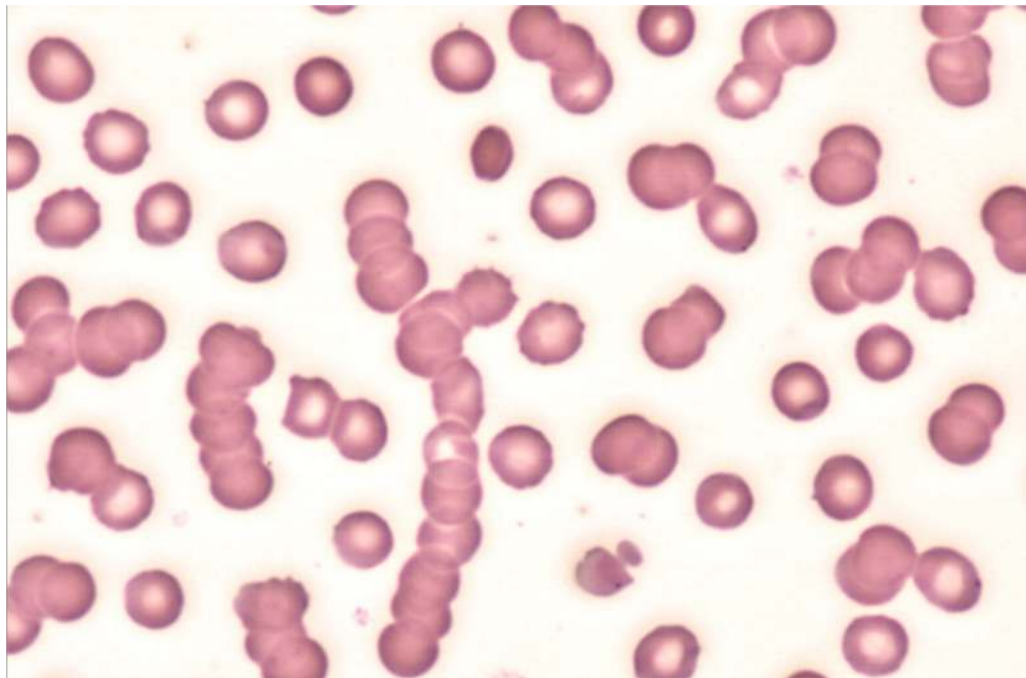


Think myelofibrosis (B symptoms, high LDH, splenomegaly, cytopenias, leukoerythroblastosis)

Smudge Cells (For RC – think **CLL**)



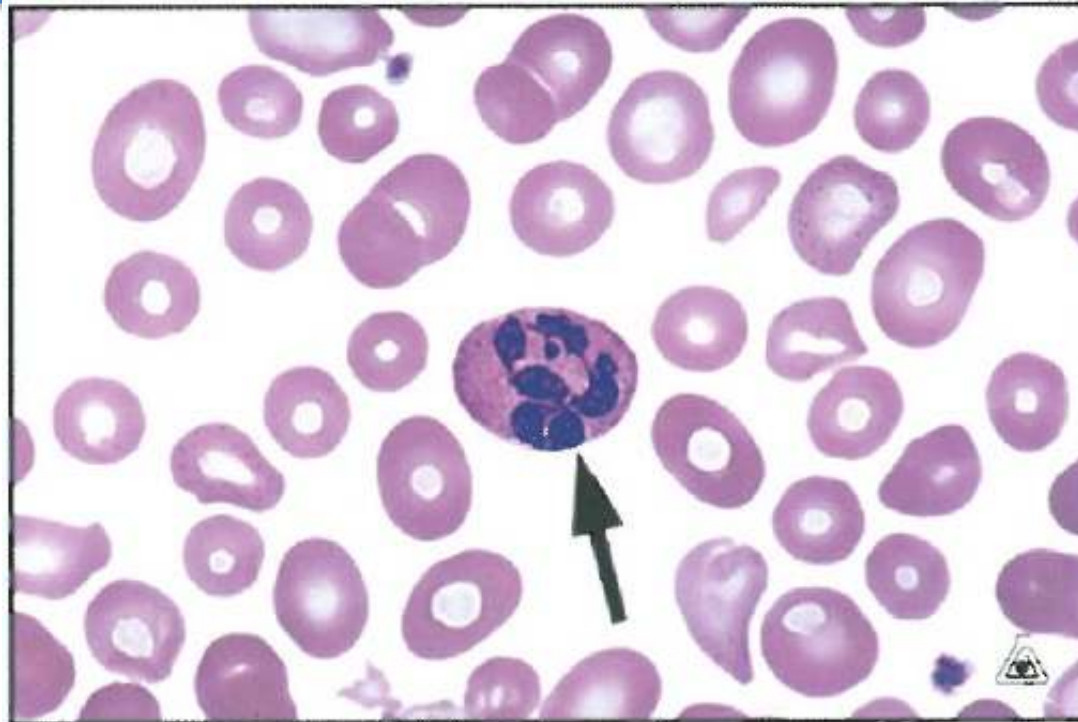
ROULEAUX



Associated with multiple myeloma as well as inflammation, pregnancy (high ESR states)

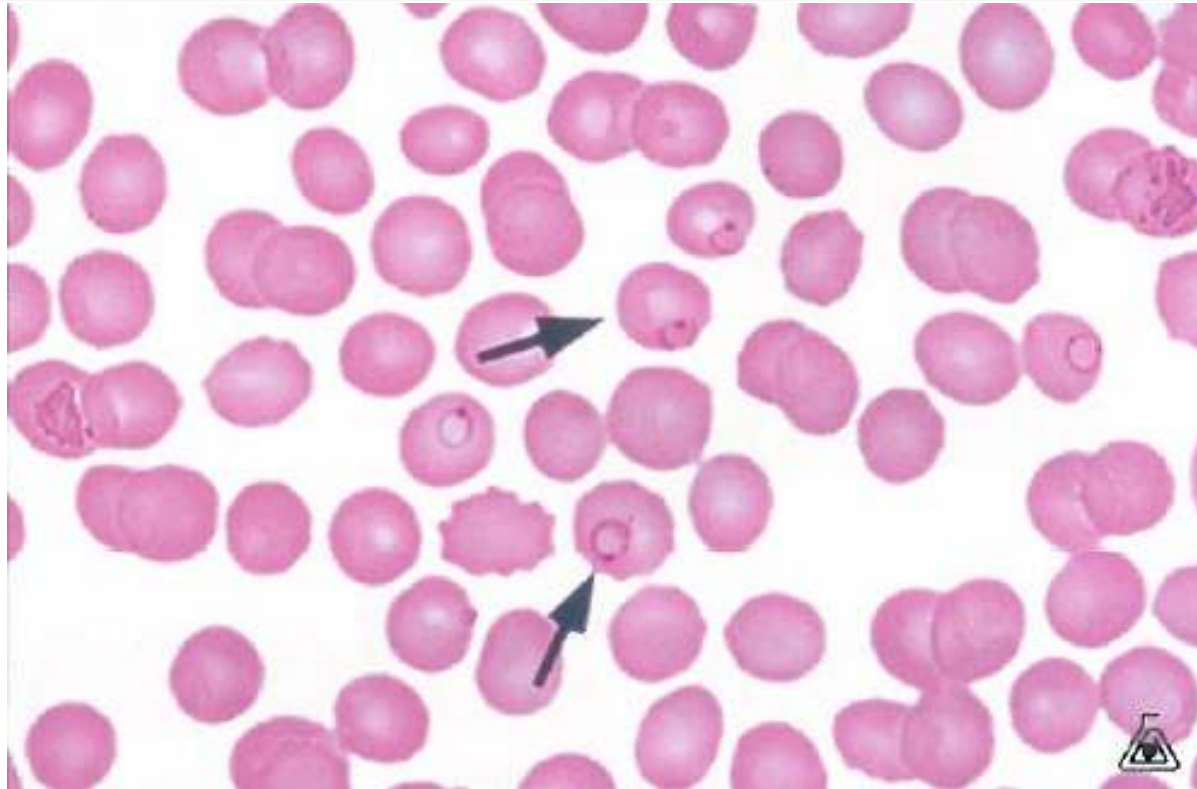
104

Hyper-segmented Neutrophil

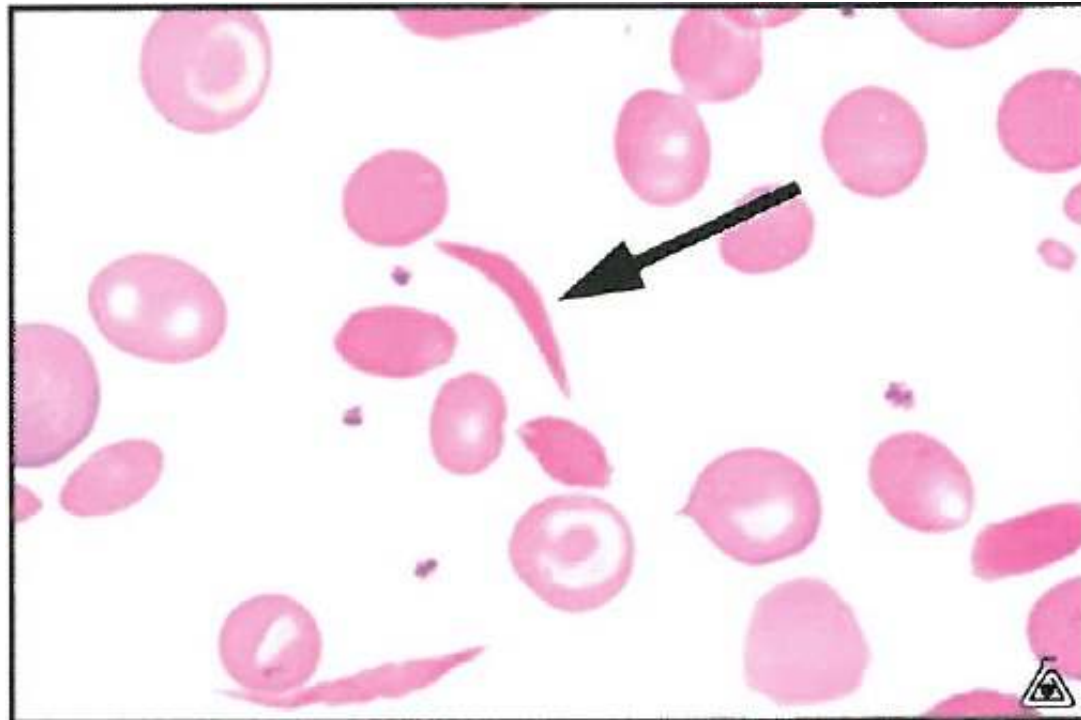


≥6 lobes = hypersegmented. Think megaloblastic anemia (e.g. folate or B12 deficiency)

Malaria



Sickle Cell



Transfusion Medicine

Recommended resource: Blood Easy 5

Available at www.transfusionontario.org

→ *review for transfusion triggers, consent, and adverse reactions*

Blood Product “Modification” (1)

Pre-storage leukoreduction

- Definition: Removes WBCs from cellular blood products (RBC/PLT)
 - Why? Reduces incidence of DHTR, CMV transmission, HLA alloimmunization, TA-GVHD
 - Standard practice in Canada since 1999 (all units leukoreduced)

Irradiation

Definition: Incapacitates residual WBCs in cellular blood products

- Why? Prevents transfusion associated GvHD (WBCs from mount immune attack)
- When? Required in immunosuppressed individuals:
 - LPDs, congenital immunodeficiency, BMT patients, after certain therapies (fludarabine, ATG)

Blood Product “Modification” (2)

Washing

- Definition: reduces plasma proteins in the blood product
- Why? minimizes allergic reactions
- When? Use in patients with history of anaphylaxis, recurrent allergic reactions, IgA deficiency (if hx of anaphylaxis & no IgA deficient products available)

CMV-negative blood products

Controversial in era of universal leukoreduction → many institutions deem blood to be “CMV-safe”

Currently recommended (by some institutions) for:

- Intra-uterine transfusions
- CMV seronegative pregnant women
- CMV seronegative recipients of allo-BMT

Blood Product “Modification”

Indications for irradiated blood - prevent transfusion associated GVHD:

- Intra-uterine or neonatal exchange transfusion.
- Congenital T-cell immunodeficiency.
- Autologous stem cell transplant recipients until 3mo post-transplant
- Allogeneic stem cell transplant until at least 6 months post-transplant (indefinite if GVHD or on immunosuppression)
- CAR-T cell until 3mo post-infusion
- All patients with Hodgkin’s Disease.
- T-cell depleting therapies

Indications for washed platelets:

- Anaphylaxis to transfusion NYD
- Recurrent/severe allergic transfusion reactions
- IgA deficiency with no IgA deficient donor available

New product (2023):

- Solvent-detergent treated plasma (better pathogen inactivation)

Choosing Wisely (ASH)

Select statements only

1. Don't transfuse more than minimum number of RBCs necessary to relieve symptoms of anemia or to return patient to a safe hemoglobin range
2. Don't test for thrombophilia in adults with VTE occurring in the setting of major transient risk factors (surgery, trauma, prolonged immobility)
3. Don't use IVC filters routinely in patients with acute VTE
4. Don't administer plasma or PCC for non-emergent reversal of VKA
5. Limit surveillance CT scans in asymptomatic patients following curative intent treatment for aggressive lymphoma
6. Don't treat with anticoagulants > 3 months in a pt with a 1st VTE with major transient risk factor
7. Don't routinely transfuse pt with SCD with chronic anemia or uncomplicated VOC
8. Don't perform baseline or routine CT scans in early, asymptomatic CLL
9. Don't test or treat for suspected HIT if low pre-test probability
10. Don't treat ITP in the absence of bleeding or very low platelet count

Choosing Wisely CA – Transfusion Medicine

1. Don't transfuse blood if other non-transfusion therapies or observation would be just as effective.
2. Don't transfuse more than one red cell unit at a time when transfusion is required in stable, non-bleeding patients.
3. Don't transfuse plasma to correct a mildly elevated (<1.8) international normalized ratio (INR) or activated partial thromboplastin time (aPTT) before a procedure.
4. Don't routinely transfuse platelets for patients with chemotherapy-induced thrombocytopenia if the platelet count is greater than $10 \times 10^9/L$ in the absence of bleeding.
5. Don't routinely use plasma or prothrombin complex concentrates for non-emergent reversal of vitamin K antagonists.
6. Don't use immunoglobulin therapy for recurrent infections unless impaired antibody responses to vaccines are demonstrated.
7. Don't order unnecessary pre-transfusion testing (type and screen) for all pre-operative patients.
8. Don't routinely order perioperative autologous and directed blood collection.
9. Don't transfuse O negative blood except to O negative patients and in emergencies for female patients of child-bearing potential of unknown blood group.
10. Don't transfuse group AB plasma to non-group AB patients unless in emergency situations where the ABO group is unknown.

Distinguishing Microcytic Anemias

- Differential diagnosis
 - Iron deficiency
 - Thalassemia
 - Advanced anemia of chronic disease
 - (Sideroblastic anemia; Lead poisoning)

	Fe deficiency	Chronic disease	Thal Trait
MCV	Low	Normal → Low	Very low (<70)
Reticulocytes	Low	Low	High
Ferritin	Decreased	Increased	Normal
TIBC	Increased	Decreased	-
Transferrin saturation	Decreased	Decreased	-
Soluble transferrin receptor	Increased	Decreased	-
RBC count	Low	Low	High
RDW	High	Normal	Normal/ High

- Thalassemia testing: * β -thal trait: hemoglobin electrophoresis ($\uparrow$ HbA2)
 - * α -thal trait: normal electrophoresis; genetic testing

Alpha and Beta Thalassemia

Biology and Genetics

- Normal adult hemoglobin (HbA) is a tetramer consisting of **2 alpha (α) globin** and **2 beta(β) globin** chains each containing an **oxygen binding heme group**.
- The β globin gene cluster exists on chromosome 16, and the α globin gene cluster exists on chromosome 11.
- Normal individuals unaffected by hemoglobinopathies inherit 2 copies of normal β globin genes and 4 copies of α globin genes.
- **Thalassemias** occur due to decreased synthesis of β (Beta-thalassemia) globin or α (alpha-thalassemia) globin genes

Epidemiology

- Common in South East Asia, Africa, Mediterranean, and Middle East populations

Definitions and Disease Subtypes

- **Thal trait** → Individuals with beta or alpha thal trait are typically asymptomatic apart from microcytosis and mild-no anemia
- **Non-transfusion dependent thalassemia (NTDT)** → Patients with thalassemia clinical manifestations, but not dependent on RBC transfusions
- **Transfusion dependent thalassemia (TDT)** → Clinical manifestations of thal + dependent on RBC transfusions

Alpha Thalassemia subtypes

- Alpha thal carrier i.e missing one copy EX (- α / α α)
- Alpha thal trait i.e missing two copies EX (- α / - α)
- HbH disease i.e missing three copies EX (- - / - α)
- Hb Barts i.e missing all 4 copies (- - / - -) *usually fatal

Beta Thalassemia subtypes → most important distinction is TDT vs. NTDT

- β^0 – mutations causing absent gene production
- β^+ – mutations causing decreased gene production

Alpha and Beta Thalassemia

Clinical Manifestations

Vary based on severity of decreased globin production
The imbalance of α and β globin gene production leads to the following disease manifestations →

- **Ineffective erythropoiesis**
 - Extramedullary hematopoiesis (skeletal changes, hepatosplenomegaly)
 - Osteoporosis
- **Chronic hemolysis**
 - Anemia
 - Gallstones
 - Venous thromboembolism
 - Pulmonary HTN
- **Iron overload - both from ineffective erythropoiesis + transfusion support**
 - Cirrhosis
 - Heart failure
 - Endocrinopathy
 - Arthralgias

Laboratory Findings / Diagnosis

- **CBC Findings** → Most patients with thalassemia will have a **microcytic** anemia. Severity depends on disease subtype
 - Alpha/beta thal traits → May have normal – mild anemia ~90g/L with very low MCVs (usually < 70)
- **Blood film** → Microcytic and hypochromic RBCs, may see features such as target cells or basophilic stippling
- Rule out iron deficiency
- **Hemoglobin electrophoresis** → see decreased levels of normal HbA, increased HbA2 and HbF or other co-occurring Hb variants (EX: HbS, HbC)
- **Hemoglobin genetics** → Needed to confirm alpha globin gene mutations

Alpha and Beta Thalassemia

Management

ALL Patients

- Prenatal genetic counselling to patients of childbearing age
- Screening and management of transfusion related and non-transfusion related complications

Supportive RBC Transfusions

- May be required intermittently or short term for patients (EX: supportive preoperatively, during aplastic crises, during periods of increased demand such as pregnancy) or lifelong.
- Typical indications for lifelong RBC transfusions include symptomatic or severe anemia (Hb < 70g/L), poor growth/failure to thrive, extra-medullary hematopoiesis

Iron Overload

- Routine measurements of ferritin
- Screening for end-organ damage from iron overload – liver and cardiac MRI for iron quantification
- Endocrinopathies → hypogonadism, diabetes, hypothyroidism
- Iron chelation (beyond GIM exam scope)

Splenectomy → May be required for massive splenomegaly, not routinely performed.

Novel Therapies (beyond exam scope, just FYI)

- Luspatercept
- Pyruvate kinase activators
- Stem cell transplant
- Gene therapy

FYI Only: CRISPR for SCD and Transfusion dependent Thalessemia

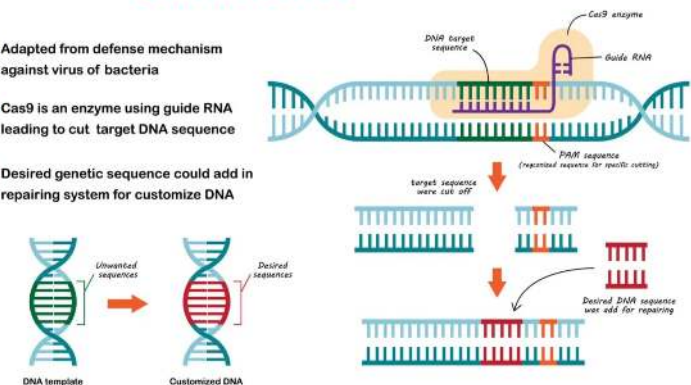
– **FYI ONLY – be aware this therapy exists, not testable!**

1. Patient's hematopoietic stem cells are mobilized (G-CSF)/ harvested and separated by apheresis
2. Single guide RNA directs CRISPR/Cas9 to the erythroid-specific enhancer region of BCL11A
 - BCL11A is a transcription factor which regulates the switch from HbF → HbA by repressing γ -globin (HbF = $\alpha 2 \gamma 2$)
 - By editing gene, the down-regulation of BCL11A causes **increased expression of HbF by edited Stem Cells**
3. Chemo (busulfan) to ablate marrow
4. Edited hematopoietic stem cells are infused and engrafted . Increased HbF = less VOC in SCD.

CRISPR = clustered regularly interspaced short palindromic repeats (CRISPR)-Cas9 nuclease system

How does CRISPR-Cas9 work?

- Adapted from defense mechanism against virus of bacteria
- Cas9 is an enzyme using guide RNA leading to cut target DNA sequence
- Desired genetic sequence could add in repairing system for customize DNA



- Downside: Cost of Gene therapy (>2 million? We don't know)
- Currently under CADTH review in Canada (to determine if provincial funding will be recommended)

Thrombocytopenia

Decreased production

- **Nutritional** (eg B₁₂)
- **Heme malignancies**
- **Marrow infiltration**
- **Marrow dysfunction** (e.g. MDS)
- **Marrow suppression** (e.g. Infection, antibiotics)

Increased consumption

IMMUNE-mediated

- **Autoimmune:**
Primary: ITP
Secondary: SLE, APLA, CVID, HIV, HCV, CLL/lymphoma, H Pylori
 - **Alloimmune:** post transfusion purpura (PTP)
- Drugs** (eg. heparin)

NON- IMMUNE-mediated

- TMA
- Bypass / ECMO
- Bleeding

Splenic sequestration

Any cause of splenomegaly

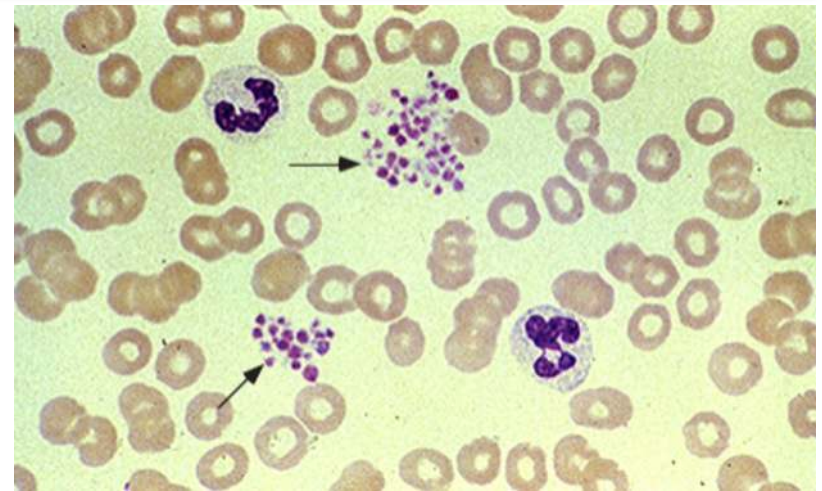
Practical Thrombocytopenia

1) Rule out pseudothrombocytopenia →

- Secondary to EDTA-associated agglutinins
- Repeat in **citrated tube**

2) Should I be worried?

- | | |
|------------------------|-------------------------------|
| – Other lineages down? | → Marrow process, MAHA, Evans |
| – Abnormal coags? | → DIC, APLA |
| – Recent heparin | → HIT |
| – Recent transfusion | → Post-transfusion purpura |
| – Is this sepsis? | |



Pancytopenia

Pancytopenia ddx

Primary marrow disorder

1. Paroxysmal Nocturnal Hemoglobinuria (PNH)
2. Aplastic Anemia
3. Leukemia
4. MDS
5. Myelofibrosis

Secondary

1. Deficiency - B12/Folate, anorexia, ↓T4
2. Infection – granulomatous, HIV, sepsis
3. Iatrogenic – chemo, rads, ETOH
4. Infiltrative – mets, amyloid, Gaucher
5. Hypersplenism

Approach to Lymphadenopathy

Generalized Lymphadenopathy				
Infectious	Inflammatory	Infiltrative	Med	Malignancy
Atypical bacterial (TB, lyme, syphilis) Viral (EBV/CMV/HIV) Fungal Parasite (Toxo)	Stills' SLE RA Castleman's	Sarcoid Amyloid	Phenytoin	Lymphoma Leukemia Mets

Vitamin B12 deficiency

Etiology:

- Dietary deficiency (vegans) or meds (PPI, H2 blocker, metformin, nitric oxide)
- Absorption issues - no intrinsic factor (pernicious anemia)
- Atrophic gastritis or Pancreatic insufficiency or Terminal ileum disease

Clinical manifestations:

- Anemia-related symptoms, mild jaundice
- Glossitis
- **Neurological: subacute combined degeneration (posterior columns & corticospinal tract), peripheral neuropathy, dementia**

Hematologic:

- CBC: macrocytic anemia, other lines can be decreased
- film: oval macrocytes, hypersegmented (≥ 5 lobes) neutrophils

Associated Labs:

- low B12, elevated methylmalonic acid & homocysteine levels
- Anti-intrinsic factor and anti-parietal cell antibodies (pernicious anemia)

Treatment:

- Daily high-dose ($\geq 1000 \mu\text{g}$) oral B₁₂ is as effective as intramuscular supplementation. Hematologic abnormalities should resolve within weeks and neuropsychiatric symptoms within months
- If profound deficiency and need to urgently replace (eg in pregnancy w symptoms) - Vitamin B12 1000 mcg IM daily x7d, then 1000mcg qweek x4, then 1000mcg IM q1month lifelong
- OR once replete, vitamin B12 1000-2000 mcg po OD

B12 deficiency or syphilis?

- Distinguish Neuro B12 deficiency from Neuro Syphilis
 - BOTH: Mental status changes/Dementia
 - Pupils: B12: Normal Syphilis: Argyle Robertson
(small, no reaction to light but does accommodate)
 - B12 = subacute combined degeneration of cord = *corticospinal tract* (UMN weakness, increased tone in legs, hyperreflexia/clonus) AND *dorsal column* (dec. vibration, proprioception)
 - Syphilis = “Tabes Dorsalis” = *dorsal column* (vibration, proprioception impaired) and dorsal roots (absent reflexes in LE). Unlike B12 deficiency, tone/power should be normal.

Both can have dorsal column findings -
Physical exam videos available on
www.internalmedicinereview.ca

Non-malignant causes of “-cytosis”

Ddx ↑ Hb:

- Cyanotic heart disease
- Chronic lung disease/OSA
- Hypoventilation
- Smoking or CO poisoning
- Renal artery stenosis
- Tumors (ex. RCC, hepatoma, uterine fibroids)
- Testosterone
- EPO
- High altitude
- SGLT2 Inhibitors

Ddx ↑ Plt:

- Iron deficiency
- Infection
- Inflammation / Burns
- CTD
- Malignancy
- LPD
- Post-splenectomy
- Surgery

Ddx ↑ WBC:

- Infections
- Inflammation
- Autoimmune disorders
- Non heme-cancer
- Steroids
- GCSF
- Trauma
- Burns
- Post seizure

Leukemoid reaction →

Total WBC > $50 \times 10^9/L$ +
left shift

Thrombosis Bonus Slides



Pre-test Probability for VTE

Scenario Pearl: Be familiar with exam for DVT for OSCE!

Well's Score for DVT (1 pt. for each) *not tested in pregnancy*

- History**
- Active CA (≤ 6 mo)
 - recent lower limb immobilization
 - recently bedridden > 3 d or surgery (w/in 4 wk)
- Physical Exam**
- tenderness along deep veins
 - evident collateral veins
 - entire leg swollen
 - calf swelling > 3 cm (10 cm below tib tub.)
 - pitting edema
 - non varicose superficial veins

*** Minus 2 for likely alternative diagnosis**

Modified Well's Score for PE

- 4 History**
- active cancer (1)
 - hemoptysis (1)
 - prior VTE (1.5)
 - recently bedridden or surgery (1.5)
- 2 Physical Exam**
- symptoms of DVT (3)
 - tachycardia (1.5)

*** No alternative diagnosis (3)**

Scoring

High probability: ≥ 3 pts
Mod probability: 1-2 pts
Low probability: 0 or less

Wells PS, et. Al. Lancet. 1997

"Modified" Scoring

High probability: > 4 pts
Low probability: ≤ 4 pts

Wells PS, et. al. Thromb Haemost. 2000

Wells Rule/DDimer threshold cutoffs by age. Ann Intern Med.
2016 Aug 16;165(4):253-61

Risk of VTE Recurrence

REMEMBER: consider indefinite anticoag in all unprovoked clots, but if reason to stop – can use risk stratification tool (e.g. HERDOO2 or DODS) to assess risk

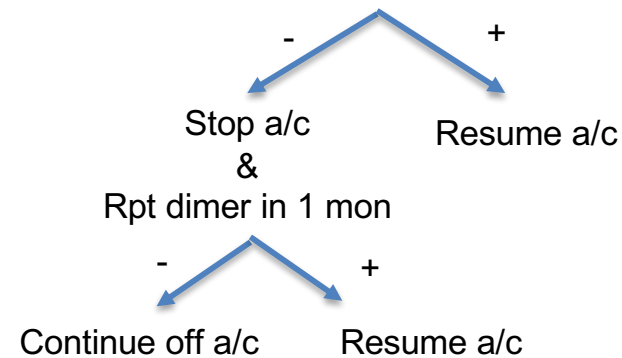
HERDOO Predictors

Hyperpigmentation or
Edema or Redness in either leg
D-Dimer ≥ 250 ug/L
Obesity, BMI ≥ 30
Older age ≥ 65 years

♀ with score $<2 \rightarrow 3\%/pt$ yr
♀ with score $2+$ & all ♂ $\rightarrow$ high risk

DODS:

D-dimer on anticoagulation (a/c)



♂ with – D-dimer x 2 $\rightarrow 7.5\%/yr$
♀ with – D-dimer x 2 $\rightarrow 3.8\%/yr$ (0.4%/yr if estrogen-associated)

Roger et al., CMAJ, 2008

DOAC Extension Studies (non-cancer associated)

**In unprovoked VTE with low-med bleed risk > 6 mos of full dose treatment,
consider lower dose-DOAC**

- Risk of recurrent VTE:
 - LOW dose (1.6%) non-inferior to FULL dose (1.4%)
 - LOW dose superior to Placebo or ASA (6.3%)
 - RENOVE – non inferiority not met (but low event rate)
 - Bleeding events:
 - Full dose 3.7% (Major bleeding RENOVE - 5.0%)
 - Low dose 2.7% (Major bleeding RENOVE – 2.1%)
 - ASA or Placebo 2.3%
 - Caution in:
 - Morbid obesity (Wt >120kg)
 - Symptomatic pulmonary hypertension
 - Index VTE event required thrombolytics
 - Active cancer
 - Thrombophilia
- Cost – low dose not always covered by drug plans for this indication

AMPLIFY – EXT (NEJM 2013)

N= 2482

Apix 5mg BID vs. 2.5 mg BID vs. placebo
Duration 12 mos

EINSTEIN CHOICE (NEJM 2017)

N = 3365

Riva 20mg daily vs. 10mg daily vs. ASA
81mg daily
Duration 12 mos

RENOVE (Lancet 2025)

N = 2768

Apix 2.5mg BID or Riva 10mg OD vs.
Apix 5mg BID or Riva 20mg OD
Duration 48mo FU

DOAC Extension Studies (cancer associated)

Extended Reduced-Dose Apixaban for Cancer-Associated Venous Thromboembolism (API-CAT) NEJM 2025

Population → Patients with **active cancer** (excluding basal cell / SCC of skin, primary CNS or CNS metastases) with **proximal DVT or PE** (segmental or large pulmonary artery) who had completed at least 6 months of therapeutic AC.

Intervention → Randomized to receive reduced dose **Apixaban 2.5mg BID** vs. Apixaban 5mg BID for 12 months.

Outcomes

VTE Recurrence (Primary outcome)

- Apixaban 2.5 mg BID **non-inferior** to Apixaban 5mg BID for VTE recurrence (2.1% vs. 2.8% with adjusted subhazard ratio 0.76; p=0.001)

Clinically Relevant Bleeding (Secondary outcome)

- Apixaban 2.5mg BID **superior** to Apixaban 5mg BID (12.1% vs. 15.6% with adjusted subhazard ratio 0.75; p = 0.03)

Cancer Associated VTE – Outpatient Prophylaxis

Parameters	Pt Characteristics	Risk
Cancer site	Very high risk (brain, stomach, pancreas)	2
	High risk (lung, lymphoma, gyne, bladder, testicular, myeloma, renal)	1
Bloodwork	Pre Chemo plt >350	1
	Hgb <100 or use of Erythropoeitin Stim Agents	1
	Pre Chemo WBC >11	1
BMI	≥ 35	1

2.5 month rate of VTE

- Low (0) < 1%
- Int (1-2) ~2%
- High (≥3) ~7%

#Khorana et al. Blood 2008

*Khorana et al. Thromb Res 2017

AVERT ,NEJM 2018

CASSINI , ASH 2018

ASCO 2020 clinical practice guideline recommends considering DVT prophylaxis (apixaban 2.5 BID, rivaroxaban 10 OD, LMWH) for high risk ambulatory cancer patients (Khorana score ≥ 2).

ASCO Clinical Practice Guidelines Updates 2020

Hypercoaguable States

Thrombophilia	Population prevalence	Mechanism	Relative risk first VTE	Relative risk recurrent VTE	Who to test?	Diagnosis
Factor V leiden (FVL)	Caucasian 5% (hetero); 0.06-0.25% (homo)	Resistance to inactivation by protein C	3-5	1.4	Controversial, require specialist advice: consider for unusual sites, if will change mgmt. See next slides for updates.	PCR
Prothrombin gene mutation	Caucasian 3% (hetero); <0.1% (homo)	Hyper-prothrombinemia	2-3	1.4		PCR
Protein C def	0.2-0.5%	Inhibit FV and FVIII	4-6.5	1.4-1.8		Activity level
Protein S def	0.1-0.7%	Co-factor protein C	1-10	1.0-1.4		Activity level
Antithrombin def	0.02%	Inhibitors thrombin (FIIa) and Xa	5-10	1.9-2.6	Venous, arterial, microvascular thrombosis; pregnancy complications	Activity level
Antiphospholipid syndrome	0-5%	Autoimmune antibodies	Triple+ highest risk; LA strongest RF	Uncertain		See Rheum Slides New ACR 2023 Diagnostic Criteria

DOACs Evidence **Bonus Slide**

DOACs vs. VKA in Acute DVT and PE

1. Preventing VTE recurrence?
 - DOACs as effective as warfarin
2. Incidence of bleeding?
 - DOACs 40% RRR in major bleeding
 - DOACs 25% RRR in non-major bleeding
3. Incidence of GI bleeding?
 - No significant difference DOAC vs. warfarin
4. Safety with renal failure?
 - DOACs ↓ risk for CNS bleed vs. warfarin in GFR 50-80
 - Apixaban ↓ major bleed vs. other DOACs in GFR < 50

DOACs in CA VTE Evidence

Clot Trial (NEJM 2003) Dalteparin 200 U/kg/day X 4 weeks then 175 U/kg/d X 5 mos

EDOXABAN - HOKUSAI Cancer (NEJM 2017) Randomized, open-label non-inferiority trial

- Dalteparin vs. Dalteparin x 5 d then Edoxaban 60 mg qd or 30 mg If wt < 60kg/CrCl 30 – 50/pGP inhib
- N = 1046, F/U 6-12 months with ~50% symptomatic PE
- VTE recurrence or major bleed in 12 mos: Edox (12.8%) non-inferior to Dalteparin (13.5%)
- Major bleeding edoxaban (6.9%) > dalteparin (4.0%) due to GI CA bleed

RIVAROXABAN - SELECT-D trial Phase 2 (Blood 2017) Prospective Randomized, open-label Pilot

- Dalteparin vs. Rivaroxaban (15mg BID X 3wk then 20mg qd) N = 406
- VTE recurrence with riva (4%) < Dalteparin (11%)
- Similar rate of major bleed (~4%). ↑ clinically relevant minor bleeds in Riva arm.

APIXABAN – CARVAGGIO trial (NEJM 2020) Randomized, open-label noninferiority trial

- Dalteparin vs. Apixaban (10mg BID X 7d then 5mg BID) N = 1170
- Recurrent VTE 5.6% in Apix vs. 7.9% in Dalteparin (HR 0.63, noninferior)
- Major bleed 3.8% in Apix vs. 4.0% in Dalteparin

Chimeric Antigen Receptor (CAR)-Therapy

Definition: immunotherapy → engineered self T cells to target cancer cells

Indications: R/RB-cell ALL, R/R aggressive lymphoma, R/R myeloma

Toxicities

1. Cytokines release syndrome (CRS)

Sx: Fever, tachypnea, headache, tachycardia, HTN, rash, hypoxia due to cytokines storm

Labs: elevated CRP/ferritin

Tx based on CRS grade: supportive care, $\pm$ steroids, $\pm$ tocilizumab

2. Immune effector cell associated neurotoxicity (ICANS) due to disruption in BBB, cytokines

Sx: aphasia, tremor, altered LOC, impaired cognition, motor weakness, seizures, cerebral edema

Tx based on ICANS grade: supportive care, CNS imaging, seizure prophylaxis/treatment; Tx concurrent CRS

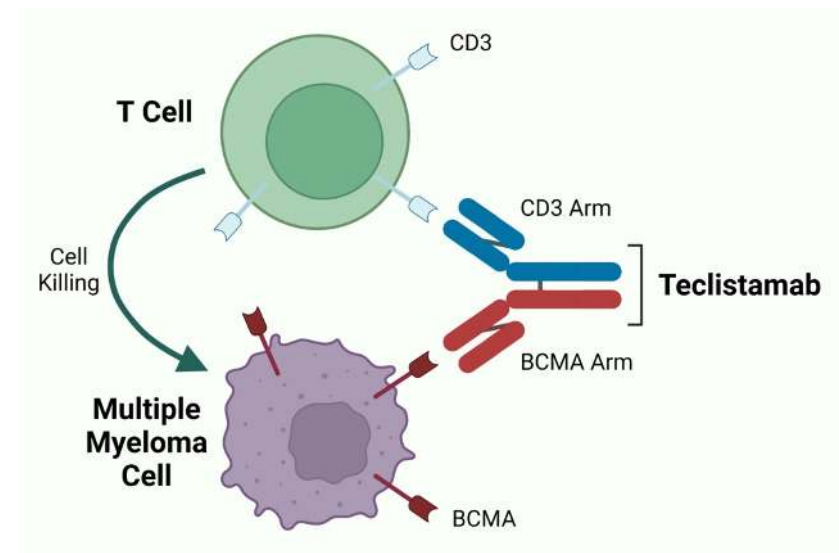
ICU if hypotensive, persistent tachy, O₂ >4L nasal cannula, ICANS neuro > grade 2

Others: infection, prolonged cytopenias, hypogammaglobulinemia, reduced vaccine response (COVID-19)

For details: <https://www.cancer.gov/about-cancer/treatment/research/car-t-cells>

Bi-specific T-cell Engagers (BiTEs)

- Monoclonal antibody with 2 targets:
 - 1 against an antigen presented on malignant cells
 - 1 against T-cells
- Brings malignant cells in proximity with cytotoxic T-cells for them to exert their effects
- Current uses: ALL, lymphoma, myeloma
- Risks: CRS and ICANS (but at lower rates than CAR-T cell therapy)
- Benefit: “off-the-shelf”



EX – Teclistamab. Anti BCMA BiTE used in treatment of MM

Oral Scenario

Obtain consent for RBC transfusion in a bleeding patient with Hb 65

1. BENEFITS: Why is transfusion required?

2. RISKS:

- **The common** (1/10-1/100): fever (FNHTR), hives, sensitization

- **The serious:**

volume overload, TRALI

anaphylaxis

bacterial infection

acute hemolytic transfusion reaction

- **The extremely rare:**

hepatitis B (1/7.5 million)

hepatitis C (1/13 million)

HIV (1/21 million)

transfusion associated GVHD

post-transfusion purpura

} Bloody Easy 5th ed.

Oral Scenario

3. ALTERNATIVES: Are there alternatives to transfusion?

- Jehovah's Witness scenario
 - Explore what blood products are acceptable (RBC, PLT, FFP, other factors, ESAs, IV Iron)
 - Risks of proceeding without transfusion
 - » Establish a power of attorney / SDM in a periop scenario
 - Options :
 - » Pre-surgical optimization: Iron, EPO, cell-saver
 - » Active bleeding: TXA

Oral Scenario

38F hx SLE, in ICU with pneumonia. Prior to admission she was awaiting an out-pt heme apt for pancytopenia. Now intubated and ventilated, off pressors for 24h. While admitted, her plt drops from $110 \times 10^9/\text{L}$ to $40 \times 10^9/\text{L}$ in 48h. She also develops a large ilio-femoral DVT on prophylactic UFH.

1. Summarize the case
2. What is your initial approach?
3. What is your differential diagnosis?
4. What additional investigations will you request?

Oral Scenario

1. **Summarize the case:** 38F in ICU w/ hx SLE, now pancytopenic + acute on chr low Plt + DVT.
2. **What is your approach?**
 - a. ABC IV MOIF + Stable vs. unstable?
 - b. Review info: meds, history, labs, rads
 - c. Assess: vitals (adrenal hemorrhage), focal neuro signs (CSVT), PE, limb vascular compromise, skin exam (petechiae/purpurae/necrosis/livido reticularis)
3. **Not to miss in differential for CLOT + low Plt**
 - DIC – sepsis
 - TMA differential – esp. APLA vs. TTP
 - Hematologic Malignancy (history of pancytopenia)
 - Pregnancy associated
 - HITT in a hospitalized patient on heparin
 - liver synthetic dysfunction and portal hypertension

Oral Scenario

Exam: BP 108/74, HR 112bpm, FiO2 0.4, Vent settings PS increased in last 24 hours from 8 to 12. Lungs: decreased air-entry bilaterally. Heart: S1 and S2 reg with systolic flow murmur, no palpable P2. Abdomen: soft, non-tender, spleen edge 5cm below costal margin. Neuro: no focal neurologic signs, Extremities: left whole leg swelling, no limb neuro-vascular compromise, no skin necrosis, livedo reticularis present.

Meds: Meropenem, Pantoprazole, Hydroxychloroquine, UFH 5000u SC bid

Labs: Hgb 78 g/L, Plt 40×10^9 , WBC 1.2, ANC 0.5, **Blasts 0.03**, Blood film – giant plt no aggregates, rare blasts, no schistocytes, toxic neut. Beta-HCG neg, AST/ALT 2xULN, INR 2.1, aPTT 63s, Fibrinogen 1.0, Creat 163 umol/L.

Imaging findings:

TTE - LVEF 40%, no valvular lesions.

CT Chest – new multiple bilateral sub-segmental PEs. Right middle lobe consolidation improved since previous scan 72h ago.

Abdominal US: ilio-femoral DVT, no SVT, no portal-vein thrombus, liver normal appearance, spleen 22 cm.

Oral Scenario

- **What is your most likely diagnosis now?**
 - =Either APLA or DIC vs. **MALIGNANCY (LGL)**
 - =Blasts in blood always need assessment
 - Heme malignancy (**LGL**, AML, MPN, MDS) vs.
 - Stress response due to critical illness
- T-cell Large granular lymphocyte leukemia (LGL) associated with autoimmune dx esp. RA. Often present like Felty-syndrome with splenomegaly and neutropenia.

Oral Scenario - 2026

An 81 year old gentleman with a history of HTN, atrial fibrillation, CAD with remote history of CABG (2004), previous colon cancer managed with resection, presents to the Emergency Department for the second time for evaluation of bruising. At his last visit he had developed a large bruise over his left thigh after bumping into the corner of his dining room table. At last ED visit he was vitally stable, and was advised to hold his aspirin and follow up with his family doctor if symptoms didn't improve. He returns now due to worsening swelling/pain at the site of his left thigh ecchymoses, and new bruising to his left flank. He cannot recall any new trauma or injuries since the development of his new bruise. You have been asked to see him for further workup and management.

1. Summarize the case
2. What is your initial approach?
3. What is your differential diagnosis?
4. What additional investigations will you request?

Oral Scenario - 2026

1. **Summarize the case:** I have been asked to see this 81 year old gentleman with multiple medical comorbidities including atrial fibrillation and coronary artery disease for worsening trauma induced and spontaneous bruising.
2. **What is your approach?**
 - a) ABC IV MOIF – Stable vs. unstable
 - b) Review info – Medications (anticoags, anti platelets, OTCs, herbals), History (previous bleeding, recent trauma/injuries, any other bleeding, EtOH), Labs (CBC, coags, fibrinogen, renal function, liver profile, lipase), Imaging – CT extremity, CT Abdo/Pelvis
 - c) Assess – Vitals (any hypotension, tachycardia), Skin – evaluate bruising *outline area to assess progression. Any other foci of bruising. CNS – any neuro sx / concerns for CNS or spinal cord hemorrhage, Abdominal exam, CV/Resp
3. **Differential Diagnosis** – Important considerations re: new bruising/bleeding
 - Medication effect – Too much anticoag or new reason for ↑ serum levels (EX: drug/drug interactions, changes in renal/hepatic function)
 - Acquired coagulopathy – EX: Acquired VWD or factor inhibitor, DIC
 - Liver failure
 - Platelet dysfunction - EX: antiplatelet, new meds, ITP
 - New hematologic malignancy

Oral Scenario - 2026

History – No new trauma or prescription / OTC medications. ASA has remained on hold since last ED visit. He does not use NSAIDs. No herbal supplements. Rare EtoH history, ~1-2 drinks/month. No other bruising/bleeding noted. No headaches. Has lost ~8 pounds over last 2 months.

Meds – Rivaroxaban 20mg OD, Candesartan 8mg OD, Bisoprolol 5mg OD.

Examination – VS T 36.9, HR 110, BP 102/68, RR 18, 97% O2 sats R/A. CN exam normal, no visible periorbital ecchymoses. Normal cardio/resp. Abdomen soft, mild diffuse tenderness non peritonitic, ~7cm in diameter dark purple ecchymosis over L flank. Large ~8cm dark purple area of ecchymosis over L upper thigh – in comparison to photo from previous, doubled in size.

Labs – Hb 98g/L (baseline 131), MCV 77, WBC 10.2, PLT 275. No abnormal cells on blood film. Cr 81. ALT 20 / ALP 159 / Tbili 12. Lipase 49. INR 1.6, aPTT 81s. 50:50 Mix – INR 1.3, aPTT 74s. Fibrinogen 1.9. D dimer > 1000.

Imaging –

- **CTA Abdomen/Pelvis** – Large ~8.5x9.1cm homogeneous collection in the retroperitoneum abutting the left kidney and spleen. No visible active extravasation. No pancreatic inflammation or visible masses.
- **CTA Lower Extremity** – Large 9x11cm homogeneous collection in along the soft tissue of the left upper thigh, no visible active extravasation.

Oral Scenario - 2026

What is your most likely diagnosis?

- **Acquired Hemophilia A** (aPTT >>> INR, new spontaneous bruising without clear triggers / medication effect, no correction with 50:50 mix).
 - Rivaroxaban may cause slight prolongation of PT/INR, but degree of PTT elevation here not consistent with anticoagulant effect.

What additional investigations will you send to confirm?

- Factor VIII level, Factor VIII inhibitor assay

Outline your acute management steps to control their bleeding

- Admit to hospital, vitals q6H, monitor for progressive ecchymosis
 - Bypassing agent (EX: FEIBA), blood product consent
 - Hematology / Hemophilia Centre Consult
 - Hold Rivaroxaban
- + once stable eval for triggers - EX GI investigations with upper and lower endoscopy to eval for malignancy – prev colorectal ca, new microcytic anemia, R/O recurrence of colorectal ca

Bonus MCQ 1 2024

A 55 year old man presents with several months of fatigue, night sweats, progressive early satiety and abdominal girth. His past medical history consists of rib fractures, in context of sport related injury, and is currently not on any medications. His social history consists of 1 pack per day smoking history for past 20 years, and 7-10 alcoholic beverages per week. On history, he mentions that he has noticed lumps and bumps in the neck for the past year. Physical exam demonstrates normal cardiorespiratory exam, mildly distended abdomen with positive Castell's sign, and adenopathy in the cervical chain, axilla, supraclavicular fossa, and inguinal region. His labs show Hb 98, plt 130, WBC 10 with ALC 3, ANC 3.3, Cr 76. CT imaging shows lymphadenopathy up to 3 cm, as well as old rib fracture. What is the most likely diagnosis?

- A. Burkitt lymphoma
- B. Multiple myeloma
- C. Chronic lymphocytic leukemia
- D. Follicular lymphoma
- E. Hodgkin lymphoma

Bonus MCQ 2 2024

A 62M with a history of carpal tunnel syndrome (recently diagnosed), presents with complaints of progressive swelling, shortness of breath and paresthesias to his toes.

OE: BP 112/63, HR 89, SpO2 94% RA, afebrile. Macroglossia, bruising, hepatomegaly, bilateral pitting edema, elevated JVP, no lymphadenopathy, normal neuro exam.

Invx: Hb 118, MCV 92, WBC 5.4 (N diff), Plt 193, INR 1.3 (ULN 1.1), aPTT 35 (ULN 32), fib. 3.2, Cr 154, 24h urine: 1.3 g

Which of the following is most likely to be found on further examination or investigation?

- A. Hypercalcemia
- B. Plasmacytoma on whole body MRI
- C. An endocrinopathy
- D. Apple green birefringence on Congo Red stain of renal biopsy

Bonus MCQ 3 2024

A 59 year old female presents with epigastric pain, nausea and vomiting, in the context of heavy alcohol use. She was found to have elevated lipase, and diagnosis of pancreatitis was made. Given worsening abdominal pain, fever, and emesis, CT abdomen was done at 48 hours demonstrating inflammation around the pancreas with a necrotic appearing collection and a small splenic vein thrombosis. Her blood work showed Hgb 140, HCT 46%, platelet 500, WBC 19 with predominant neutrophilia (ANC 15) without left shift, Cr 120, INR 1.2, aPTT 32. What further investigations would you order?

- A. No further testing
- B. JAK2 molecular test
- C. Antiphospholipid syndrome testing
- D. Thrombophilia testing for inherited hypercoaguable states

Bonus MCQ 4 -2024

A 32 year old female is G1P0, currently GA 34 weeks, presents for evaluation with a bleeding history. Her history consists with mucosal bleeding with epistaxis requiring nasal packing in emergency department 3 years ago, wisdom tooth extraction complicated by prolonged bleeding for which she required dental repacking, and menorrhagia requiring changing of menstrual product every hour. Her current examination does not demonstrate petechiae, and no active mucocutaneous bleeding. Her investigations show Hb 109, MCV 82, platelet 170, WBC 6. She did have von Willebrand disease testing two weeks ago which showed normal platelet aggregation testing, and vWF antigen level 60%, vWF:activity 70%, and factor VIII level 110%. What is the next best step?

- A. Mixing study
- B. No further testing
- C. INR/APTT
- D. Genetic testing
- E. Repeat vWD testing after pregnancy

Bonus MCQ 5 2024

You are asked to see a 72yM POD#12 from a CABG (LIMA to LAD, left SVG to OM1, RCA). He has had a prolonged hospital stay due to a sternal wound infection. He has been mobilizing with physiotherapy but complained of calf pain and is found to have bilateral thrombus of the superficial femoral vein. BP 110/60, wt 100 kg, HR 90, afebrile, 94% RA.

Current meds: ASA 81mg daily, ramipril 10 mg daily, bisoprolol 5 mg daily, atorvastatin 80 mg daily, heparin 5000 u sc BID.

Labs reveal: Hgb 107, WBC 11, Plt 118, Cr 122. In comparison to his admission CBC with Hb 132 / WBC 8 / PLT 288, Cr 118.

Which of the following would not be an appropriate empiric anticoagulant choice?

- a) Enoxaparin 100 mg sc BID
- b) Rivaroxaban 15 mg po BID
- c) Fondaparinux 10 mg sc OD
- d) Apixaban 10 mg po BID

Bonus MCQ 6 - 2024

A 35 year old female with history of scleroderma presents with epistaxis and found to have platelet of 3, diagnosed with ITP, initiated on corticosteroid therapy with dexamethasone, and received IVIG 1 g/ kg. Two days later, she develops headache, nausea, and emesis. She is blood group A+. Repeat bloodwork shows Hgb 110 from 123 on admission, platelet increased to 40, WBC 7, Cr 65, with normal INR/PTT.

What is the most likely diagnosis?

- A. TTP
- B. DIC
- C. Aseptic meningitis
- D. Scleroderma renal crisis
- E. Cerebrovascular event

Bonus MCQ 7 2024

A patient is referred to your clinic for microcytic anemia. They are a 32 year old lady who has been feeling well and on routine blood work has had persistent microcytic anemia despite iron replacement. She has never had admissions to hospital, and has never had a blood transfusion. They are on no medications. She has been diagnosed by her ophthalmologist as having non-proliferative retinopathy. Her investigations show Hemoglobin 92 g/L, MCV 68, WBC 8.0, Platelets 220. Her blood film shows a large number of target cells and several dense contracted and folded red cells. What is your suspected diagnosis:

- a) HbSS disease
- b) HbSC disease
- c) G6PD deficiency
- d) Sickle cell trait

Bonus MCQ 8 2024

A 67-year-old man is referred to your clinic for fatigue and abnormal blood work. He has complained of fatigue and early satiety. Physical examination is remarkable for hepatosplenomegaly. Investigations show a Hemoglobin of 68 g/L, White Blood Cell Count of 14 with a normal differential, Platelets of 58. Electrolytes, creatinine, and liver enzymes are normal. LDH is elevated at 510. Blood film shows myelocytes, metamyelocytes, and teardrop cells. There is no evidence of dysplasia. What is the most likely diagnosis?

- a) CMML
- b) Essential Thrombocythemia
- c) Myelofibrosis
- d) Lung cancer

Bonus MCQ 9 2024

A 40-year-old female presents for evaluation of bilateral, distal lower limb paresthesias. Initial work-up is unremarkable aside from an IgM protein of 5 g/L on screening serum protein electrophoresis. Hemoglobin, creatinine, and calcium are normal. What would be the next best step?

- A. Bone marrow biopsy and aspirate
- B. Observe with annual SPEP
- C. Send for serum free light chains
- D. Send for 24-hour urine protein
- E. Initiate empiric Vitamin B12

Bonus MCQ 10 2024

A 63-year-old female with history of ER/PR positive breast cancer treated with chemotherapy and radiation 2 years prior presents with fatigue, worsening exercise capacity including dyspnea on exertion, and overall decreased functional status. She continues on hormonal therapy currently. Her exam demonstrates no breast masses, and no localizing bony pain. She appears pale, and a grade 3/6 systolic ejection murmur can be heard over left sternum border, and no other pertinent exam findings. Her blood work shows Hgb 72, MCV 108, WBC 2 with ANC 0.8, platelet 75, no blasts seen with blood film demonstrating hypolobulated neutrophils with poor granulation. What is the most likely cause of the patient's presentation?

- A. Acute myeloid leukemia
- B. Myelofibrosis
- C. Myelodysplastic syndrome
- D. Metastatic breast cancer to the bone
- E. Vitamin B12 deficiency

Bonus MCQ 11 2024

A 31-year-old lady G3P3 receives 2 units of red blood cells for a post-partum hemorrhage. 1 week later she presents with fatigue and dyspnea. On examination, she is pale with mild scleral icterus and appears jaundiced. Her hemoglobin is 72 g/L (hemoglobin prior to delivery of 112 g/L), WBC 9, Platelets 244. Blood film shows polychromasia with occasional spherocytes. LDH is elevated at 623, total bilirubin is 42, and haptoglobin is undetectable. What is the most likely cause of her presentation?

- a) Delayed hemolytic transfusion reaction from ABO incompatibility
- b) Delayed hemolytic transfusion reaction from Rh incompatibility
- c) Idiopathic autoimmune hemolytic anemia
- d) Post-partum TTP

Bonus MCQ 12 2024

A 26-year-old lady with sickle cell disease (HbSS) presents with right limb weakness that progresses within hours to right sided hemiplegia. An urgent CT head confirms a new left ischemic stroke. She has a history of recurrent pain crises and had an episode of acute chest syndrome during an admission 5 years ago. Her hemoglobin is 97 g/L, white blood cell count of 9 and platelets of 490. She has been started on ASA and DVT prophylaxis, and morphine for pain control. What should be done in addition to her current management?

- a) 3L of normal saline
- b) Hydroxyurea
- c) Exchange transfusion
- d) 2 units RBCs

Bonus MCQ 13 - 2025

A 24-year-old male presents to the Emergency Department with several hours history of worsening right knee swelling and pain. This started after exercising at the gym and feeling a “pop” in his knee. They have a history of moderate hemophilia A with baseline factor VIII level of 8%. On examination the right knee is swollen, tender, and warm to touch. Hb 135, PLT 250, WBC 5.5, MCV 85, INR 1.0, PTT 42 s (ULN 35 s). What is the next best immediate step in management?

- a) Arrange for urgent right knee ultrasound
- b) Administer empiric Factor VIII product
- c) Administer 2 units of platelet
- d) Perform arthrocentesis

Bonus MCQ 14 - 2025

A 19-year-old female presents with 3-week history of worsening bruising and menorrhagia. She has also noticed increased fatigue with dyspnea on exertion. No recent infectious symptoms. Examination is notable for sporadic ecchymoses. Hb 84, WBC 1.2, PLT 44, Neutrophils 0.6, reticulocytes 5, Vitamin B12 400, ferritin 50. No palpable hepatosplenomegaly or lymphadenopathy. What is the most likely diagnosis?

- a) Pure red cell aplasia
- b) Parvovirus B19 infection
- c) Hodgkin lymphoma
- d) Aplastic anemia

Bonus MCQ 15 - 2026

A 67 year old male presents to your office for evaluation of lower extremity skin ulcers over the left and right shins. His medical history includes CAD with previous MI, HTN, and Polycythemia Vera. His home medications include Hydroxyurea 1g M-F, 1.5g SS, Candesartan 8mg daily, ASA 81mg daily. His last CBC was a Hb of 121 / Hct 42% / WBC 10 / PLT 199. Bilateral ABI were 1.2 (normal). An arterial duplex US demonstrated no significant lower extremity vessel stenosis.

What is the best next step?

- a) CT-A lower extremities
- b) Skin biopsy
- c) Stop Hydroxyurea
- d) Start therapeutic anticoagulation

Bonus MCQ 16 - 2026

A 38 year old G3P1A1 female at 29+4 weeks gestation is seen in follow up for anemia and iron deficiency in pregnancy. Her ferritin levels throughout her first and second trimester were <3 and 7 respectively and she has been managed on oral iron supplementation with 1 tablet of ferrous fumarate daily . Her Hb levels from 1 week ago were 91g/L with MCV of 73. She is experiencing ongoing fatigue, difficulty with concentration, and finding it increasingly difficult to care for her other children at home.

What is the best next step in treatment - .

- A. Arrange for a 1U PRBC transfusion
- B. Switch ferrous fumarate to every other day
- C. Arrange for 3 doses of IV Venofer
- D. Increase dose of ferrous fumarate

Bonus MCQ 17 - 2026

A 54 year old female is brought to the ED for acute onset of a severe headache, vomiting, and a decreased level of consciousness after she slipped and fell on the icy pavement earlier today. A stat CT head is performed which demonstrates a large left sided subdural hematoma with evidence of mass effect. She is awaiting emergent Neurosurgical OR. Her medical history includes HTN, T2DM, and a previous mechanical mitral valve repair for rheumatic heart disease. She has been on Warfarin for > 20 years and is on a stable dose of 4mg daily. Her INR has come back at 3.2.

What is your next step in management?

- A. Administer 10mg IV Vitamin K
- B. Administer 2000U of PCC and 10mg IV Vitamin K
- C. Stop Warfarin. Let INR decrease without reversal agents
- D. Administer 1000U of PCC

Bonus MCQ 18 - 2026

A 26 year old male presents to the ED with fatigue, fever, cough, and shortness of breath. He is previously healthy, and just returned home after travelling to Vancouver for a business trip. On arrival his temp is 38.3 and O2 sats are 88% on room air. He appears to be in mild respiratory distress and has coarse breath sounds bilaterally. Exam is otherwise normal apart from a reticular rash to the arms and legs. His CXR reveals bilateral patchy opacities. Bloodwork – Hb 62 / WBC 11.4 / MCV 98 / PLT 450. Na 134 / K 4.5 (on repeat – initial sample hemolyzed)/ Cr 82. ALT 22 / ALP 118 / Tbili 57. CRP 45. A rapid COVID/respiratory viral PCR is negative.

What is the most likely explanation of his anemia?

- A. VB12 deficiency
- B. MDS
- C. Cold Agglutinin Disease
- D. Anemia of Inflammation/Critical Illness

Bonus MCQ 19 - 2026

A 24 year old female is referred for evaluation of anemia. 6 months ago she underwent workup for heavy menstrual bleeding and fatigue and was found to be anemic with a Hb of 92 / MCV 62 / WBC 8 / PLT 518. Her TSH was normal, and she was found to be iron deficient with a ferritin of 3. INR/PTT were normal, and her VWF Ag and Activity levels were in the normal range. She was started on oral iron supplementation with ferrous gluconate, and her menorrhagia has improved with insertion of an IUD. Her repeat bloodwork demonstrated a Hb of 111 / MCV 64 / WBC 7.5 / PLT 400. Ferritin has improved to 101. What is the most likely explanation for her bloodwork abnormalities?

- A. Iron refractory iron deficiency anemia (IRIDA)
- B. Thalassemia
- C. Sick cell trait
- D. Anemia of chronic disease

Bonus MCQ 20 - 2026

A 22 year old female presented to the ED with spontaneous bruising and small red dots appearing over her upper and lower extremities. She has had no headaches and no other bleeding symptoms. She has a remote history of a similar presentation in 2021 managed at that time as ITP. She was treated with a 4 day course of Dexamethasone and has had normal PLT counts since. Today in the ED her CBC reveals a Hb of 121, MCV 81, WBC 6.6, PLT 7. Blood film comments on some giant platelets, no other abnormalities. Cr is 71. Her blood group is O+. Her weight is 61 kg. Which of the following is the least appropriate next step in management?

- A. Dexamethasone 40mg PO daily x 4 days
- B. Prednisone 60mg PO daily
- C. 1U PLT transfusion
- D. IVIG 1g/kg IV x 1 dose

Bonus MCQ 21 - 2026

A 27 year old male with diagnosis of AML is transferred to the ICU for fevers and respiratory distress. He is intubated and receiving treatment with broad spectrum antibiotics and anti-fungal agents. His last dose of chemotherapy was 10 days ago with high dose cytarabine as part of his consolidation. Since his admission to hospital he has required regular blood products due to cytopenias. In the ICU alone he has required 8 units of PLTs over the last 72 hours. His most recent CBC reveals a Hb of 69, WBC 4.3, ANC 1.8, PLT 3. His INR is 1.3, PTT 32s, Fibrinogen 2.3. He receives a unit of RBCs and PLTs. Repeat CBC performed 1 hour post transfusion to assess his response demonstrates a Hb of 78, WBC 4.2, ANC 1.9, PLT 5. Which of the following is the most likely explanation for his platelet response?

- A. Septic shock
- B. DIC
- C. Myelosuppression from chemotherapy
- D. Platelet alloimmunization

Bonus MCQ 22 - 2026

A 37 year old male was recently admitted to hospital 14 days ago post a motor vehicle collision during which he sustained multiple traumatic injuries including a left femur and left ankle fracture requiring operative repair. He also sustained left sided rib fractures, and a left clavicular fracture. He received DVT prophylaxis with Enoxaparin in hospital, had no postoperative complications, and was discharged to rehab 3 days ago. He was transferred to the ED with new chest pain and dyspnea. His left lower extremity is enlarged > R, is well perfused with pulses palpable bilaterally. He is alert and oriented, but complaining of ongoing chest discomfort. VS on latest assessment BP 104/82, HR 110, RR 20, and O2 sats of 94% on room air. Weight today is 79 kg. Initial labs include Hb 116, PLT 383, Cr 80, ALT 22, Tbili 8, INR 1.2, Troponin 32. Repeat is pending. His ECG demonstrates R axis deviation and some ST depressions in leads II, III, and aVF. He undergoes a CTPA which demonstrates acute left sided segmental and subsegmental pulmonary emboli as well as enlargement of the pulmonary artery and RV suggestive of R heart strain. He is pending a Doppler US of his extremities. What is your suggestion for next steps in management?

- A. Start Enoxaparin 80mg SC BID
- B. Start Apixaban 5mg BID
- C. Administer systemic thrombolysis
- D. Consult IR for IVC filter insertion

Bonus MCQ 23 – 2025

A 58yM presents with 2 day history of progressive RUQ pain to the ER. He has a history of alcohol use disorder but reports he is sober x 8 yrs, denies ETOH, nausea and vomiting. He has mild scleral icterus and tender hepatomegaly on exam. No leg edema or asterixis. BP 156/78, HR 90, Afebrile.

Labs: Hgb 72, MCV 97, Plt 118, Total Bili 52 (unconj 46), Cr 110, LDH 682, retics 123, DAT negative, Haptoglobin <0.01 (low), INR 1, Alb 38, AST 72, ALT 78, ALP 82. Urinalysis is bland. US Liver demonstrates mild perihepatic ascites, hepatosplenomegaly with thrombosis of the Right hepatic vein.

Which of the following tests is most likely to yield diagnosis:

- a) Serum ethanol level
- b) Lipase and CT to rule out pancreatitis
- c) Peripheral blood for flow cytometry
- d) sPEP with immunofixation
- e) Bone marrow aspiration

BONUS MCQ Answers

1) D – follicular lymphoma. Clinical presentation not in keeping with aggressive lymphoma given duration of adenopathy and symptoms. No features of myeloma. Insufficient lymphocytosis to meet criteria for CLL. Hodgkin lymphoma tend to present as cervical/mediastinal adenopathy with anatomical spread to adjacent nodes. Most likely is follicular lymphoma, indolent course - many patients do not require treatment at diagnosis (treatment indications based on GELF criteria).

2) D - AL Amyloidosis: diagnosed with tissue (fat pad biopsy or biopsy of affected organ) showing apple green birefringence on Congo Red stain. Syndrome not compatible with myeloma, plasmacytoma or POEMS.

Note: AL amyloid can cause an acquired FX deficiency à abnormal coags.

3) A - Thrombosis Canada Guidelines recommend no thrombophilia testing for provoked VTEs; elevated HCT and thrombocytosis reactive to intercurrent illness; APLA not likely/indicated with otherwise provoked VTE, normal aPTT.

No cirrhosis on CT, so not recommended from ASH guidelines.

4) E - vWF level and activity increases during pregnancy and need to be repeated following pregnancy in individuals with + BAT (bleeding assessment tool: MCMDM-1 bleeding assessment tool score > 4)

5) A - This may be a case of HIT. Even not knowing his original plt count, his 4T score is 4 based upon thrombosis and timing (CABG = HIGH doses of heparin while on pump). Enoxaparin / LMWH would be contraindicated while doing HIT workup.

6) C - Adverse events associated with IVIg are usually mild and transient and include changes in blood pressure, tachycardia, mild flu-like symptoms, and headache. More serious adverse events are rare and include acute renal failure, aseptic meningitis, acute anaphylactic reactions, and hemolytic anemia (group A, and group AB particularly).

BONUS MCQ Answers

- 7) B - Answer: B - HbSC disease is characterized by less anemia and pain crises, and more ocular and bone complications. The blood film of target cells with 'folded' red blood cells is consistent with HbSC. In patients with HbSS the blood film would show sickle cells (even if not in a crisis), and patients typically have more encounters with the healthcare system. In oxidative hemolysis from G6PD deficiency other findings are seen (i.e. bite cells) and there would be a trigger. In patients with sickle cell trait target cells are not typical, and ocular complications are uncommon.
- 8) C – Myelofibrosis - Answer: Myelofibrosis – classic blood film and high LDH with mild pancytopenia. CMML would have some evidence of dysplasia and differential showing high monocytes. ET can progress to MF, but by itself should have thrombocytosis. Metastatic lung cancer is a possibility, but in the absence of symptoms on the stem this is less likely.
- 9) A - Bone marrow biopsy and aspirate – this presentation is concerning for potential amyloidosis with peripheral nervous system involvement. The next best test would be BM to confirm presence of clonal plasma cell population and look for amyloid deposits. IgM is higher-risk so annual SPEP alone is inappropriate. Serum free light chains and 24-hour urine protein may be useful (the latter particularly if renal involvement) but a BM is the most high-yield.
- 10) C – Myelodysplastic syndrome - Presentation consistent with secondary MDS given history of radiation and chemotherapy, and pancytopenia with dysplastic features in neutrophils - "hypolobulated" and "poor granulation". Myelofibrosis would present with leukoerythroblastosis on blood film. No blasts seen to raise concern for AML.
- 11) Answer: B Delayed hemolytic transfusion reaction from Rh incompatibility – this is the best answer, as non-ABO antigen groups are common causes of delayed hemolytic transfusion reactions. ABO incompatible transfusions cause acute hemolytic transfusion reactions. Idiopathic AIHA is possible, but less likely from the stem. There is no thrombocytopenia to suggest TTP.

BONUS MCQ Answers

- 12) Answer: C Exchange transfusion – an acute ischemic stroke in SCD is an indication for exchange transfusion, and with a Hb >90 this can definitely be done. Saline is helpful but not the most important (but will help dilute HbS percentage). Hydroxyurea can be a long-term management (though this patient should start a chronic exchange transfusion program).
- 13) B – this hemophilia patient is presenting with potential hemarthrosis with right knee swelling. “Factor First” should be a guiding principle whenever a patient presents with hemophilia – administer factor concentrate with any suspected bleeding and subsequently order investigations. Platelets have no role in hemophilia management unless patient is thrombocytopenic. Arthrocentesis can be considered in large/symptomatic effusions but is controversial and would not be the first step.
- 14) D – pancytopenia with severe reticulocytopenia in an otherwise young, healthy patient suggests a bone marrow failure syndrome. Isolated anemia and severe reticulocytopenia may be in keeping with PRCA (primary or secondary to parvovirus B19 infection) but would not affect other lineages. Hodgkin lymphoma often presents with lymphadenopathy and constitutional symptoms but not pancytopenia.
- 15) C – Patient presenting with lower extremity skin ulcers, no clear other etiology likely secondary to Hydroxyurea. Difficult to manage side effect. Must stop hydroxyurea. Refer for wound care. Compression stockings. Indication to switch to alternate agent (EX: JAKi) given need for ongoing cytoreduction. Already had Doppler US thus CTA less informative, no new clot to be anticoagulated, and skin biopsy less likely to be informative given known drug association.
- 16) C – Arrange for 3 doses of IV Venofer. Ongoing symptoms + anemia, indication for intravenous iron. Little benefit in switch to EOD dosing at this point or dose increase of oral. Given anemia 2/2 IDA, preferred first line treatment = IV iron over transfusion.

BONUS MCQ Answers

17) B – Patient on Warfarin presenting with life threatening bleed + emergent procedure meets indication for reversal of anticoagulation. IV Vitamin K alone too slow (reversal of anticoagulant effect may take hours vs. PCC may be effective within 30 minutes). INR of 3.2, correct dose of PCC 2000U + concurrent administration of IV Vitamin K.

18) C – This patient presents with clinical features suggestive of pneumonia that is likely bacterial given the negative COVID/Resp Viral PCR (possibly *Mycoplasma*). Tbili is elevated, and previous bloodwork was hemolyzed suggestive of hemolysis. In this setting + rash (livedo reticularis), clinical picture most suspicious for a cold autoimmune hemolytic anemia. MCV elevated which in this case may be reflective of reticulocytosis in setting of hemolysis. Degree of anemia and acuity less likely VB12 deficiency alone. Other cell lines normal, and acuity also less suggestive of MDS. Anemia chronic inflammation/critical illness more likely to be normocytic, and would not cause elevations in hemolysis parameters.

19) B – IRIDA is a rare form of inherited iron deficiency that impairs iron absorption. Patients may present as being non responsive to oral iron, and typically achieve better improvement with IV therapy. In this case, our patient's Hb and ferritin appear to have responded well to oral therapy making this less likely. SC trait typically non anemic and normal MCV. Anemia of chronic disease similarly normocytic. MCV in the 60s more suggestive of possible thalassemia. Alpha thal requires confirmation by genetic testing.

20) C – This patient presents with a likely recurrence of ITP, previously steroid responsive. Appropriate first line treatment options here include steroids or IVIG. Would prefer steroids in this case in absence of significant bleeding, but IVIG also reasonable and her risk of hemolysis as a group O patient less high. No indications for PLT transfusion in ITP in the absence of severe life threatening bleeding.

BONUS MCQ Answers

21) D. Thrombocytopenia in a critically ill leukemia patient may be multifactorial, and a lack of appropriate PLT response i.e platelet refractoriness, also has many possible causes. Key point in this scenario is a lack of appropriate platelet response (< 10 – 15 point increment) within 1 hour of PLT transfusion which is highly suggestive of an immune mediated cause of PLT refractoriness like platelet alloimmunization secondary to anti-HLA or anti-HPA antibodies. This patient is at risk given his multiple transfusions throughout admission. DIC, sepsis, splenic sequestration are all examples of non-immune mediated causes, but should still see some initial immediate response to transfusion in these cases. Absence of low fibrinogen and deranged coags less suggestive of DIC. Myelosuppression as well should have an initial response rather than immediate refractoriness within one hour.

22) A. Start Enoxaparin 80mg SC BID. Enoxaparin is dosed 1mg/kg BID or 1.5mg OD – based on weight of 79kg, 80 SC BID appropriate upfront dosing. In this scenario, R heart strain and elevated trop in the absence of hemodynamic instability not an indication for thrombolysis and risks likely outweigh benefits (+ recent surgery/trauma within last 3 weeks at least a relative contraindication for thrombolysis). While he may have evidence of a DVT, he has no clear contraindications to immediate anticoagulation that would warrant IVC filter insertion at the moment. DOAC not necessarily contraindicated, though some may suggest up front LMWH for large volume VTE. Apixaban requires 10mg BID x 7d up front dosing for acute VTE.

23) D – Peripheral blood for flow cytometry. He has a DAT-negative Hemolytic Anemia with acute hepatic vein thrombosis. Hemolysis + thrombosis suggests PNH even though there is no hemoglobinuria. While alcohol can cause acute marrow suppression and hepatitis it wouldn't explain hemolysis. Pancreatitis is not suggested by stem nor would it explain hemolysis. Bone Marrow can be considered in the workup and diagnosis of PNH but flow cytometry is required to demonstrate loss of CD55, CD59.

Thank You to our Generous Sponsors!

GOLD Sponsors



Aryaehr.com

EMR Software built by doctors for doctors



johnstonshawinc.com

Medical Professional Insurance
& Financial Service Solutions